

REFERENCE POINTS AS INFORMATION^{*}

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Abstract

We develop a model of reference dependence based on the idea that reference points provide *comparative information*. When people are uncertain how much they value an option, they may still be able to tell whether it is better or worse than a reference point on each dimension, thus obtaining ordinal comparative information. We show that value uncertainty and comparative information generate familiar regularities such as thinking in gain-loss categories; avoiding losses; income targeting and bunching; a preference for advantages over tradeoffs; behavioral attenuation; simultaneous overreaction to small and underreaction to large changes; and—when allowing for caution regarding value uncertainty—the endowment effect. The model predicts that reference effects are most pronounced when value uncertainty is high and the reference point well-known. We develop a theory of endogenous reference point formation in which the reference point is given by the option that delivers the most informative comparison for valuation. This yields several novel predictions, including when the reference point is determined by the status quo, expected outcomes, or similarity-based memory. Finally, we formulate a theory of endogenous choice bracketing, showing narrow brackets can be optimal when they produce sufficiently precise comparative information.

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1 Introduction

A central insight of behavioral economics is that valuation and choice are often reference dependent. In the dominant class of models used to explain this phenomenon, people know how much utility they get from each option, and the reference point affects that utility.

This paper proposes a model in which reference dependence arises because reference points provide *comparative information*. We start from the premise that people are often uncertain about the utility value of options or actions. For instance, they may not know how to translate attributes into utility, they may be uncertain about the characteristics of a good, or the good may be too complex to evaluate. However, even when absolute utility assessments are difficult, relative assessments—comparisons with reference points—may still be feasible, in particular ordinal ones. For instance, I may be unable to assess the absolute utility of \$90 today, but I know that it is less than the utility of the \$100 I typically make. Such partially ordinal comparisons provide information that—in the presence of value uncertainty—generates reference dependence.

To illustrate, consider the classic Tversky and Kahneman (1991) improvements vs. tradeoffs effect. I must choose between two new jobs. Relative to my current one (the reference), Job A pays more for fewer hours (two improvements); Job B pays even more, but for more hours (tradeoff). While I am unsure of the utility I will derive from these different attributes, I do know the utility of my current wage and hours. Conducting an ordinal comparison of Job A's uncertain utilities to the known ones allows me to rule out low-utility realizations: in both dimensions, the distribution of possible utility realizations gets truncated from below. This increases my posterior evaluation of Job A in both dimensions—a “boost.” For Job B, instead, the tradeoff means my utility beliefs get truncated from below in one dimension but from above in the other, pulling the posterior evaluation in opposite directions. This increases the probability of choosing Job A over B. That positive and negative dimension-by-dimension comparisons raise or lower utility evaluations is the hallmark characteristic of reference dependence, yet here it arises purely through an informational channel—the decision maker is uncertain about utility but can generate an (at least partially ordinal) comparison of each option with the known reference point.

We show that analyzing reference points through the lens of value uncertainty and comparative information generates many other classic reference dependence effects, such as thinking in gain-loss categories, avoiding losses, income targeting, and bunching. We argue that this modeling approach has several advantages. First, it offers an explanation for why reference effects might exist in the first place. Second, it highlights a tight relationship between reference dependence and value uncertainty, with associated new comparative statics predictions. Third, the model links reference dependence to a range of other behavioral phenomena—from con-

trast effects to behavioral attenuation—within the same framework. Fourth, treating reference points as information allows us to develop a new theory of reference point formation: in our model, the endogenous reference point is the option that delivers the most informative comparisons for subsequent decisions; this generates sharp predictions about when the reference point is given by expected outcomes, the status quo, similarity-based memory, or other options. Fifth, an informational view of reference points allows us to endogenize the choice bracket (narrow vs. broad). Finally, this approach also allows simple welfare calculations.

Our information-based approach does not rule out reference-dependent feelings. In our model, we deliberately leave out such preferences to isolate the explanatory power of information alone, and we show that some of its predictions coincide with those of traditional models while others do not. We discuss the relationship between the two approaches in Section 8.

Why uncertainty and comparative ordinal information? Our approach to studying the implications of value uncertainty and (partly ordinal) comparative reasoning has a direct analogue in a long line of work in cognitive psychology. As we review in Appendix A, a core insight in cognitive psychology is that people generally exhibit large uncertainty about absolute magnitudes (such as how warm, long, loud or bright a stimulus is) and implicitly use comparisons with reference objects to inform their judgments, suggesting that cognition is optimized for detecting differences rather than encoding absolute levels (Laming, 1984, 2009; Braida and Durlach, 1972; Garner, 1953). For example, according to famous visual illusions, a circle looks smaller when it is surrounded by big circles than by small ones. Importantly, this body of work shows that while comparisons are easier than absolute evaluations, ordinal comparisons are especially precise (Luce and Green, 1974; Parducci, 1965; Laming, 1997). For example, we generally cannot perfectly tell *how much* larger one circle is compared to another one but we can often easily tell which is bigger. This evidence has led to the view that even ostensibly absolute cardinal judgments are often constructed from underlying (largely ordinal) comparisons (Braida and Durlach, 1972; Stewart et al., 2005, 2006).

Our model builds on these key ideas. Just like people exhibit uncertainty in the perception of absolute magnitudes, they may also exhibit uncertainty in the evaluation of choice options, and just like visual and sensory systems make use of ordinal comparisons with reference objects to judge absolute magnitudes, people’s internal valuation system may also make use of ordinal comparisons with known reference points. For example, it is much easier to assess that following a pay raise one’s utility must increase than it is to estimate the precise absolute utility consequences from the raise. Key for our argument is not that comparative judgments are necessarily all ordinal—as we discuss below, what matters is that they are *also* ordinal (or especially precise around the reference point), even if comparisons are partly cardinal in nature.

Setup. The model rests on three core premises. First, the decision maker (DM) is uncertain about how to value option attributes. We are agnostic about whether this uncertainty reflects preference uncertainty, aggregation uncertainty, or something else. The DM holds a prior over the utility contribution of each attribute. As is standard in the literature, the DM observes a cardinal absolute signal about the utility of the option in each dimension.

Second, the DM has access to a reference point, the utility of which is fully or partially known. For example, the DM may have experienced an outcome, and thus (possibly imperfectly) remembers its utility. We assume that in each dimension, the DM receives a *comparative ordinal signal* about whether the outcome is better or worse than the reference point. This information implies that our DM effectively partitions outcomes into gain-loss categories. We allow this ordinal signal to be noisy such that the DM gets the correct signal with higher probability when the utility difference between the option and the reference point is large. This implies that, on average, our DM responds not only to ordinal comparisons but also to cardinal differences.

Third, our DM is Bayesian about value uncertainty and how information is used. Our model thus builds on the “Bayesian brain” literature in cognitive science (Anderson, 2013; Oaksford and Chater, 2007; Gershman, 2021; Griffiths et al., 2024). However, what matters for our model is not so much that the DM is literally Bayesian, but rather that they combine imperfect utility information with comparative signals, and recognize the informational value of reference points.

Reference dependence, contrast and attenuation. Focusing on Normally distributed priors and signals, we show that posterior means admit a simple structure: the sum of an *absolute* utility evaluation of the outcome and a *relative*, reference-dependent component that depends on the distance between the outcome and the reference point as well as uncertainty terms. This reference-dependent term captures the positive or negative informational boost from the comparison.

This structure parallels classic models of reference dependence, in particular those combining absolute and relative utility (e.g. Sugden, 2003; Kőszegi and Rabin, 2006; Wakker, 2010). Yet in our framework such reference dependence emerges purely through informational mechanisms—the weight the DM places on the reference dependent component is a function of their uncertainty about different objects. First, a DM with no value uncertainty exhibits no reference dependence. This predicts that reference effects will be more pronounced when a decision is difficult or unfamiliar. Second, the weight decreases when there is limited information about the utility value of the referent. This formalizes the widespread intuition that reference points must “sink in” to affect behavior.

Relative to the benchmark of no value uncertainty, the DM’s perceived utility is excessively sensitive around the reference point—where gain-loss categorization shifts—but is otherwise

insufficiently sensitive to variation in outcomes. Therefore, our DM overreacts to small changes around the reference point but underreacts to large ones. Intuitively, uncertainty induces belief compression toward the prior, leading to behavioral attenuation, except around the reference point, where ordinal information produces a contrast effect. The model, therefore, naturally accounts for the sometimes-puzzling empirical evidence on the coexistence of insensitivity and overreaction (e.g., Augenblick et al., 2025; Graeber et al., 2025).

Avoiding losses, income targeting, and bunching. While our model features no asymmetry between gains and losses, it nonetheless predicts a class of behaviors traditionally attributed to loss aversion. Intuitively, losses receive an evaluation penalty because negative comparisons with the reference point reduce beliefs about the option’s value through the comparative boost.

The simple insight that crossing the threshold from gains to losses produces a disproportionately large effect on utility evaluations implies that our informational model explains various empirical regularities that emphasize the importance of losses. This includes the improvements-vs.-tradeoffs effect and systematic choice reversals that occur when an attribute crosses the threshold into the loss domain. The common thread that runs through these contexts is that a reference point is manipulated (or introduced) in such a way that the number of gains and losses changes across choice options.

Another example is income targeting in labor supply decisions (e.g., Camerer et al., 1997; Abeler et al., 2011) or, more generally, bunching at reference points (e.g., Pope and Schweitzer, 2011; Allen et al., 2017). Intuitively, our DM doesn’t really know the utility of different outcomes but the ordinal comparative signals imply that they do know whether something is better or worse than the reference point whose utility they know. Thus the DM’s perceived marginal utility is higher around the reference point than elsewhere. Hence, over a range of wages, our DM will work “just enough” to exceed the reference income level and avoid the negative informational boost from getting less. This mechanism produces bunching around the reference point and a locally downward-sloping labor supply curve—key stylized facts documented in the literature. This mechanism is not too dissimilar from how Prospect Theory generates targeting and bunching, except that here it has a purely informational origin.

Links to other phenomena. We further show that the same mechanism—value uncertainty and comparative ordinal signals—can give rise to a range of well-established behavioral phenomena which are not traditionally viewed as related to reference dependence. These include the asymmetric dominance effect and tallying heuristics, which come about through a desire to avoid options that are perceived as losses. They also include various contrast effects which reflect people’s ability to clearly interpret ordinal comparative information, as in studies on sequential contrast effects, the evaluability effect, or the “coherence” part in coherent arbitrariness.

Which reference point? Because in our model reference points only provide information, we propose that individuals use as reference points the alternative that is, on average, most useful for the current decision through the comparative information it facilitates. We understand this as an as-if account: individuals need not consciously choose reference points; rather, just as we typically bring to mind memories relevant to the situation at hand, we also bring to mind useful reference points. For example, when assessing the value of a high-quality wine, we naturally compare it to other high-quality wines we have tried, rather than to the worst ones. Our objective is not to propose a new type of reference point but, rather, to make novel predictions about which classic reference point is used when.

Formally, this becomes a problem of optimal information choice. The key tension is a bias-variance tradeoff: all else equal, the comparative information from reference points is most useful when they are close to the expected outcome (else they always produce the same comparative signal, which is not very helpful) and when they are well-known (else the information contained in the comparison is weak). This tradeoff implies that when all potential reference points are equally well-known, the most informative reference point is the expected outcome, akin to Gul (1991) or Kőszegi and Rabin (2006). On the other hand, when the options near the expected outcome are not well known, a well-known option, like the status quo, can be more informative.

We also link our view of reference points as information to work on similarity-based memory. We consider environments in which the outcome distribution is itself unknown, such as when the DM does not know the average utility of the objects in a store. In this setting, we prove that with Normally distributed signals, the most informative reference point in the presence of a contextual cue is given by a formula that exactly matches the canonical similarity-based memory functional form used in psychology and economics (Kahana, 2012; Bordalo et al., 2020). This shows that the assumptions of value uncertainty and informative comparisons provide a possible microfoundation for work on memory-based reference points.

Endogenous choice brackets. When do people bracket narrowly (e.g., financial wealth) vs. broadly (total wealth)? As we did for reference points, we propose that people adopt the bracket that is most informative for choice, on average. While narrow brackets lead the DM to neglect correlations across dimensions, they can nonetheless be superior for choice when they produce very precise comparative information. For instance, precise comparative information about how our current financial wealth compares to a reference point is easy to discern from a Vanguard account, while generating a comparative signal about how today's total wealth compares to last year's total wealth is more difficult and noisy. Similarly, comparative information may be more precise under narrow bracketing because the quantities involved are smaller. In this spirit, we

show that the most informative bracket can be narrow when the resulting comparative information is sufficiently more precise. This prediction is supported by the experimental evidence in Oprea et al. (2026) who show that when given a direct choice between narrow and broad brackets the vast majority of experimental subjects choose the narrow bracket and explain this choice by pointing to the greater simplicity of the resulting comparison.

Caution. Finally, we extend our model to incorporate *caution*—uncertainty aversion with respect to one’s own value uncertainty, akin to a form of ambiguity aversion. With this additional assumption, our model captures the endowment effect and other forms of status quo bias.

Welfare. Our approach dramatically simplifies welfare calculations. The behavioral welfare literature has reached no consensus on whether to include the gain-loss component of reference-dependent utility models. The central question is whether it reflects *experienced utility*, that is, true “pleasure” and “pain” that should be part of welfare, or merely what is sometimes called *decision utility*, an aspect that matters for choosing but does not affect the enjoyment of options (Kahneman et al., 1997). Such difficulties do not arise with our approach. In our model, the utility function has an unambiguous interpretation, and reference points only provide information—they only affect decision utility.

Related literature. Our paper contributes to the literature on the cognitive foundations of behavioral economics. We connect cognitive noise and uncertainty to reference dependence. Relative to existing models of cognitive noise (e.g., Gabaix, 2014, 2019; Woodford, 2020; Kőszegi and Matějka, 2020; Khaw et al., 2021; Frydman and Jin, 2022; Vieider, 2024), we add reference points as information. The closest antecedent is Villas-Boas (2024), who models a DM who receives an ordinal signal relative to a reference point in the context of risky choice, but does not examine the applications or endogenous reference points we consider. Our approach is also similar to that of Natenzon (2019) and Shubatt and Yang (2024) who model DMs that learn from other choice options through comparisons. In Augenblick et al. (2025), people understand the direction in which they need to update beliefs, but not the magnitude. In Woodford (2012), efficient coding yields a steeper slope of perceived utility around the prior, yet his model does not study reference points. We also relate to the extensive literature on incomplete preferences or preference imprecision and their implications.¹

Second, we build on the enormous literature on reference dependence (e.g. Barberis, 2013), including reference point as expectations (Loomes and Sugden, 1986; Gul, 1991; Kőszegi and

¹Among recent contributions, see Masatlioglu and Ok (2005, 2014); Sagi (2006); Butler and Loomes (2007, 2011); Ok et al. (2015); Chakraborty (2021); Nishimura and Ok (2026); Cerreia-Vioglio et al. (2024).

Rabin, 2006) or similarity-based memory (Bordalo et al., 2020). A standard reading of this literature is that reference points affect utility. For example, Kőszegi and Rabin (2006) note that “a person’s utility depends not only on consumption . . . but also on a ‘reference’ consumption level.” We study how far a purely informational model can go in explaining reference phenomena. As noted, this does not rule out reference-dependent preferences or loss aversion. Interestingly, classic papers already allude to the presence of a broader cognitive mechanism. Kahneman and Tversky (1979) note that “the perceptual apparatus is attuned to the evaluation of changes or differences,” pointing to the role of comparative cognition. Similarly, while their classic paper on the endowment effect adopts a hedonic interpretation of reference effects, Kahneman et al. (1990) also point to the importance of value uncertainty, noting that “The valuation ambiguity produced by [...] lack of commensurability is necessary [...] for both loss aversion and a buying-selling discrepancy.” Recent experimental evidence is consistent with the broad idea that reference effects increase with value uncertainty. Graeber & Enke (2026) show that contrast and assimilation with respect to comparison points both weaken when decision uncertainty is experimentally reduced.

The paper proceeds as follows. Section 2 illustrates the core intuition of our model. Sections 3 and 4 present the general model and apply it to different regularities. Section 5 endogenizes the reference point. Section 6 studies endogenous bracketing, Section 7 caution, and Section 8 concludes. The appendix includes additional results and all proofs.

2 An Illustrative Example

To build intuition, we begin with a sketch of a simple special case. Suppose a DM evaluates the utility of a one-dimensional option, $U(x) = u_x$. They are uncertain about the value of u_x , and are Bayesian with a Normally distributed prior, $u_x \sim \mathcal{N}(\tilde{u}_x, \sigma_p^2)$. This is represented by the dashed line in the top left panel of Figure 1.

The DM receives two types of information about u_x . First, through deliberation, a cardinal signal, $s = u_x + \varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$. Here, σ_ε^2 reflects the DM’s uncertainty in assessing the utility. Second, the DM has a reference point r with utility u_r , and receives an ordinal signal o that tells him whether the alternative is better or worse than the reference point: $u_x - u_r \gtrless \nu$. For now, assume that the utility of the reference point is known and that $\nu = 0$, meaning that the ordinal comparison is noiseless.

From the cardinal signal s , the DM obtains a posterior that follows the standard Normal-Normal shrinkage formula, with mean $\mathbb{E}[u_x|s] := \tilde{u}_x^s = \lambda s + (1 - \lambda)\tilde{u}_x$ and variance σ_x^2 . This posterior is illustrated in grey in the left panel of Figure 1. What happens with an ordinal signal?

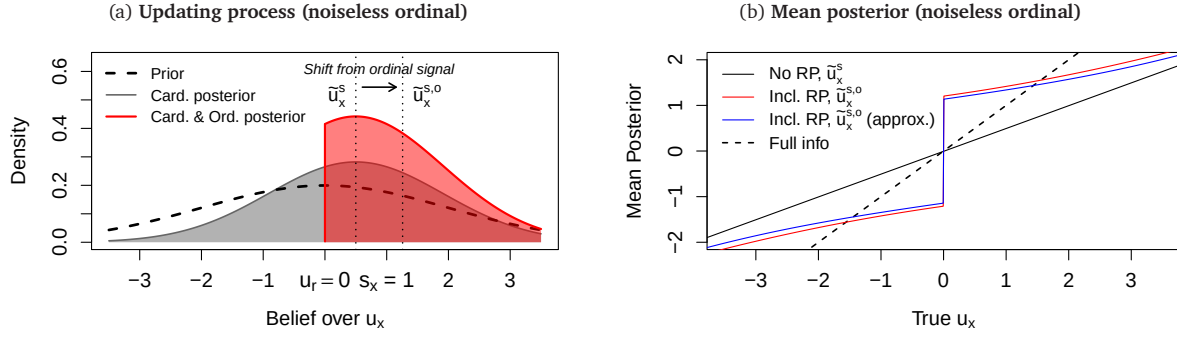


Figure 1: Panel (a) shows an example of a prior (dashed black); posterior after cardinal signal $s_x = 1$ (grey distribution, mean at left dotted line); and posterior after adding the ordinal signal ($u_x > u_r$), which truncates the distribution (in red). The arrow at the top illustrates the resulting ‘boost’ to the mean, which shifts from $\tilde{u}_x^s$ to $\tilde{u}_x^{s,o}$. Panel (b) plots the mean posterior as a function of u_x fixing $u_r = \tilde{u}_x = 0$. The figure compares the mean posterior with only the cardinal signal (in black), including the ordinal signal (in red), a piece-wise linear approximation (in blue), and the full-information benchmark (dashed).

The belief gets truncated, from above or from below depending on the ordinal signal. In case of a positive ordinal signal, this gives us the density depicted in red in the top left panel of Figure 1. Naturally, this “boosts” the posterior mean, which is given by

$$\mathbb{E}[u_x | s, o] = \underbrace{\tilde{u}_x^s}_{\text{Ref.-indep. component}} + \underbrace{\sigma_x}_{\text{Weight of ref.-dep. comp.}} \underbrace{\psi\left(\frac{\tilde{u}_x^s - u_r}{\sigma_x}, o\right)}_{\text{Gain-loss component}} \quad (1)$$

While the exact formula for the gain-loss component $\psi(\cdot)$ involves the inverse Mills ratio, it admits a very good piecewise linear approximation:

$$\psi\left(\frac{\tilde{u}_x^s - u_r}{\sigma_x}, o\right) \approx \begin{cases} \max\left\{\sqrt{\frac{2}{\pi}} - \frac{2}{\pi\sigma_x}(\tilde{u}_x^s - u_r); 0\right\} & \text{if } u_x > u_r \\ -\max\left\{\sqrt{\frac{2}{\pi}} + \frac{2}{\pi\sigma_x}(\tilde{u}_x^s - u_r); 0\right\} & \text{if } u_x < u_r. \end{cases} \quad (2)$$

The top right panel of Figure 1 plots the average posterior mean as the true value of u_x varies. Two key forces are evident in this figure: *attenuation* away from the reference point, and a *comparative boost* (contrast effect) around the reference point. This reflects how the posterior mean in Eq. (1) is the sum of two components. First, an *absolute* (reference-independent) term that reflects Bayesian shrinkage, combining the cardinal signal with the prior. This generates the standard Bayesian attenuation of perceived utility to variation in the option’s objective value. Second, a *relative* or *comparative*, reference-dependent component. Intuitively, a signal that the

utility of the option is above that of the reference point boosts the mean belief because it rules out low utility values. The weight of the gain-loss component endogenously reflects the DM’s value uncertainty—in our model, there is no reference dependence when the DM perfectly knows u_x .

The model sketched here assumes that the utility of the reference point is perfectly known and that the ordinal signal is always correct. When we relax these assumptions below, the mean posterior becomes smoother, but the patterns of contrast and attenuation remain.

Which reference point? We propose that the endogenous reference point is the option that is *most informative* for choice. This is an as-if formulation—clearly, individuals do not consciously choose their reference points. But just as people are known to retrieve memories that are roughly useful for the situation at hand, they may also retrieve reference points that are, on average, most useful for assessing the options they face.

As an example, suppose the DM knows they will want to assess the utility u_x of an alternative x and minimize the Mean Squared Prediction Error. The value of u_x is drawn from a Normal distribution with known mean $\tilde{u}_x$ and variance σ_x^2 . Two reference points are available. First, the status quo r_{SQ} , which has a known utility value but is biased—its value differs from the expected outcome $\tilde{u}_x$ by some amount b . Second, an alternative reference point r_A , whose average value coincides with the expected outcome $\tilde{u}_x$, but its utility is not perfectly known, with variance $\sigma_{r_A}^2$. Assume for now that the DM expects no further information and no ordinal noise (both assumptions are relaxed below). Which reference point provides more information? The status quo is the most informative reference point if and only if its bias satisfies

$$b \leq b^* \approx \frac{3}{2}\sigma_x \sqrt{\log \left(1 + \frac{5}{4} \cdot \frac{\sigma_{r_A}^2}{\sigma_x^2} \right)}.$$

The status quo is the more informative reference point whenever its utility is not too far from the expected outcome, and when the uncertainty about the expected outcome is sufficiently large. Conversely, when the expected outcome has no uncertainty attached to it, it is always the most informative reference point. The intuition is simple. Reference points that are too low or too high will almost always yield the same ordinal comparison, which would be uninformative. On the other hand, uncertainty about the reference point’s utility diminishes its informational value. These two forces create a bias-variance tradeoff, captured by the formula above.

Moreover, we show below that once the decision maker does not know the distribution of outcomes but has access to a contextual cue, a well-known formula from research on similarity-based memory turns out to characterize the informationally optimal reference point.

3 Formal Framework and Reference Points as Information

3.1 Framework: Objects of Uncertain Values

A decision maker (DM) faces one or more alternatives defined by n dimensions. Each option x returns a total utility of

$$U(x) = \sum_{i=1}^n u_{x,i},$$

where $u_{x,i}$ denotes the utility of x from dimension i , for $i = 1, \dots, n$. For example, the item x in question may be a car, and the various $u_{x,i}$ may be the utility from speed, seat comfort, *etc.*

The DM knows this structure, but is uncertain about each $u_{x,i}$.² This uncertainty can have different sources. First, the DM may be uncertain about the actual characteristics of the car, such as its true speed. Second, the DM may be uncertain about the utility from a given characteristic: they may know that the car’s maximum speed is 280 km/h, but may be unsure of the utility this provides. Third, the DM may struggle to aggregate across dimensions. We remain agnostic about the source of the uncertainty.

The individual is Bayesian and holds priors about each unknown variable. Throughout the paper, for any random variable z , denote by $\tilde{z}$ the mean belief and $\tilde{z}^A$ the posterior mean after information A . For tractability, we assume that the prior about each $u_{x,i}$ is distributed according to $\mathcal{N}(\tilde{u}_{x,i}, \sigma_{p,i}^2)$ and that values are independent across dimensions and across goods ($u_{x,i}$ and $u_{y,j}$ are independent for all i, j and all x, y that are distinct).

In addition to the knowledge implicit in the prior, the DM also receives *cardinal* information about the utility in each dimension, which can be interpreted as the outcome of a deliberation process. For each $i = 1, \dots, n$, the DM receives a signal $s_{x,i} = u_{x,i} + \varepsilon_{x,i}$, where $\varepsilon_{x,i} \sim \mathcal{N}(0, \sigma_{\varepsilon,i}^2)$.³ When no confusion arises, we will drop the subscript x .

Before proceeding, three remarks are in order. First, we focus on Normal priors and signals for simplicity. As will be clear, while the exact functional forms of our results rely on Normality—mostly, because truncated Normal distributions are well-behaved—the core intuition generalizes. Second, we assume additively separable utility across dimensions, again for simplicity; our results naturally extend, suitably adapted, for example to a multiplicative structure. Third, in this paper we focus on deterministic multiattribute choice; the Conclusion discusses an extension to choice under risk.

²For evidence on such uncertainty see, among many others, Butler and Loomes (2007); Cubitt et al. (2015); Agranov and Ortoleva (2017); Enke and Graeber (2023); Enke et al. (2027); Halevy et al. (2023).

³Each $\varepsilon_{x,i}$ is assumed to be independent of all utilities and other primitive errors.

3.2 Reference Points as Information

We study situations in which the DM evaluates option x while having (at least mental) access to another object, r , with utility $u_{r,i}$ in the same n dimensions. Here, we treat r as the reference point; later, we show how the same framework can be used to model situations in which r is something else, such as another option in the choice set. We maintain reference points as exogenous for now and endogenize them in Section 5.

We consider both situations in which the utility of r in each dimension is fully known, and when it is not. We assume that the DM believes $u_{r,i} \sim \mathcal{N}(\tilde{u}_{r,i}, \sigma_{r,i}^2)$ for all i . This uncertainty may arise from limited experience with the reference point or from memory imperfections. Prior to receiving any further information, $u_{r,i}$ and $u_{x,i}$ are treated as independent. We discuss the case of correlation in Section 4.

Our innovation is that the DM receives *ordinal* signals, in each dimension, about whether it is the reference point r or the item at hand x that has higher utility. We allow these signals to be noisy, and assume that the DM obtains signal $o_{x,i}$ with values

$$o_{x,i} = \begin{cases} + & \text{if } u_{x,i} - u_{r,i} \geq v_{x,i} \\ - & \text{if } u_{x,i} - u_{r,i} < v_{x,i} \end{cases} \quad (3)$$

where $v_{x,i} \sim \mathcal{N}(0, \sigma_{o,i}^2)$ for $i = 1, \dots, n$.⁴ (We will drop the subscript x when no confusion arises.) The parameter $\sigma_{o,i}^2$ captures the noise in the ordinal signal, which we assume is independent of everything else. As we discuss below, a feature of this form is that the probability of receiving the correct ordinal signal increases in the distance between the true utility of the option and the reference point. Not only does this correspond to the Probit model, it also aligns with standard principles in psychology and economics on just noticeable differences. In general, we should expect the noise in ordinal comparisons to vary across dimensions: little in comparing car MPGs, more in qualitative dimensions like seat comfort.

To sum up, our model encompasses three types of uncertainty. Continuing with the car example, and assuming one relevant dimension (seat comfort): You have a prior about the utility of owning a car and get a cardinal signal by looking at the new car, both of which have uncertainty ($\sigma_{p,i}^2$ and $\sigma_{\varepsilon,i}^2$). You additionally have access to the utility from a comparison car—for example the one you currently own—and that you think you like pretty well, but you’re not entirely sure how well ($\sigma_{r,i}^2$). Moreover, you think that the target car has higher seat comfort than your current

⁴One may also want to consider the event in which $u_{x,i} - u_{r,i} = v_{x,i}$ as separate and associated to a signal “=.” Since this is a probability-zero event (given our absolutely continuous distributions), to avoid extra notation and since nothing changes, for simplicity, we leave this possibility aside and associate this event with the signal “+.”

car, but again, you're not fully certain ($\sigma_{o,i}^2$).

Cardinal comparative information? In our model, comparative information is ordinal: the DM can tell whether $u_{x,i} \geq u_{r,i}$ but not by how much. While this resonates with a large literature in psychology emphasizing that ordinal reasoning is especially simple and precise (see Appendix A), there are clearly also contexts in which people have a sense for *how much* better or worse an option is compared to a reference quantity. We now discuss to what extent our model already captures this intuition, and how it can easily be extended to capture it in full.

As we mentioned, in our model the probability of a correct ordinal signal increases in $|u_x - u_r|$. Therefore, average posterior beliefs already vary with the cardinal utility difference in the same way as in a model with a cardinal comparative signal. This aspect of our formulation is deliberate, to allow for cardinal differences to affect comparative information in a simple and tractable way without giving up the importance and tractability of ordinal information.

Should one wish to incorporate a direct cardinal comparative signal into our setup, this is straightforward as well. Concretely, suppose that in addition to the ordinal signal, the DM also receives a cardinal comparative signal, $d_i = u_{x,i} - u_{r,i} + \zeta_i$, where $\zeta_i \sim \mathcal{N}(0, \sigma_{d,i}^2)$. It turns out that our current model is informationally *equivalent* to one in which the cardinal comparative signal d_i is also received, simply by adapting the parameters, under mild assumptions on the relevant covariance matrix. It is also straightforward to extend our results to incorporate an unrestricted version of this signal in a simple and tractable way. We discuss both points in Appendix B.1. The bottom line is that the results developed below do not rely on the assumption that *all* comparative information is ordinal, only that *some* of it is.

3.3 Perceived Utility with Reference Points

Given the additively separable structure of $U(x)$ and independence across dimensions, we can focus on each dimension separately, as the multi-dimensional case is simply the sum across dimensions; therefore, we suppress the subscript indicating the dimension (we write u_x for $u_{x,1}$).

A standard result in Bayesian updating gives us that, if $\sigma_p^2 + \sigma_\varepsilon^2 > 0$, the posterior after the cardinal signal is $\mathcal{N}(\tilde{u}_x^s, \sigma_x^2)$, with $\tilde{u}_x^s = \lambda s + (1 - \lambda)\tilde{u}_x$, where $\lambda := \frac{\sigma_p^2}{\sigma_p^2 + \sigma_\varepsilon^2}$, and $\sigma_x^2 := \frac{\sigma_\varepsilon^2 \sigma_p^2}{\sigma_\varepsilon^2 + \sigma_p^2}$. How does the ordinal signal affect it? The following proposition characterizes its effects on posterior means.⁵ In what follows, let ϕ and Φ denote the standard normal density and cumulative distribution function, respectively.

⁵Proposition 11 in Appendix B.2 characterizes the posterior variance. It shows that, holding variances fixed, its average over ordinal signals, using the DM's predictive probabilities conditional on s , is *increasing* in $|\tilde{u}_x^s - \tilde{u}_r|$, strictly so if $\sigma_x^2 > 0$.

Proposition 1. Suppose $\sigma_x^2 + \sigma_r^2 > 0$. Then,

$$\mathbb{E}[u_x|o, s] = \tilde{u}_x^s + \frac{\sigma_x^2}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}} \psi\left(\frac{\tilde{u}_x^s - \tilde{u}_r}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}, o\right) \quad (4)$$

$$\text{where } \psi(t, o) := \begin{cases} \frac{\phi(t)}{\Phi(t)} & \text{if } o = + \\ -\frac{\phi(-t)}{\Phi(-t)} & \text{if } o = -. \end{cases} \quad (5)$$

To illustrate, consider the case of a “+” ordinal signal. Then,

$$\mathbb{E}[u_x|+, s] = \underbrace{\tilde{u}_x^s}_{\text{Ref.-indep. component}} + \underbrace{\frac{\sigma_x^2}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}}_{\text{Weight of ref.-dep.}} \underbrace{\frac{\phi\left(\frac{\tilde{u}_x^s - \tilde{u}_r}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}\right)}{\Phi\left(\frac{\tilde{u}_x^s - \tilde{u}_r}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}\right)}}_{\text{Gain-Loss component}} \quad (6)$$

The posterior mean after the cardinal and ordinal signals is the sum of two components. First, an *absolute* (reference-independent) term that reflects the DM’s mean belief combining the cardinal signal with the prior through a Bayesian shrinkage, giving rise to attenuation.⁶ Second, a *relative* or *comparative*, reference-dependent component: a “boost” in beliefs obtained from the “+” ordinal signal, where the weight of the boost depends on the DM’s uncertainty. This formula is the generalization of the one used in our example in the previous section, Eq. (1), allowing for uncertainty about u_r and noise in the ordinal signal, and giving the correct expression of the boost rather than an approximation.

Through the comparative boost, the mean belief is contrasted away from the reference point. Formally, this boost is given by the so-called inverse Mills ratio that has attracted much attention in the econometrics literature on correcting for sample selection bias (Heckman, 1979). Boosts have the expected signs: it is immediate to show that $\psi(t, +) > 0$ and $\psi(t, -) < 0$ for all t . As the formula shows, however, how much this happens depends on the various uncertainties—about u_x , u_r , and about the ordinal signal.

Proposition 1 describes beliefs conditional on signal realizations. While this makes the mechanics clear, signal realizations may not be observed by the researcher. Perhaps a more empirically relevant metric is average beliefs across possible signal realizations: Figure 2 illustrates the average mean posterior and average reference-dependent component for different levels of σ_x , fixing the reference point and varying u_x . (Proposition 12 in Appendix B.2 provides a full

⁶We understand attenuation as $\partial \mathbb{E}_{s,o} [\mathbb{E}[u_x|s, o]] / \partial u_x < 1$.

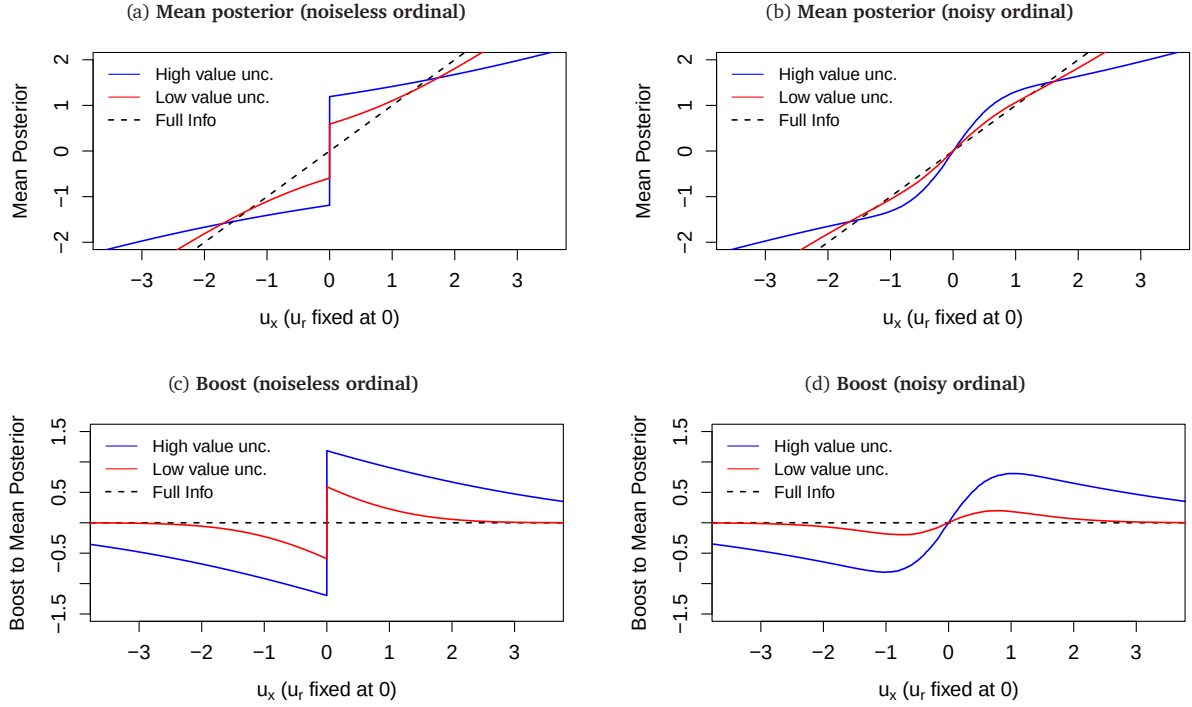


Figure 2: Illustration of the mean posterior (top panels) and its reference-dependent boost component (bottom panels) as a function of value uncertainty σ_ε , separately for noiseless and noisy ordinal signals. We set $\tilde{u}_x = u_r = \tilde{u}_r = 0$ and then vary u_x (and thus s_x). All panels compare a ‘low value uncertainty’ DM ($\sigma_\varepsilon = 0.75$, red line) with a ‘high value uncertainty’ DM ($\sigma_\varepsilon = 2$, blue line), where we hold constant $\sigma_p = 2$. The left panels (a, c) illustrate the case with a *noisy reference, noiseless ordinal* signal ($\sigma_r = 0.25, \sigma_o = 0$). The right panels (b, d) illustrate the case with a *noisy reference, noisy ordinal* signal ($\sigma_r = 0.25, \sigma_o = 0.5$). The top row (a, b) plots the mean posterior, $\tilde{u}_x^{s,o}$, while the bottom row (c, d) plots the reference-dependent ‘boost’, $\psi(\cdot)$. Black dashed lines indicate the full information benchmark. Simulation conducted over 10,000 draws.

characterization of average beliefs.) We will now study how the DM’s mean posterior (which we also refer to as “evaluation”) varies as a function of u_x and the different types of uncertainty in the model, $\sigma_x^2, \sigma_r^2, \sigma_o^2$; we discuss these results at an intuitive level and then provide formal statements. Before doing so, however, we briefly discuss a useful approximation.

Applying the Model: Approximations. While the formula in Proposition 1 involves the Inverse Mills ratios, it also admits a very simple approximation based on a first-order Taylor approximation around $t = 0$, very useful in applications:

$$\psi(t, o) \approx \psi^a(t, o) = \begin{cases} \max \left\{ \sqrt{\frac{2}{\pi}} - \frac{2}{\pi} t ; 0 \right\} & \text{if } o = + \\ \min \left\{ -\sqrt{\frac{2}{\pi}} - \frac{2}{\pi} t ; 0 \right\} & \text{if } o = -. \end{cases} \quad (7)$$

This approximation appears to work remarkably well. Appendix C.1 derives this approximation as well as an approximation of average boosts across signal realizations, and shows simulations of their performance.

Varying the value of the option. Increasing the value of the option impacts beliefs through two channels: by changing the distribution of the cardinal signal received about u_x , and by changing the probability of receiving a positive ordinal signal.

Conditional on a positive ordinal signal, as we increase u_x , $\tilde{u}_x^s$ is (on average) higher, but the boost is (on average) *lower*. This is because $\psi(t, +)$ is decreasing in t , and is also intuitive: the impact of learning that u_x is above u_r should be smaller if you already know that u_x is likely above u_r . Symmetrically, with a negative signal, increasing u_x increases the magnitude of a negative boost. Panel (c) of Figure 2 illustrates these effects on the boost absent ordinal noise. Below u_r , an increase in u_x increases the magnitude of the negative boost. At u_r , there is a discontinuous jump as the ordinal signal switches from negative to positive. Above u_r , further increases reduce the magnitude of the positive boost.

With ordinal noise, the effect of an increase in u_x is smoother, as illustrated by panel (d) of Figure 2. In general, the maximal average boost occurs strictly above u_r : too close to u_r , incorrect ordinal signals are more likely, so the average boost is smaller.⁷

The total effect of changes in u_x combines the effect on the boost with that on the reference-independent portion of beliefs. The combination can be seen in the top panels of Figure 2: while average beliefs about u_x are strictly increasing in its true value, beliefs are attenuated when far from the reference point and excessively sensitive close to it, where the implicit change in gain-loss categorizations through the ordinal signal produces a contrast effect.

Taken together, this combination gives rise to systematic over- or under-estimation, depending on how close u_x is to u_r . Suppose $u_x > u_r$. As long as the ordinal noise is not too big and priors are aligned, the average posterior belief will be strictly above the true u_x when u_x is close to u_r , as the boost dominates. As the difference between them increases, the mean belief will eventually go below the true value—the boost becomes weaker and attenuation dominates.

Varying uncertainty. Proposition 1 makes it clear that the reference-dependent component in our model also depends on (i) uncertainty about the option, σ_x^2 , which depends on both prior (σ_p^2) and signal (σ_ε^2) uncertainty; (ii) uncertainty about the utility of the reference point, σ_r^2 ; and (iii) how noisy the ordinal information is, σ_o^2 . This highlights a central implication of our approach and a key difference from classic models of reference dependence: the strength of the

⁷Another way to illustrate this is through a heatmap of the full distribution of the mean posterior for a given value of $u_x - u_r$; this is included in Figure 7 in Appendix B.2.

gain-loss component depends on uncertainty. On the other hand, uncertainty also affects the reference-independent component, as it changes the Bayesian shrinkage coefficient λ .

The red and blue lines in Figures 2.(c) and (d) illustrate how value uncertainty about the option modulates the effects of the reference-dependent boost. A key insight is that the boost (positive or negative) is more pronounced when x is less well-known. As a result, the degree of reference dependence—such as the strength of the contrast effect around the reference point—*increases* in value uncertainty. This is a central prediction of our model.

As discussed above, our model typically predicts overestimation of u_x when it is right above the reference point and underestimation when it is far above. Changes in uncertainty σ_x^2 have two effects on this over- and underestimation. First, as we just saw, it increases the boost. Second, the Bayesian attenuation becomes more pronounced. As a result, an increase in value uncertainty will generally *push up* the posterior mean close to the reference point, yet *push it down* far away from the reference point. That is, it amplifies both the contrast *and* attenuation effects. This is illustrated by the red and blue lines in the top panels of Figure 2.

While the strength of reference effects *increases* in value uncertainty about the option, it typically *decreases* in uncertainty about the reference point (σ_r^2) or about the ordinal signal (σ_o^2). Intuitively, if I barely know what the utility of the reference point is, it also doesn't help me much to know that the option is better; and if I know that a signal is noisy, I react less. More formally, observe from Eq. (4) how σ_r^2 and σ_o^2 affect the boost in exactly the same way conditional on a given ordinal signal. If the ordinal signal is unexpected ($\tilde{u}_x^s - \tilde{u}_r$ and o have different signs) or if there is at least some uncertainty ex-ante ($|\tilde{u}_x^s - \tilde{u}_r|$ not too large), then the boost is decreasing in both; this is very intuitive, as greater uncertainty yields smaller updating.

Maybe more surprisingly, when the ordinal signal is confirmatory and widely expected ($|\tilde{u}_x^s - \tilde{u}_r|$ is large and o has the same sign), the boost can be *increasing* in both σ_r^2 and σ_o^2 . As we explain in Appendix B.4, this is because, if $\tilde{u}_x^s$ is a long way above $\tilde{u}_r$ and there is no noise in the ordinal signal, a positive signal is barely informative—the DM already knows that x is better than r . However, the less r is known, the more uncertain its values are, and the more informative the signal is. A similar logic applies to σ_o^2 : when averaging over signal realizations, increasing the ordinal noise σ_o^2 also increases the probability that contradictory signals are received. This “flattens out” the average boost, as can be seen when comparing panels (c) and (d) of Figure 2.

Formal statements of comparative statics. We conclude by stating formally the comparative statics discussed above; Appendix B.4 provides additional comparative static results for the reference dependent boost in Eq. (4), both conditional on and averaging across signal realization.

Proposition 2. Assume $\tilde{u}_x = \tilde{u}_r = u_r$ and $\sigma_p^2, \sigma_\epsilon^2 > 0$.

1. (Over and under-sensitivity w.r.t. u_x .) $\mathbb{E}[\tilde{u}_x^{s,o}]$ is strictly increasing in u_x . Moreover, there exists $0 < \underline{\sigma} < \bar{\sigma}$:

(a) if $0 < \sigma_o < \underline{\sigma}$, there exist $0 < \varepsilon < \varepsilon'$ such that

$$\underbrace{\frac{\partial \mathbb{E}[\tilde{u}_x^{s,o}]}{\partial u_x} > 1 \quad \text{if } |u_x - u_r| < \varepsilon,}_{\text{Oversensitivity near RP}} \quad \underbrace{0 < \frac{\partial \mathbb{E}[\tilde{u}_x^{s,o}]}{\partial u_x} < 1 \quad \text{if } |u_x - u_r| > \varepsilon'}_{\text{Attenuation away from RP}}$$

(b) if $\sigma_o \geq \bar{\sigma}$, $0 < \frac{\partial}{\partial u_x} \mathbb{E}[\tilde{u}_x^{s,o}] < 1$ for every u_x .

(c) If $\sigma_o = 0$, $\mathbb{E}[\tilde{u}_x^{s,o}]$ is discontinuous at $u_x = u_r$, while $0 < \frac{\partial}{\partial u_x} \mathbb{E}[\tilde{u}_x^{s,o}] < 1$ (attenuation) for every $u_x \neq u_r$.

2. (Over- and under-estimation.) With the same $\underline{\sigma}$ as in Part 1, if $0 < \sigma_o < \underline{\sigma}$, there exists $\Delta > 0$ such that

$$\underbrace{\mathbb{E}[\tilde{u}_x^{s,o}] > u_x \text{ for } u_x \in (-\infty, u_r - \Delta) \cup (u_r, u_r + \Delta),}_{\text{Overestimation right above RP or much below}} \quad \underbrace{\mathbb{E}[\tilde{u}_x^{s,o}] < u_x \text{ for } u_x \in (u_r - \Delta, u_r) \cup (u_r + \Delta, \infty).}_{\text{Underestimation right below RP or much above}}$$

If $\sigma_o \geq \bar{\sigma}$, then $\mathbb{E}[\tilde{u}_x^{s,o}] < u_x$ if $u_x > u_r$, and $\mathbb{E}[\tilde{u}_x^{s,o}] > u_x$ if $u_x < u_r$.

3. (Effects of uncertainty.) Assume $u_x > u_r$; signs are reversed when $u_x < u_r$.

(a) If $0 \leq \sigma_o < \underline{\sigma}$ there exists $\Delta_r > 0$ such that

$$\frac{\partial E[\tilde{u}_x^{s,o}]}{\partial \sigma_r^2} < 0 \quad \text{if } 0 < u_x - u_r < \Delta_r, \quad \frac{\partial E[\tilde{u}_x^{s,o}]}{\partial \sigma_r^2} > 0 \quad \text{if } u_x - u_r > \Delta_r;$$

(b) $\frac{\partial E[\tilde{u}_x^{s,o}]}{\partial \sigma_p^2} > 0$ and there exists $\Delta_\varepsilon \geq 0$ such that

$$\frac{\partial E[\tilde{u}_x^{s,o}]}{\partial \sigma_\varepsilon^2} < 0 \quad \text{if } u_x - u_r > \Delta_\varepsilon;$$

Varying the reference point. Appendix B.5 discusses comparative statics with respect to the reference objects in the model, u_r (the true utility of the reference point) and $\tilde{u}_r$ (the belief about the utility of the reference point). Unlike in Prospect Theory, these two objects can be separated in our model and exhibit opposite effects on utility evaluations: they decrease in u_r (a contrast effect), yet increase in $\tilde{u}_r$ (an assimilation effect).

3.4 Multiple Comparators

We focused thus far on a single option evaluated in the presence of a reference point. However, in practice, many contexts will give rise to multiple comparisons. For example, when multiple options are present, with or without a reference point, the DM may be able to generate comparative signals also among the available options. Perhaps more intriguingly, people may make use of multiple reference points. Most models of reference-dependent preferences posit a single reference point. However, Kahneman (1992) already argued that outcomes can be evaluated relative to distinct reference points not integrated into a single benchmark, and experimental evidence supports this possibility (Boles and Messick, 1995; Sullivan and Kida, 1995; Ordóñez et al., 2000; Koop and Johnson, 2012; Lu et al., 2015; Terzi et al., 2016). A small literature, mostly in psychology, has developed prospect-theoretic models in which multiple reference points jointly shape choice (Wang and Johnson, 2012; Lu et al., 2015). One advantage of our approach to reference points as sources of information is that it naturally extends to multiple reference points: the DM then receives multiple ordinal signals and updates according to Bayes' rule.

When comparators are known. Suppose first the DM noiseless binary ordinal signals relative to a finite set C of perfectly known comparators, and $\sigma_x > 0$. Focus on a specific dimension. If $u_r < u_x$ or $u_r > u_x$ for all $r \in C$, then the situation collapses to that in which the DM has a single comparator equal to the largest or smallest u_r , respectively; we can then simply use the formulas derived above. Otherwise, let $u_* := \max\{u_r : r \in C, o_{x,r} = +\}$ and $u^* := \min\{u_r : r \in C, o_{x,r} = -\}$, and the posterior is simply $N(\tilde{u}_x^s, \sigma_x^2)$ truncated to $[u_*, u^*]$. Thus

$$\mathbb{E}[u_x | o, s] = \tilde{u}_x^s + \sigma_x \frac{\phi\left(\frac{u_* - \tilde{u}_x^s}{\sigma_x}\right) - \phi\left(\frac{u^* - \tilde{u}_x^s}{\sigma_x}\right)}{\Phi\left(\frac{u^* - \tilde{u}_x^s}{\sigma_x}\right) - \Phi\left(\frac{u_* - \tilde{u}_x^s}{\sigma_x}\right)}.$$

Comparators that are not perfectly known. When the utilities of comparators are not perfectly known or when ordinal information is noisy, formulas become more complex. In Balakrishnan and Dean (2025), we derive the solution for two comparators; we include the precise expressions in Appendix B.6. Independently of the specific formulas, however, posteriors with multiple comparators have certain clear regularities. Most importantly, as the following Proposition makes clear, adding a reference point above or below the value of two ranked alternatives compresses their perceived value difference, while adding one between their two values increases it (the ordinal signals push them further apart).

Proposition 3. *Let u_x, u_y, u_z be independent and normally distributed with strictly positive vari-*

ances. Then

$$\begin{aligned}\mathbb{E}[u_x - u_y | u_x > u_y] &> \mathbb{E}[u_x - u_y | u_x > u_y > u_z] \\ \mathbb{E}[u_x - u_y | u_x > u_y] &> \mathbb{E}[u_x - u_y | u_z > u_x > u_y] \\ \mathbb{E}[u_x - u_y | u_x > u_y] &< \mathbb{E}[u_x - u_y | u_x > u_z > u_y].\end{aligned}$$

4 Applications

We now discuss several implications of our approach to reference dependence. Our model involves no gain-loss asymmetry. However, because ordinal comparative signals make beliefs about utility values substantially more sensitive near the reference point than far from it, we will show that our DM exhibits various behaviors traditionally attributed to loss aversion. Intuitively, our DM will not pick options that involve “losses,” or negative comparisons, simply because they give rise to lower beliefs about the resulting utility. We show that this same mechanism can also generate several other well-documented phenomena in behavioral economics. We begin with three classes of behaviors: Avoiding losses in discrete choice, bunching, and contrast effects.

4.1 Avoiding Losses in Discrete Choice

A hallmark of classic reference dependence is that, when asked to choose between two alternatives, people favor the one that does not involve losses. Our model generates the same effect, and so replicates many paradigmatic examples of reference dependence in discrete choice, albeit with new comparative static predictions.

Shifting Attributes Into the Loss Domain. Perhaps the cleanest example of ‘loss avoidance’ is that a DM becomes less likely to choose an alternative when a change in the decision environment turns one of its attributes from a gain into a loss, holding relative values fixed. Consider two two-dimensional options, x and y , with $u_{x,1} > u_{y,1} > u_{r,1}$ and $u_{y,2} > u_{x,2} > u_{r,2}$, so that both options initially lie above the reference point in both dimensions. Suppose that subjects initially choose x and y with roughly equal frequency. Now raise the reference point in dimension 1 so that y becomes a loss in that dimension, with $u_{x,1} > u_{r',1} > u_{y,1}$. A robust finding in the reference-dependence literature is that y is then chosen less often than x .

Our model trivially generates this “loss penalty” effect as long as the ordinal signal is not too noisy. To build an intuition, consider the simplest case: no ordinal noise, perfectly known utility of the reference point, and that the DM uses only the ordinal information stemming from the

reference point (i.e., they do not use ordinal information from the comparison of the two choice alternatives); we generalize all of these later.

First, note that changing the reference point from r to r' must *decrease* the perceived value of y . The reason is that changing the reference point only affects the gain-loss component of perceived utility, by turning a positive boost into a negative one. Using the approximations in (7), the change in utility evaluations of y is roughly $\sigma_{y,1} \left(-2\sqrt{\frac{2}{\pi}} + \frac{2}{\pi} \left(\frac{u_{r',1} - u_{r,1}}{\sigma_{y,1}} \right) \right)$.

The effect on x is instead *positive*. Since x continues to be above the reference point, its evaluation continues to receive a positive boost, which is now even stronger—recall that boosts are stronger the closer the item to the reference point (see Figure 2.c). Again using our approximations, the valuation increases by about $\frac{2}{\pi}(u_{r',1} - u_{r,1})$.

Taken together, the perceived value of y decreases while that of x increases for any cardinal signal realization, thus y is chosen less often. The following proposition shows that this intuition generalizes, assuming that ordinal noise is not too large. For any option x and point r , let $o_{x,r}$ denote the vector of ordinal signals comparing x and r in each dimension.⁸

Proposition 4. *Suppose $n = 2$, and consider two options x, y and two reference points r, r' such that*

$$u_{r,1} < u_{y,1} < u_{r',1} < u_{x,1}, \quad u_{r',2} = u_{r,2} < u_{x,2} < u_{y,2}.$$

Suppose x and y have strictly positive prior and cardinal-signal noise variances in every dimension, reference-point priors are correctly centered and have the same variances, and all ordinal signals have identical variances σ_o^2 . Then there exists $\bar{\sigma}_o^2 > 0$ such that, for every $\sigma_o^2 < \bar{\sigma}_o^2$,

$$\Pr \left[\mathbb{E}[U(x) \mid s, o_{x,r}] > \mathbb{E}[U(y) \mid s, o_{y,r}] \right] < \Pr \left[\mathbb{E}[U(x) \mid s, o_{x,r'}] > \mathbb{E}[U(y) \mid s, o_{y,r'}] \right].$$

The probabilities are over signal realizations, holding the stated true utilities fixed.

In Appendix B.7 we show that the same holds true when the DM also takes into account ordinal comparisons between x and y on each dimension.

Improvements versus Tradeoffs. The same mechanism also gives rise to the classic ‘Improvements versus Tradeoffs’ effect. As discussed in the Introduction, a robust empirical regularity is that, starting from a reference point—such as a status quo—individuals exhibit a preference for options that imply improvements rather than tradeoffs relative to that reference point (e.g.,

⁸We assume that the DM treats each ordinal comparison distinctly - i.e. they compare r to x along each dimension, when evaluating x , then compare r to y along each dimension, when evaluating y . This ignores the fact that, if the utility of r is known with noise, then comparing x to r will update beliefs about r which should be taken into account when evaluating y . While we believe this set up to be psychologically plausible, the result remains true if this channel is switched on, as well as if all cross boosts are taken into account, as we show in Appendix B.7.

Tversky and Kahneman, 1991). For instance, suppose that a DM chooses between two two-dimensional options, x and y . There are two potential reference points, r and r' . From r , option x implies improvements in both dimensions, whereas option y implies tradeoffs (one improvement and one loss). Suppose that from r' the pattern reverses. Then, the empirical regularity is that the DM is more likely to choose x when the reference point is r than when it is r' .

Our model can easily generate this, for purely informational reasons, with an argument similar to the one above. Intuitively, the change in the reference point again only affects the boosts, with two improvements giving rise to two positive boosts and tradeoffs producing conflicting boosts. A direct corollary of Proposition 4 gives the required result.

Corollary 1. *Suppose $n = 2$. Maintain the information assumptions of Proposition 4, and suppose*

$$u_{x,1} > u_{r,1} > u_{y,1} > u_{r',1}, \quad u_{y,2} > u_{r',2} > u_{x,2} > u_{r,2}.$$

Then there exists $\bar{\sigma}_0^2 > 0$ such that, for every $\sigma_0^2 < \bar{\sigma}_0^2$,

$$\Pr[\mathbb{E}[U(x) \mid s, o_{x,r}] > \mathbb{E}[U(y) \mid s, o_{y,r}]] > \Pr[\mathbb{E}[U(x) \mid s, o_{x,r'}] > \mathbb{E}[U(y) \mid s, o_{y,r'}]].$$

Notably, our model predicts an improvements vs. tradeoffs effect also when gain-loss asymmetries in preferences play no role, such as in perceptual tasks. In line with these predictions, Trueblood (2015) shows that people exhibit the improvement-versus-tradeoffs effect also in magnitude estimation problems, where reference points cannot affect preferences.

Asymmetric Dominance. While the improvements-vs.-tradeoffs-effect says that choice is affected by *moving* a reference point such that options asymmetrically dominate it, the asymmetric dominance effect says that choice is affected by *adding* a comparison point that is asymmetrically dominated. There are two dimensions, and subjects choose either between x and y , or between x , y , and a third option z , dominated by x in both dimensions but better than y in one dimension. Empirically, adding z increases the probability of choosing x . While the asymmetric dominance effect is not typically thought of as an example of reference dependence, our model naturally gives rise to this behavior.⁹ The channel is the same as that described above: The dominated alternative z provides two positive boosts for the dominating alternative x , but one positive and one negative boost for the alternative y . Indeed, this is a simple corollary of Proposition 3; in Appendix B.8 we state it formally.¹⁰

⁹Natenzon (2019) and subsequently Shubatt and Yang (2024) propose explanations for the asymmetric dominance effect that are conceptually very close, though in their models the signals are not dimension-by-dimension.

¹⁰Our model unambiguously predicts the asymmetric dominance effect because adding the decoy produces a negative boost for y but not for x . Notably, this asymmetry is absent in a second classic violation of the independence

Tallying Heuristics. The above argument makes it clear that the DM in our model favors options that do not involve losses. We now show that, if other sources of information become too noisy, this is the *only* criterion the DM will use: they will choose the option that has the highest number of gains, or lowest number of losses—the so-called tallying (counting) heuristic or Dawes’ Rule (Dawes, 1979; Gigerenzer and Goldstein, 1996).

The DM faces a binary choice between x and r , which may or may not be the reference point. Suppose that the cardinal signals are sufficiently noisy that the DM’s choice is effectively entirely based on the comparative signals. Denote by $D_x \subseteq \{1, \dots, n\}$ the dimensions where x dominates and D_r those in which r dominates. Consider the natural special case in which ordinal signals are noiseless, and the two options have, within each dimension, identical prior and noise variances. This leads to a simple formula based on counting advantages.

Observation 1. Suppose $\tilde{u}_{x,i} = \tilde{u}_{r,i}$ and $0 < \sigma_{p,x,i}^2 = \sigma_{p,r,i}^2 =: \sigma_{p,i}^2 \forall i$ and $\sigma_{o,i}^2 = 0$. Then, for any $u_x, u_r \in \mathbb{R}^n$ such that $u_{x,i} \neq u_{r,i} \forall i$ and $\sum_{i \in D_x} \sigma_{p,i} \neq \sum_{i \in D_r} \sigma_{p,i}$, there exists $\bar{\sigma}_\varepsilon^2$ such that if $\sigma_{\varepsilon x,i}^2 = \sigma_{\varepsilon r,i}^2 > \bar{\sigma}_\varepsilon^2 \forall i$,

$$\mathbb{E}_s[\mathbb{E}[U(x) \mid s, o]] > \mathbb{E}_s[\mathbb{E}[U(r) \mid s, o]] \Leftrightarrow \sum_{i \in D_x} \sigma_{p,i} > \sum_{i \in D_r} \sigma_{p,i}.$$

If $\sigma_{p,i}^2 = \sigma_{p,j}^2 \forall i, j$, the condition on the right is equivalent to $|D_x| > |D_r|$.

When the noise in the cardinal signals is high enough, average evaluations rank the two options by “counting” the number of relative advantages and disadvantages, weighted by the prior uncertainty in that dimension. Intuitively, if priors are the same and no cardinal signals are given, the DM is then left with a collection of positive and negative boosts. If prior uncertainty is the same in each dimension, then the *optimal* choice of a rational Bayesian agent boils down to counting which option has more advantages.

4.2 Income Targeting and Bunching

A canonical implication of reference dependence is bunching right at, or just above, a reference point, such as when workers’ accumulated earnings bunch at an income target (Camerer et al., 1997; Abeler et al., 2011; DellaVigna et al., 2017, 2022) or when the performance of athletes bunches at salient goals or reference points (e.g., Pope and Schweitzer, 2011; Allen et al., 2017).

of irrelevant alternatives: the compromise effect. In it, introducing a decoy produces one additional positive and one additional negative comparison for both target options, x and y . Rather than asymmetrically adding advantages and disadvantages, the compromise-effect decoy makes x the “middle option” in both dimensions. Consistent with the intuition that our model explains decoy effects that reflect asymmetric advantages/disadvantages, our model does not generally give rise to the compromise effect; see Appendix B.8.

We now illustrate that our model also leads to bunching, focusing on the standard example of labor supply; the same intuition holds more generally.

A DM chooses hours a to work at wage $w > 0$. Utility is $U(a, w) = u(aw) - a^2/2$ with $u(aw) = \beta aw$. In general, the DM may exhibit information-based reference dependence in both the earnings and cost of effort dimensions. We focus on the former, which could reflect either more uncertainty in that dimension (uncertainty in the utility of money, in the wage, or in computing total earnings), or that the reference point is more precise in that dimension.¹¹

For each a , the DM has a prior about $u(aw)|a \sim \mathcal{N}(\overline{\beta w a}, \sigma_p^2)$ and receives independent cardinal signals, $s_a = u(aw) + \varepsilon^a$, $\varepsilon^a \sim \mathcal{N}(0, \sigma_\varepsilon^2)$, where $\overline{\beta w}$ could be interpreted as average marginal benefit in the past. We assume the prior mean varies with the action so that the agent already knows, as is realistic, that working more hours delivers higher utility in earnings. The DM has a reference point given by earnings of r , knows that the utility of this earnings level is $u_r = \beta r$, and receives a noiseless ordinal signal o_a about how each $u(aw)$ compares with u_r . (Introducing ordinal noise leads to similar results, though ‘bunching’ is less sharp.)

Since our DM understands monotonicity between the reference point and his earnings, they should also understand the earnings ranking of actions: posterior beliefs should be such they know that working six hours delivers higher utility from earnings than working five. To ensure this, we assume that the DM also receives additional ordinal signals relative to both higher and lower actions; without these, independent error realizations may lead to non-monotone posteriors, which is unrealistic. This is our baseline model. However, computing Bayesian posterior means for this rich signal structure becomes analytically intractable because the many ordinal signals across actions introduce interdependent truncations. Fortunately, a simple tractable alternative closely approximates the posterior mean utility of each action.

A simpler problem. Consider the problem above, but suppose that the DM only receives ordinal signals about how the earnings level associated with each action compares to the reference point but not about how the different actions compare to each other. Instead of these infinitely many ordinal signals, suppose that the error term in the cardinal signals is perfectly correlated across actions— $s'_a = u(aw) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ —yet the DM computes posterior beliefs ignoring this correlation.

Because the error term is the same, posterior beliefs respect monotonicity. Crucially, Appendix C.3 reports finite-grid comparisons of posterior-mean curves under the two specifications. The average posterior mean is remarkably similar to that induced by the original model

¹¹If we assumed the opposite we would obtain effort targeting instead of income-targeting, just like standard loss-aversion models predict effort targeting when loss aversion applies to effort but not to money.

with all ordinal signals, including the boost around the reference point.¹² We will therefore use this alternative, which can be solved using standard tools, to highlight the core features of our original model. For a given realization of ε , the DM's maximization problem is given by

$$\max_{a \in \mathbb{R}_+} \lambda(\beta w a + \varepsilon) + (1 - \lambda)\overline{\beta w a} + \sigma_x \psi \left(\frac{\lambda(\beta w a + \varepsilon) + (1 - \lambda)\overline{\beta w a} - \beta r}{\sigma_x}, o_a \right) - \frac{a^2}{2} \quad (8)$$

where $o_a = +$ if $aw \geq r$, and $o_a = -$ otherwise.¹³ Denote $a^*(w)$ its largest maximizer for any $w > 0$. This is a standard maximization problem except for the discontinuous, positive jump at $a = \frac{r}{w}$. It will be useful to compare its solution to the solution of the equivalent problem without a reference point, call it $a^{\text{no rp}}(w)$.¹⁴

Proposition 5. Suppose $\beta, r, \overline{\beta w}, \sigma_p^2, \sigma_\varepsilon^2 > 0$. For a given $\overline{\beta w}$ and ε , there exist $0 < w_1 < w_2$ such that:

1. (Bunching and downward-sloping labor supply). $a^*(w) = \frac{r}{w}$ and $\frac{\partial a^*(w)}{\partial w} = -\frac{r}{w^2} < 0$ for all $w \in (w_1, w_2)$.
2. (Bunching of earnings). $a^*(w)w$ is strictly increasing for $0 < w < w_1$, has an upward discontinuous jump at w_1 , and is then constant and equal to r for $w \in (w_1, w_2)$.
3. (Comparison with reference-free). $a^*(w) < a^{\text{no rp}}(w)$ for $0 < w < w_1$ and $w > w_2$, $a^*(w) > a^{\text{no rp}}(w)$ for $w_1 < w < w_2$ and $a^*(w_2) = a^{\text{no rp}}(w_2) = \frac{r}{w_2}$.

Figure 3 illustrates these results. Without reference point, the optimal action is strictly increasing in w , with standard behavioral attenuation. Adding reference effects (i) lowers the perceived utility of earnings below r and (ii) raises it at or above r . When wages are low, the negative boost reduces the perceived marginal benefit of earnings, so the DM exerts less effort than without reference points—a discouragement effect. This continues until wages reach w_1 , at which point it becomes worthwhile to “push” to the reference point: both effort and earnings jump upward to attain the earnings target. As wages further increase, however, optimal earnings stay put at r : as long as we are above the reference point, marginal benefits of effort drop

¹²It is intuitive that this is a good approximation, at least on a finite grid: the models are identical except that the former has i.i.d. noise but ordinal signals across actions, and the latter has perfectly correlated noise but no ordinal signals across actions. While ordinal signals of the former model yield monotone posteriors, the latter model achieves them automatically. But then, calibrating the noise terms will lead to a very similar distribution of posteriors outside boundary actions—precisely what the simulation shows.

¹³Recall our simplifying assumption that the DM receives a positive ordinal signal in the case of equality.

¹⁴That is, $a^{\text{no rp}}(w)$ is the solution to $\max_{a \in \mathbb{R}_+} \lambda(\beta w a + \varepsilon) + (1 - \lambda)\overline{\beta w a} - \frac{a^2}{2}$, meaning $a^{\text{no rp}}(w) = \lambda\beta w + (1 - \lambda)\overline{\beta w}$.

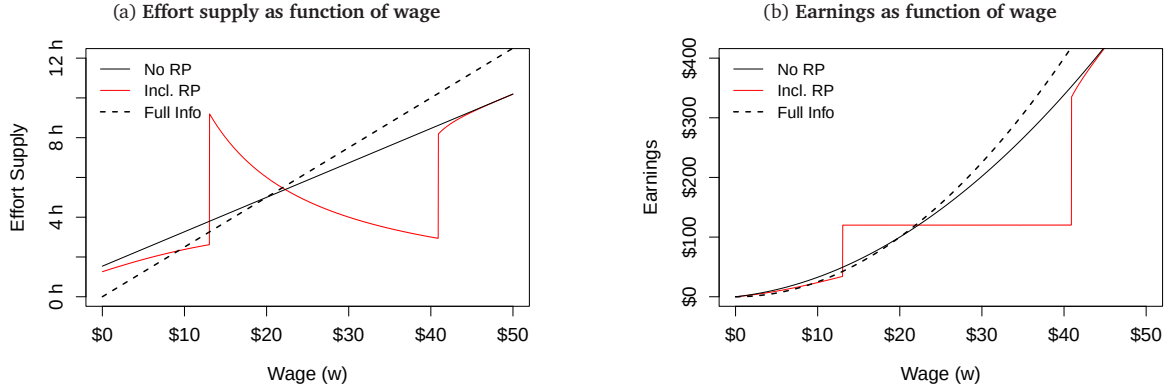


Figure 3: Effort supply and earnings as a function of the wage, with and without reference point. Parameters are $\bar{\beta} = 1$, $\beta = 1$, $\bar{w} = 20$, $\sigma_\varepsilon = 80$, $\sigma_p = 120$, and reference earnings of \$120. Both panels show a single DM with noise realization $\varepsilon = 0$.

and stay low. Hence, $a^*(w) = r/w$ decreases as the wage rises, generating a locally downward-sloping labor supply curve. This continues until wages grow enough that increasing the action is again optimal. For wages far enough away from the reference point, these effects disappear and the action converges to $a^{\text{no rp}}(w)$. Note that, while Figure 3 plots the actions and earnings of a single DM with a given noise realization, averaging across multiple DMs with independent noise realizations would give smoother patterns.

4.3 Contrast Effects

A final set of applications relates to the difference between joint and separate evaluations and the well-documented regularity of contrast effects when options are evaluated jointly (either simultaneously or temporally close).

Consider two options x, r with $u_x > u_r$. The DM has independent identical priors $u_x, u_r \sim \mathcal{N}(\tilde{u}, \sigma_p^2)$, with $\sigma_p^2 > 0$, and observes s_x and s_r , which have the same strictly positive variance σ_ε^2 . When the options are evaluated separately, no ordinal signal is given, and the average valuation difference is

$$\mathbb{E}_{s_x, s_r} [\mathbb{E}[u_x | s_x] - \mathbb{E}[u_r | s_r]] .$$

In joint evaluation the DM also observes the (possibly noisy) ordinal comparison o . The average gap is

$$\mathbb{E}_{s_x, s_r, o} [\mathbb{E}[u_x - u_r | s_x, s_r, o]] .$$

This second gap is strictly larger. To see why, let $D := u_x - u_r$. Its prior is $\mathcal{N}(0, 2\sigma_p^2)$, and $s_x - s_r$ is a cardinal signal about D with noise variance $2\sigma_\varepsilon^2$. Under the identical-prior and identical-noise

assumptions, $s_x + s_r$ is independent of the difference and its signals. The ordinal signal compares D with the known reference value zero. For the fixed positive true difference, the average ordinal correction is strictly positive by Proposition 15 in Appendix B.4. This yields the strict comparison of the two average gaps. This simple insight helps to shed light on three empirical regularities in the literature.

Within- vs. between-subjects experiments. It is well-understood that in within-subjects experiments subjects are more responsive to treatment variations than in between-subjects experiments. For instance, in a review, Charness et al. (2012) note that this is because “subjects have a reference or comparison point when responding to the second question.” Our model formalizes this form of contrast effect.

Evaluability in joint vs. separate evaluation. Suppose an experimental subject is asked to state their WTP for two two-dimensional goods, where each option is strictly better on one dimension. One dimension is very difficult to value (e.g., the units are very unfamiliar) and one relatively easy to value. A canonical result is that, in separate evaluations, the difference in WTP between the two options is largely determined by the easy-to-value dimension; when options are evaluated jointly, the difficult-to-value dimension has a substantial effect as well (Hsee, 1996).

Our model naturally captures this phenomenon. In separate evaluations, the DM only receives cardinal absolute signals in each dimension, and because they are much more noisy in one dimension than the other (producing Bayesian shrinkage to the prior), differences in valuations across the two goods are largely driven by the easy-to-value (less noisy) dimension.

In joint evaluations, the DM evaluates each option by using the other option as a comparison. The ordinal signals will have a larger impact on final valuations in the difficult-to-value dimension because—as shown in Proposition 14 in Appendix B.4—the boost increases in value uncertainty. Intuitively, our DM does not know how to cardinally value the difficult-to-evaluate dimension (meaning it has little impact in separate evaluations), but they do know which option is ordinal better, which helps them understand relative values.

Sequential contrast effects. Much research in psychology and economics documents contrast effects in sequential evaluations, according to which an item is judged as higher-quality (or bigger, brighter, and so on) when the immediately preceding item had relatively low quality; see, for example, Radbruch and Schiprowski (2025) and Bhargava and Fisman (2014) on interviewing, Chen et al. (2016) on judges, and Hartzmark and Shue (2018) on financial markets.

Assume that when the DM evaluates option x , they obtain an ordinal signal comparing it with the previous option r where $u_r < u_x$. Our model trivially generates the implication that a positive comparison raises the posterior mean of x relative to the evaluation based on cardinal

information alone. Furthermore, with positive prior, cardinal-signal, and ordinal-noise variances, increasing the true value u_r while holding the posterior mean $\bar{u}_r$, the average posterior mean of x (a contrast effect), as shown in Proposition 16.

Coherence despite arbitrariness. A final related regularity is the famous observation due to Ariely et al. (2003) that even when subjects’ absolute valuations of an option are fairly arbitrary, their subsequent decisions are internally coherent in the sense that higher prices lead to lower demand, or higher quality to higher willingness-to-pay. Our model provides a natural reason for why evaluations are both noisy (in terms of levels) but coherent in terms of changes: the ordinal signal implies that even when absolute evaluations are highly uncertain, stated valuations will appear coherent because they respect the ordinal ranking implied by the option attributes, assuming that past evaluations serve as a reference point.

4.4 Assimilation and Contrast

Our model assumes that $u_{r,i}$ and $u_{x,i}$ are independent. This is often unrealistic: for instance, learning that the utility of eating 10 apples (the reference bundle) is 1 billion tells the DM that the utility of eating 5 apples is probably also high, for example because they really like apples.

Our model can be easily adapted to accommodate correlation. Suppose that the DM receives a cardinal signal about $u_{r,i}$ and uses this signal to update also about $u_{x,i}$ because the two are correlated. Then, the DM’s mean belief before the ordinal signal ($\tilde{u}_x$) will be given by a standard combination of the prior, the cardinal signal about $u_{x,i}$ and the cardinal signal about $u_{r,i}$. This mean belief will still be normal. Moreover, it is straightforward to adjust the boost from the ordinal signal for the correlation, see Proposition 13 in the Appendix. The impact of an ordinal signal will now depend on the posterior variance of $u_{x,i}$ relative to the covariance between $u_{x,i}$ and $u_{r,i}$. A positive comparison raises $u_{x,i}$ ’s posterior mean if the former is bigger than the latter, leaves it unchanged if the two are equal, and lowers it if the covariance is higher than the variance. The directions reverse for a negative comparison. Positive covariance alone therefore does not imply the usual positive-boost direction.

Extending the model in such a way is not only natural, it also highlights that reference points can produce both contrast and assimilation effects. As discussed above, the comparative signal that our paper focuses on produces a contrast effect—a higher reference point decreases valuations.¹⁵ In contrast, the signal about $u_{r,i}$ in combination with correlation gives rise to an assimilation effect—a higher belief about the utility of the reference point increases valuations of

¹⁵This remains true if $u_{x,i}$ and $u_{r,i}$ are correlated, as long as the variance of $u_{x,i}$ is higher than the covariance between $u_{x,i}$ and $u_{r,i}$

x . The Appendix discusses how this prediction is consistent with recent experimental evidence in neuroscience and economics that separately measures the reference point and beliefs about it.

5 Which Reference Point?

Empirical studies have shown that many alternatives can serve as reference points, including the status quo, expectations, memory, or the outcomes of others. Our model offers a novel approach to predicting which reference point will be used in a given situation. Since we understand reference points as sources of information, it is natural to look at which one is *most informative*, that is, reduces value uncertainty through comparative information and thereby improves subsequent choices. This provides a simple, tractable criterion for endogenizing the reference point, grounded in information theory.

In following this approach, our argument is not that people can literally “choose” a reference point. However, reference point selection need not be a conscious choice; it may instead operate as an unconscious background process, as is common in decision sciences. Much as it is widely accepted that people often unconsciously bring to mind pertinent memories for each situation, they may also use a reference point most relevant to a given choice situation. For instance, when evaluating a high-quality wine, we arguably compare it with high-quality wines we have previously tasted rather than with the worst wine we have ever tried.

We are agnostic about the nature of the mental construct that may serve as a reference point. Past experiences and memories may play this role, but reference points may also be formed through mental simulation or imagination. Formally, unfamiliar or weakly imagined reference points might then be characterized by large reference-point uncertainty.

5.1 The Informational Value of a Reference Point

We study a setting in which the DM knows they will face option x with utility $u_x \sim \mathcal{N}(\tilde{u}_x, \sigma_p^2)$. They could use a reference point r , of which they know $u_r \sim \mathcal{N}(\tilde{u}_r, \sigma_{p,r}^2)$. They also know that, at the time of evaluation, they will receive (i) an ordinal signal o about u_x vs. u_r with noise σ_o^2 and (ii) cardinal signals, $s_x = u_x + \varepsilon_x$ and $s_r = u_r + \varepsilon_r$, with $\varepsilon_x \sim \mathcal{N}(0, \sigma_{\varepsilon_x}^2)$, $\varepsilon_r \sim \mathcal{N}(0, \sigma_{\varepsilon_r}^2)$.¹⁶

¹⁶We focus on one-dimensional objects for simplicity; extending to multiple dimensions is immediate. By considering $\sigma_p^2, \sigma_{p,r}^2$ separately from $\sigma_{\varepsilon_x}^2$ and $\sigma_{\varepsilon_r}^2$, we have a general treatment that allows us to modulate how much additional information is revealed after the choice of reference point. This allows for a general treatment of whether the evaluation of the reference point occurs before memories are accessed (high prior variances $\sigma_p^2, \sigma_{p,r}^2$, low signal noise $\sigma_{\varepsilon_x}^2, \sigma_{\varepsilon_r}^2$), or after they are (i.e., no further signals, $\sigma_{\varepsilon_x}^2, \sigma_{\varepsilon_r}^2 \rightarrow \infty$). Consistent with our notation, let σ_r^2 denote the posterior variance of the belief about r after s_r .

After observing all signals, the DM infers u_x and chooses action a to minimize the Mean Squared Prediction Error (MSPE), as is common in information theory. We can think of a as the agent's estimate of u_x . We then define the benefit of reference point r as the difference in the expected MSPE with and without the ordinal signal it generates (the normalization simplifies the algebra below). That is,

$$V(r) := \underbrace{\mathbb{E}_{s_x} \left[\min_{a \in \mathbb{R}} \mathbb{E}[(u_x - a)^2 | s_x] \right]}_{\text{Expected MSPE without } r} - \underbrace{\mathbb{E}_{s_x, s_r, o} \left[\min_{a \in \mathbb{R}} \mathbb{E}[(u_x - a)^2 | s_x, s_r, o] \right]}_{\text{Expected MSPE using } r}.$$

We now show that this measure admits a particularly simple characterization.

Lemma 1. *Assume $\sigma_x^2 > 0$. Then, the informational value of reference point r in evaluating x is*

$$V(r) = \frac{\sigma_x^4}{\sigma_x^2 + \sigma_r^2 + \sigma_o^2} \mathbb{E}_{s_x, s_r} \left[h \left(\frac{\tilde{u}_r^{s_r} - \tilde{u}_x^{s_x}}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}} \right) \right] \quad (9)$$

where $h(x) := \frac{\phi^2(x)}{\Phi(x)(1-\Phi(x))}$.

The Lemma shows that the informational benefit of a reference point is captured by a simple function that, unsurprisingly, is formally related to that of our boosts—indeed, h is the product of their magnitude in both directions. Note that when no further information is expected (that is, $\sigma_{\varepsilon_x}^2, \sigma_{\varepsilon_r}^2 \rightarrow \infty$), Lemma 1 further simplifies to

$$V(r) = \frac{\sigma_x^4}{\sigma_x^2 + \sigma_r^2 + \sigma_o^2} h \left(\frac{\tilde{u}_r - \tilde{u}_x}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}} \right). \quad (10)$$

5.2 The Most Informative Reference Point

The simple functional form in Lemma 1 leads to a first comparative static on the information benefits of a reference point.

Proposition 6. *If $\sigma_x^2 > 0$, $V(r)$ is strictly decreasing in $|\tilde{u}_r - \tilde{u}_x|$. If, in addition, $\sigma_{p,r}^2 > 0$, $V(r)$ is strictly decreasing in $\sigma_{\varepsilon_r}^2$.*

This result provides a simple rule of thumb to assess the information value of reference points. First, reference points are more informative the more information the DM will receive about them—a small $\sigma_{\varepsilon_r}^2$. This is immediate, as it implies more informative ordinal comparisons.¹⁷

¹⁷This is also the case for prior variance $\sigma_{p,r}^2$, but only if $|\tilde{u}_r - \tilde{u}_x|$ is not too large; see Proposition 7 and ft. 18 below.

Second, reference points are more informative the closer they are to the *mean* of the distribution of u_x , that is, with a small *bias* $|\tilde{u}_r - \tilde{u}_x|$. To see why, note that when the reference point is centered exactly at $\tilde{u}_x$, we have an equal chance of either ordinal signal, and both ordinal signals are moderately informative. If, instead, $\tilde{u}_r$ is far above $\tilde{u}_x$, the ordinal signal $u_x < u_r$ is almost guaranteed and (correspondingly) not very informative, while the ordinal signal $u_x > u_r$ would be very informative, but also very unlikely.

The two criteria in evaluating a reference point—how well-known and how well-centered it is—imply the existence of a bias-variance tradeoff. When all potential reference points are equally well-known, the criteria also have a simple implication: the expected outcome is the most informative reference point. This connects to theories in which the reference point is given by average outcomes or expectations, except that here it arises endogenously and purely for informational reasons.

Expected outcome vs. status quo. The bias-variance tradeoff has direct implications for when the status quo will be the most informative reference point. Consider a DM looking to buy a car. As a reference point, they could use the car they currently own, the status quo (SQ). They know this car well ($\sigma_{p,r_{SQ}}^2 = 0$), but it is likely to be significantly worse than the ones in the store ($\tilde{u}_{r_{SQ}}$ is well below $\tilde{u}_x$). Alternatively, they can use a car that a friend recently bought (A), which they know less well ($\sigma_{p,r_A}^2 > 0$), but is more representative of the cars in the store ($\tilde{u}_{r_A} = \tilde{u}_x$).

The DM has a choice between a backward-looking reference point—the status quo, known well but far from what they expect to have to evaluate—or a forward-looking one—known less well but close to the expected outcome. Our results give a precise rule for which is more informative. The optimal reference point is the status quo if it is not *too* far from what they expect to see in the store. There is an exact, unique threshold: Once $\tilde{u}_{r_{SQ}}$ and $\tilde{u}_x$ are sufficiently far apart, the friend's car A will be more informative. Similarly, the better the alternative reference point A is known (lower σ_{p,r_A}^2), the lower is this threshold. This is formalized below in a simple formula, a direct implication of Lemma 1.

Observation 2. Suppose $\sigma_x^2 > 0$, and consider r_{SQ} and r_A such that r_{SQ} is biased ($b_{SQ} := |\tilde{u}_{r_{SQ}} - \tilde{u}_x| > 0$) but perfectly known ($\sigma_{p,r_{SQ}}^2 = 0$) while r_A is unbiased ($\tilde{u}_{r_A} = \tilde{u}_x$) but imperfectly known ($\sigma_{p,r_A}^2 > 0$). Suppose for simplicity that no further information is expected and that there is no ordinal noise for either ($\sigma_{\varepsilon_x}^2, \sigma_{\varepsilon_{r_A}}^2 \rightarrow \infty, \sigma_{o_{SQ}}^2 = \sigma_{o_A}^2 = 0$). Then, $V(r_{SQ}) > V(r_A)$ if and only if its bias is small enough relative to the uncertainty about x , that is,

$$V(r_{SQ}) > V(r_A) \Leftrightarrow b_{SQ} < b^* \quad \text{where} \quad b^* := \sigma_x h_+^{-1} \left(\frac{2}{\pi} \frac{\sigma_x^2}{\sigma_x^2 + \sigma_{r_A}^2} \right) > 0$$

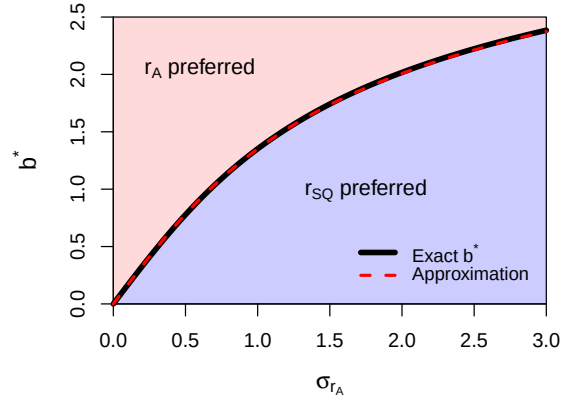


Figure 4: Bias-Variance tradeoff in most informative reference point. r_{SQ} is biased ($b_{SQ} := |\tilde{u}_{r_{SQ}} - \tilde{u}_x| > 0$) but perfectly known ($\sigma_{p,r_{SQ}}^2 = 0$) while r_A is unbiased ($\tilde{u}_{r_A} = \tilde{u}_x$) but imperfectly known ($\sigma_{p,r_A}^2 > 0$). It assumes $\sigma_p^2 = 1$, no further information expected ($\sigma_{\varepsilon_x}^2, \sigma_{\varepsilon_{r_A}}^2 \rightarrow \infty$), and no ordinal noise ($\sigma_{o_{SQ}}^2 = \sigma_{o_A}^2 = 0$).

and h_+^{-1} is the inverse of h restricted to $[0, \infty)$. Moreover, b^* can be well-approximated by

$$b^* \approx \frac{3}{2} \sigma_x \sqrt{\log \left(1 + \frac{5}{4} \cdot \frac{\sigma_{r_A}^2}{\sigma_x^2} \right)}.$$

Figure 4 plots the correct and approximated threshold.

One implication of our results concerns the role of the “tenure” of the status quo. Assuming an option becomes better known the longer it serves as the status quo, our model predicts that tenure increases the likelihood that it will be used as a reference point. Central to this argument is the information value: a “just-adopted” status quo, that is less known, is also a less informative reference point.

Our results naturally speak to the experimental literature on reference points. A frequently made informal argument in this literature is that reference points (in particular expectations-based ones) need to “sink in” to affect behavior (see Heffetz, 2021, for a literature review). From our model’s perspective, the simple explanation is that reference points are adopted only when the DM has accumulated sufficient information about their utility values.

The role of ordinal noise. An additional comparative static result pertains to ordinal noise (and prior variance):

Proposition 7. *Suppose $\sigma_x^2 > 0$. Then there exists $\bar{\delta} > 0$ such that, if $|\tilde{u}_r - \tilde{u}_x| < \bar{\delta}$, $V(r)$ is strictly decreasing in σ_o^2 and in $\sigma_{p,r}^2$.*

As long as the bias is not too big, the DM prefers a reference point with smaller ordinal

noise (and a smaller prior variance). This is intuitive, as less noise in the ordinal signal implies better information.¹⁸ This means that DMs will tend to adopt reference points that allow precise ordinal comparisons, for example, because they are nearby or because they are presented in similar formats and units.

Discussion. In our model, a different reference point can be selected for each choice situation. While theoretically convenient, it may be more realistic for a DM to use the same reference point across a *class* of related choice situations. Our approach could easily be extended.

Second, we assume that reference points can be used at zero or equal cost. In some cases, however, some reference points could be made more easily available (making them more likely to get adopted), for example, in an experiment. Our model could easily be extended by making different reference points differentially costly to use or retrieve.

Third, and related, throughout we restrict to the selection of a single reference point. In general, however, our DM would benefit from using several. Presumably, however, cognitive or other costs limit how many reference points can be effectively used; we see this as a promising direction for future work. Interestingly, the problem becomes similar to the well-studied Scalar Quantization problem, with algorithmic solutions proposed by Lloyd (1982) and Max (1960).

5.3 Cues and Similarity-Based Memory Reference Points

A long literature in psychology formalizes memory through similarity-based recall (e.g. Tulving and Thomson, 1973; Shepard, 1987; Kahana, 2012): essentially, the probability that a memory is recalled is proportional to its similarity to current circumstances. Building on this work, Bordalo et al. (2020) model a DM whose reference point is given by similarity-based memory. In this literature, the reference point is often specified as a function of the exponential of the distance between a contextual cue and a memory trace, $e^{-q(c-m)^2}$, for some constant q , memory trace m and cue c . We now show that our model provides a microfoundation for this formulation.

As noted above, when all potential reference points are equally well-known, the most informative one coincides with the expected (average) outcome. Oftentimes, however, the DM may be unsure of the distribution of outcomes they face and try to infer it from “cues” in the environment. For example, entering a new store, the DM may not know the average utility of

¹⁸Notably, this holds only if the bias is not too large (it is sufficient, for example, that it lies below $\sqrt{\sigma_p^2 - \sigma_x^2}$): when the bias is very large, then the DM may actually prefer some ordinal noise or prior variance about r . The reason is the same as that of the non-monotonicity of the boost w.r.t. σ_r^2 and σ_o^2 discussed above. When the bias is large, the DM has little doubt about the ordinal signal they will get, meaning that they expect little information; however, adding ordinal noise makes this expected ordinal signal *more* informative, because it lowers the (relative) chance of signals from values which are very close.

clothes in this store, but they may observe contextual cues—how fancy is the store, are there mice on the floor? From these cues, the DM infers “what type of store” it is, and, thus, which reference point will be most informative to retrieve. The idea is that the customer has different experiences stored in memory, and the cue provides information about which one will be most useful as a comparator.

Memory cues. Suppose the DM believes $u_x \mid \tilde{u} - x \sim N(\tilde{u}_x, \sigma_x^2)$, where the average outcome is uncertain itself, $\tilde{u}_x \sim N(\hat{\tilde{u}}_x, \sigma_\mu^2)$ with $\sigma_\mu^2 > 0$. Suppose that there is a cue c that is perfectly informative about the average outcome $\tilde{u}_x$, that is $c = \tilde{u}_x$ (this is just for simplicity and easily generalizes). The analyst observes this cue, while the DM observes it with some noise—he receives $c_o = c + e$, where $e \sim N(0, \sigma_e^2)$ with $\sigma_e^2 > 0$. The DM chooses the optimal reference point r^* based on this information.

Imagine that the DM has a wide set R of potential reference points stored in memory—a reference point is available for all posterior means with a common noise level. The DM retrieves the V -maximizing reference point—denote it r^* . Given a cue c , what is the conditional density that they retrieve r^* with $\tilde{u}_{r^*} = m$ from memory?

Proposition 8. *Suppose reference points are available at all possible means and at a common noise level: that is, $\sigma_{p,r}^2$, $\sigma_{\epsilon_r}^2$, and σ_o^2 are the same for all $r \in R$; and for every $z \in \mathbb{R}$, there exists $r \in R$ with $\tilde{u}_r = z$). Let $f(m \mid c)$ denote the conditional density of $\tilde{u}_{r^*}$ at m given cue c , and let $f(m)$ denote its unconditional density. Then*

$$f(m \mid c) = \frac{e^{-\tau(c-m)^2} f(m)}{\int_{\mathbb{R}} e^{-\tau(c-m')^2} f(m') dm'}, \quad (11)$$

where $\tau := \frac{\sigma_\mu^2 + \sigma_e^2}{2\sigma_\mu^2 \sigma_e^2}$.

This result says that the probability of a particular memory becoming the reference point is determined by two terms, both of which are familiar in the memory literature. The numerator is the similarity between the cue and the memory, times the unconditional density. The denominator captures interference—holding fixed the similarity between the cue and the memory of interest, the probability of recalling it decreases in the similarity between the cue and other memories. Thus, any given memory is more likely to be used as a reference point the higher its cue similarity and the lower interference.¹⁹

¹⁹For intuition, notice that, if all potential reference points are available and have the same uncertainty, Lemma 1 tells us that the agent chooses the reference point equal to their posterior beliefs. Introducing a cue c increases the probability of posteriors close to that c —Bayes’ rule and the normal PDF then deliver the formula above.

Remarkably, this functional is precisely the rule commonly used for recall or memory-based reference points in psychology (Kahana, 2012) and economics (Bordalo et al., 2020). The only difference is that here this is derived as optimal for a decision maker who has value uncertainty and uses the informational value of reference points.²⁰

Summary. Viewing reference points through the lens of the most informative comparator generates several natural predictions and links to earlier literature:

1. When the DM knows the distribution of outcomes and all potential reference points are equally well known, the most informative reference point is as close as possible to the expected outcome.
2. When the DM knows the distribution of outcomes, yet some potential reference points are better known than others, a bias-variance tradeoff exists. If, for example, the status quo is known much better than the expected outcome, then it is the most informative reference point if it is not too biased.
3. When the DM does not know the (mean of the) distribution of utility outcomes and all potential reference points are equally well known, the most informative reference point is determined by similarity-based recall.
4. In addition to the above, the DM will choose reference points that are easier to compare—for example, they are expressed in the same units, or in similar formats, or adjacent.

6 Endogenous Choice Bracketing

A large literature in behavioral economics and finance emphasizes the importance of choice brackets and asks at what level of aggregation people compare outcomes with reference points (Read et al., 1999; Rabin and Weizsäcker, 2009). The typical finding is that people *bracket narrowly*, evaluating outcomes against reference points one at a time even when they are inter-related. We now study what the approach of viewing reference points as information has to say about the optimal selection of choice brackets. A main insight is that narrow brackets can improve choice when the comparative signals they produce are more precise than the comparative signals that would be generated under broad bracketing. For example, it is typically easy for investors to discern precise comparative signals about individual assets or portfolios from their brokerage account, as these prominently display how much assets have gained or lost in value. In contrast, assessing changes in total wealth (which is arguably what is ultimately relevant for the

²⁰In this sense, our model builds on an earlier literature in psychology that emphasizes that similarity-based recall is sometimes optimal based on Bayesian principles (e.g., Anderson, 2013; Malmendier and Wachter, 2024).

investor) is less transparent. In these situations, relying on the narrow bracket can deliver better information and lead to better decisions. Our approach of endogenizing choice brackets—which are mostly treated as exogenous in the literature—connects to the rational inattention model of mental accounting developed by Kőszegi and Matějka (2020) and the narrow thinking model of Lian (2021).

Suppose the DM obtains utility from two dimensions—two expenditure buckets, two investments, two characteristics of a good, *etc.* Formally, total utility is $U(x) = u_{x,1} + u_{x,2}$, where $u_{x,1}$ and $u_{x,2}$ need not be independent. The agent also has a fixed reference bundle r with known utility components $u_{r,1}$ and $u_{r,2}$, so $U(r) = u_{r,1} + u_{r,2}$. This formulation allows correlated utility components and can also represent nonadditive utility over underlying payoffs.²¹

As in the rest of the paper, we assume Gaussian priors,

$$\begin{pmatrix} u_{x,1} \\ u_{x,2} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \tilde{u}_{x,1} \\ \tilde{u}_{x,2} \end{pmatrix}, \begin{pmatrix} \tau_1^2 & \rho \tau_1 \tau_2 \\ \rho \tau_1 \tau_2 & \tau_2^2 \end{pmatrix} \right), \quad \tau_1, \tau_2 > 0, \quad \rho \in (-1, 1),$$

and, for tractability, focus on the reference-centered homoskedastic case in which $\tilde{u}_{x,i} = u_{r,i}$ and $\tau_1 = \tau_2 = \tau$.²²

We consider two types of bracketing. Under *Broad Bracketing*, the agent receives an ordinal signal o^B comparing $U(x)$ with $U(r)$, subject to Gaussian ordinal noise with variance σ_B^2 . Under *Narrow Bracketing*, the agent receives two ordinal signals o_1^N and o_2^N , comparing $u_{x,1}$ with $u_{r,1}$ and $u_{x,2}$ with $u_{r,2}$, respectively, each subject to Gaussian ordinal noise with variance σ_N^2 . All noise terms are independent of each other and of $(u_{x,1}, u_{x,2})$. No additional cardinal signals are

²¹For example, suppose total utility is $V(z_1 + z_2)$, where V is strictly increasing. Let $z_{r,1}$ and $z_{r,2}$ be fixed payoff reference levels and write $z_r := z_{r,1} + z_{r,2}$. The value $V(z_r)$ is known, although V need not otherwise be known. For any fixed utility reference components satisfying $u_{r,1} + u_{r,2} = V(z_r)$ and any $g > 0$, define

$$u_{x,i} := u_{r,i} + \begin{cases} \frac{V(z_1 + z_2) - V(z_r)}{z_1 + z_2 - z_r} (z_i - z_{r,i}), & z_1 + z_2 \neq z_r, \\ g (z_i - z_{r,i}), & z_1 + z_2 = z_r, \end{cases} \quad i = 1, 2.$$

Then $u_{x,1} + u_{x,2} = V(z_1 + z_2)$. Moreover, the coefficient multiplying each payoff deviation is strictly positive, so

$$u_{x,i} > u_{r,i} \iff z_i > z_{r,i}, \quad i = 1, 2,$$

and

$$U(x) > U(r) \iff z_1 + z_2 > z_r.$$

Thus the construction preserves total utility and the signs of componentwise and aggregate comparisons. It does not generally preserve Gaussian priors or additive Gaussian comparison noise; these remain additional assumptions in the analysis below.

²²Appendix B.9.2 presents the corresponding results for the general Gaussian case with arbitrary means, arbitrary reference points, and different variances.

received.

The broad bracketer is Bayesian and reports $\mathbb{E}[U(x) \mid o^B]$. The narrow bracketer is naive: the agent updates each bracket separately, ignoring ρ , and reports $\mathbb{E}[u_{x,1} \mid o_1^N] + \mathbb{E}[u_{x,2} \mid o_2^N]$. We focus on the naive version both to follow the bracketing literature and because it makes our point sharpest: narrow bracketing can be optimal even in this case. Considering Bayesian Narrow Bracketing only strengthens the conclusions; we return to it below and solve it in Appendix B.9.1.

The following proposition characterizes when Narrow Bracketing is better than Broad Bracketing by comparing their expected MSPE. (For Narrow, we compute the MSPE under the true distribution.) Let

$$\Gamma_N(\rho) := 1 + 2\rho - \frac{2}{\pi} \arcsin \left(\frac{\tau^2}{\tau^2 + \sigma_N^2} \rho \right).$$

Proposition 9 (Narrow versus Broad Bracketing). *If $\Gamma_N(\rho) > 0$, Narrow Bracketing has weakly lower expected MSPE than Broad Bracketing if and only if*

$$\sigma_B^2 \geq \bar{\sigma}_B^2(\rho) := \frac{2(1 + \rho)^2(\tau^2 + \sigma_N^2)}{\Gamma_N(\rho)} - 2\tau^2(1 + \rho).$$

If $\Gamma_N(\rho) \leq 0$, Broad Bracketing has strictly lower expected MSPE for every finite σ_B^2 .

Whether it is optimal to bracket narrowly or broadly depends on the features of the environment: the noises σ_B^2 and σ_N^2 , and the correlation ρ . The left panel of Figure 5 illustrates this: fixing σ_N^2 , Narrow dominates when the noise in Broad is higher and the correlation positive. The role of noises is intuitive: The noisier the ordinal signal in the broad case, the more a decision maker favors narrow bracketing. Without correlation, narrow is better if $\sigma_B^2 > 2\sigma_N^2$.

More subtle is the role of correlation. Under the reference-centered homoskedastic normalization and natural restrictions on precision (e.g., $\sigma_B^2 = 2\sigma_N^2$, and $\sigma_N^2 \leq \tau^2$), positive correlation favors Narrow Bracketing, while negative correlation favors Broad Bracketing. This may appear counterintuitive, since our naive version of Narrow Bracketing neglects correlation, so how can positive correlation favor it? The reason is that two forces are at play: an information effect and a naivete effect. To separate them, it is useful to consider also *Bayesian Narrow Bracketing*, which uses the same two ordinal signals but aggregates them with the correct joint distribution. The right panel of Figure 5 plots $-\text{MSPE}$ for the three cases assuming $\sigma_B^2 = 2\sigma_N^2$.

Under this normalization, Bayesian Narrow is better than Broad with positive ρ , and worse with negative ρ . This is because Broad has only one ordinal signal, on the variable of interest $U(x)$, whereas Bayesian Narrow has two signals, on its components. Bayesian Narrow can thus also obtain mixed signals which, with positive correlation, provide evidence that the aggregate

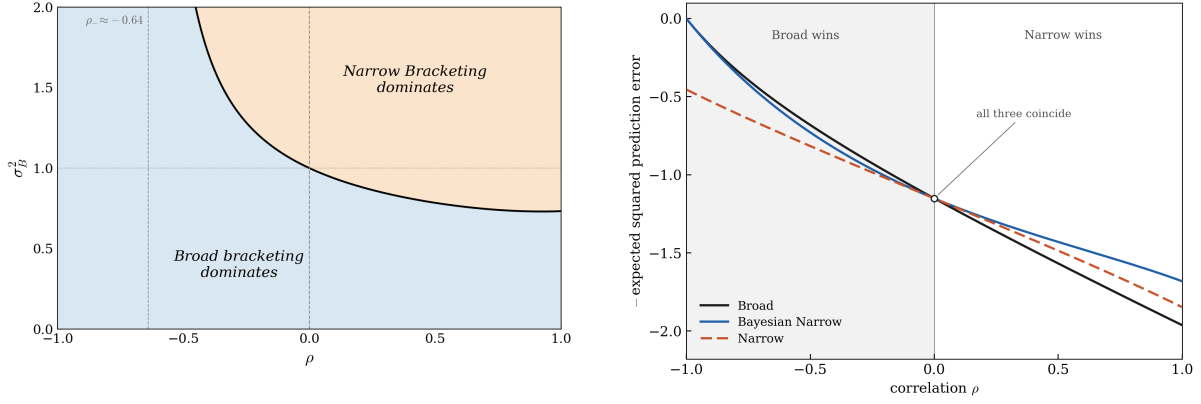


Figure 5: Narrow versus Broad Bracketing assuming $\tilde{u}_{x,i} = u_{r,i} = \mu$ for $i = 1, 2$, $\tau_1 = \tau_2 = \tau = 1$, and $\sigma_N^2 = 1/2$. Panel (a) plots the threshold of Proposition 9. Above it, naive Narrow Bracketing has lower true expected squared prediction error than Broad Bracketing. For sufficiently negative ρ , $\Gamma_N(\rho) \leq 0$, and Broad Bracketing dominates for every finite σ_B^2 . Panel (b) fixes $\sigma_B^2 = 1 = 2\sigma_N^2$ and plots $-\text{MSPE}$ (higher is better) as ρ varies, adding the Bayesian Narrow benchmark.

deviation $U(x) - U(r)$ is close to zero. With negative correlation, instead, mixed signals are likely but less informative. This *information effect*, which has the same sign as ρ , highlights one core informational advantage of Narrow Bracketing with positive correlation—generating multiple signals that may be more informative. To our knowledge, this is new.

At the same time, correlation neglect induces the *naivete effect*: naive Narrow is strictly worse than Bayesian Narrow whenever $\rho \neq 0$. This reduces the appeal of Narrow Bracketing. Crucially, however, under the normalization $\sigma_B^2 = 2\sigma_N^2$ and when $\sigma_N^2 \leq \tau^2$, as in Figure 5, the information effect dominates for every positive ρ , making even naive Narrow better than Broad. Thus, under these conditions, positive correlation favors Narrow Bracketing, while negative correlation favors Broad Bracketing.

Taken together, these results show that Narrow Bracketing may be optimal even when the DM approaches it naively, and this is particularly the case when (i) it provides more precise ordinal signals and (ii) outcomes are positively correlated.

A recent experiment by Oprea et al. (2026) provides support for our theory. In their experiment, subjects choose between lotteries in either a narrow bracketing or a broad bracketing format with lifetime wealth integrated. When subjects are directly asked to choose between the two presentation formats, they overwhelmingly prefer the narrow brackets. In written advice sent to other subjects, they explain that this gives rise to simpler displays and comparisons, and choice data indeed show considerably more dominance violations under broad brackets.

7 Caution

Our model thus far assumed that the DM chooses the option with the highest expected value without accounting for its uncertainty. However, a growing body of evidence suggests that people are averse to their own value uncertainty (e.g., de Clippel et al., 2025), which can be modeled through *caution* (Cerreia-Vioglio et al., 2015, 2024) or *complexity aversion* (Mononen, 2026; Puri, 2025; Gabaix, 2026). This attitude is both empirically and theoretically related to the broader notion of ambiguity aversion, of which it could be seen as a specific instance. We now extend our model to allow for caution, and show how this has immediate implications, such as generating the endowment effect.²³ For brevity, this section contains only a sketch; Appendix B.10 develops it in detail.

Modeling Caution. We model aversion to uncertainty using a classical approach: adding a concave transformation, which yields aversion to uncertainty like the concavity of the utility yields risk aversion in Expected Utility. We assume that the DM evaluates option x by integrating over the possible values of $U(x)$ using a concave transformation of CARA form, that is, $-e^{-\alpha U(x)}/\alpha$ for $\alpha > 0$, where α captures the degree of aversion to value uncertainty. In Appendix B.10, we prove that in this case, evaluations following cardinal and ordinal signals are given by:

$$\bar{U}(x|s, o) := \sum_i^n \text{CE}(u_{x,i}|s_i, o_i), \quad (12)$$

$$\text{CE}(u_{x,i}|s_i, o_i) = \underbrace{\tilde{u}_{x,i}^s}_{\text{Ref-indep.}} - \underbrace{\alpha \frac{\sigma_{x,i}^2}{2}}_{\text{Variance penalty}} + \underbrace{\bar{\psi}_\alpha \left(\frac{\tilde{u}_{x,i}^s - \tilde{u}_{r,i}}{\sqrt{\sigma_{x,i}^2 + \sigma_{r,i}^2 + \sigma_{o,i}^2}}, \frac{\alpha \sigma_{x,i}^2}{\sqrt{\sigma_{x,i}^2 + \sigma_{r,i}^2 + \sigma_{o,i}^2}}, o_i \right)}_{\text{Caution-adjusted ref-dep. boost}}, \quad (13)$$

where the adjusted boost is given by

$$\bar{\psi}_\alpha(m, \alpha k, o) := \begin{cases} \frac{1}{\alpha} (\ln \Phi(m) - \ln \Phi(m - \alpha k)) & \text{if } o = + \\ -\frac{1}{\alpha} (\ln \Phi(-m + \alpha k) - \ln \Phi(-m)) & \text{if } o = -. \end{cases}$$

This shows that evaluations admit a tractable representation even adding caution. First, we can represent evaluations as the sum of the dimension-by-dimension *certainty equivalents* $\text{CE}(u_{x,i}|s_i, o_i)$, each of which admits a simple functional form. Adding caution introduces the variance penalty

²³In contrast to risk aversion (defined over prizes), caution can be understood as aversion to uncertainty about the utility (Cerreia-Vioglio et al., 2015, 2024). Cerreia-Vioglio et al. (2024) show how caution can generate the endowment effect but the setup of our paper is substantially different, as is the treatment of reference-dependence (since we do not have reference-dependent valuation but ordinal information).

term—as the DM dislikes value uncertainty—and changes a bit the expression for the reference-dependent boost, though it maintains the core properties that we identified for mean beliefs above (e.g., it is strictly positive following a positive comparison).

Endowment effect and status quo bias. Consider the endowment effect. In what follows, we only sketch how our model generates it; Appendix B.10 develops the calculations in full. For a given object x of true utility $u_x \in \mathbb{R}_{++}$, the DM needs to determine the price p_{WTP} they are willing to pay to acquire the object when they do not own it and the price p_{WTA} they are willing to accept to sell the object if they own it. We follow the assumptions we held throughout the paper: the agent has a prior $\mathcal{N}(\tilde{u}_x, \sigma_{p,x}^2)$ for u_x , of $\mathcal{N}(\tilde{u}_n, \sigma_{p,n}^2)$ for the utility u_n of not having it; receives cardinal signals with variance σ_ε^2 on all unknowns; receives all ordinal signals in each dimension, with ordinal noise σ_o^2 . For now, assume that the utility of monetary exchanges is known and equal to the identity function.²⁴

We assume that the DM has additional knowledge about the reference point, which we take to be the endowment. Formally, we assume that the DM receives an additional cardinal signal about the value of the endowment with noise σ_ε^2 , which can be due to more attention to the reference object than to alternatives, or that being endowed with an object may trigger memories of similar ones, resulting in a more precise evaluation of its value.

For transparency of the calculations, and to highlight which effects are due to boosts and caution, assume that priors are well calibrated ($\tilde{u}_x = u_x$, $\tilde{u}_n = u_n$), the utility of not having the object is 0, the priors of owning or not owning an object have the same variance ($\sigma_{p,x}^2 = \sigma_{p,n}^2$), and there is no ordinal noise ($\sigma_o^2 = 0$). (It is easy to see how our results hold more generally.) Denote by $\hat{\sigma}_x^2$ and σ_x^2 the posterior variance about u_x when it is the endowment or not, respectively, after cardinal signals alone.

Under these assumptions, a WTP-WTA gap follows straightforwardly from our model, using the formulas above. Indeed, computing p_{WTA} and p_{WTP} when cardinal signals are received at their expected value, accounting for all ordinal signals, yields

$$p_{\text{WTP}} = u_x - \frac{\alpha}{2} (\sigma_x^2 - \hat{\sigma}_x^2) + \frac{1}{\alpha} \ln \frac{\Phi\left(\frac{u_x + \alpha \hat{\sigma}_x^2}{\sqrt{\hat{\sigma}_x^2 + \sigma_x^2}}\right)}{\Phi\left(\frac{u_x - \alpha \hat{\sigma}_x^2}{\sqrt{\hat{\sigma}_x^2 + \sigma_x^2}}\right)}, \quad p_{\text{WTA}} = u_x + \frac{\alpha}{2} (\sigma_x^2 - \hat{\sigma}_x^2) + \frac{1}{\alpha} \ln \frac{\Phi\left(\frac{u_x + \alpha \sigma_x^2}{\sqrt{\sigma_x^2 + \hat{\sigma}_x^2}}\right)}{\Phi\left(\frac{u_x - \alpha \sigma_x^2}{\sqrt{\sigma_x^2 + \hat{\sigma}_x^2}}\right)},$$

²⁴Appendix B.10 develops the general case in which the utility changes induced by monetary gains/losses are also unknown. It shows that in that case, our model correctly predicts not only the endowment effect, but also that p_{Ch} —the price that makes the agent indifferent between the object and that amount when no object is the endowment—lies strictly in between p_{WTA} and p_{WTP} , in line with experimental evidence.

giving us

$$EE := p_{WTA} - p_{WTP} = \underbrace{\alpha (\sigma_x^2 - \hat{\sigma}_x^2)}_{\text{Caution penalty}} + \underbrace{\frac{1}{\alpha} \ln \frac{\Phi\left(\frac{u_x + \alpha \sigma_x^2}{\sqrt{\sigma_x^2 + \hat{\sigma}_x^2}}\right) \Phi\left(\frac{u_x - \alpha \sigma_x^2}{\sqrt{\hat{\sigma}_x^2 + \sigma_x^2}}\right)}{\Phi\left(\frac{u_x + \alpha \hat{\sigma}_x^2}{\sqrt{\hat{\sigma}_x^2 + \sigma_x^2}}\right) \Phi\left(\frac{u_x - \alpha \hat{\sigma}_x^2}{\sqrt{\sigma_x^2 + \hat{\sigma}_x^2}}\right)}}_{\text{Comparative-Boost Adjustment}}.$$

Intuitively, the ordinal comparisons between owning and not owning the mug generate comparative boosts in several directions (with a negative total effect), but our DM dislikes value uncertainty and their utility evaluation of the reference bundle is more precise ($\sigma_x^2 > \hat{\sigma}_x^2$). Indeed, it is immediate to show that, as long as $\alpha > 0$, $\sigma_x^2 > 0$, and $\sigma_e^2 > 0$, the endowment effect above is strictly positive.

At the same time, in the absence of value uncertainty ($\sigma_x^2 = 0$), our model predicts no endowment effect, reminiscent of the conjecture by Kahneman et al. (1990), already noted in the Introduction, that “lack of commensurability is necessary.” This conjecture is now supported by a broad body of evidence that shows that the endowment effect is highly susceptible to manipulations of (what we can think of as) value uncertainty, including experience or knowing the object’s market price (see Cerreia-Vioglio et al., 2024, for a literature review).

Finally, while the calculations above show how our model can generate the endowment effect, very similar ones show how it also generates status quo bias, following an identical logic.

8 Conclusion

This paper has argued that the combination of value uncertainty and reference points as information provides a framework for thinking about multiple phenomena at the core of behavioral economics, including classic reference dependence phenomena, a new theory of endogenous reference points, and a model of endogenous choice bracketing. Building on the present paper, in Dean et al. (2026) we study how a similar mechanism—ordinal information from one’s own past actions—gives rise to an information-based model of habit formation and reasoning by analogy. For instance, in the presence of value uncertainty, knowing that I chose an option yesterday provides information and increases my posterior evaluation of it.

Reference-dependent hedonics and / or comparative information? In our opinion, both introspection and empirical evidence suggest that two statements are true. First, reference-dependent hedonics and sensations exist. For example, it seems uncontroversial that a lukewarm shower feels more pleasurable after a cold shower than after a hot one. To the extent that people anticipate these hedonics, they drive reference-dependent choice. Second, as already anticipated

by Kahneman et al. (1990), people sometimes react more strongly to reference points when they exhibit greater value uncertainty, such as when the problem is more difficult or less familiar (e.g. List, 2003; Graeber & Enke, 2026; Jin et al., 2024).

Assuming this is true, there are two potential readings of our paper, and we do not know ourselves which is correct. A first is that reference-dependent hedonics (in terms of “experienced utility”) and comparative information in the presence of value uncertainty (a view of “decision utility”) both exist, that they are orthogonal to each other, and that future empirical work might test which mechanism is more important in which setting. For example, future work could more systematically manipulate the informativeness of reference points or the decision maker’s degree of value uncertainty to investigate whether this shapes reactions to reference points.

A second possibility is that the information-based and the hedonics-based views are linked through the idea that information generates feelings. For example, if a decision maker is uncertain about the utility of a lukewarm shower and they receive a negative comparative signal because they just had a hot shower, then the decision maker’s feelings may reflect this change in beliefs. After all, classic reference-dependence papers such as Kőszegi and Rabin (2006, 2009) are often interpreted through the lens of news utility. If this second possibility is true, there would be no point in empirically “separating” the hedonic and information-based perspectives because one would produce the other. However, even in this second case our model would formalize new comparative statics predictions with respect to uncertainty (complexity, experience, informativeness of the reference point *etc.*) that are absent in classic models of reference-dependent preferences.

Extensions and applications. Several extensions are natural. On the one hand, we abstract from other important cognitive components, like selective attention allocation and salience (Bordalo et al., 2022), or rational inattention (Maćkowiak et al., 2023). Integrating these forces could be very fruitful.

On the other hand, several applications to specific domains would be particularly interesting. Perhaps most obviously, we have not applied our model to decisions under risk, a standard environment for reference dependence. Such a model could predict, for purely informational reasons, why people dislike lotteries with positive expected values but with some chance of losses, which may speak to the equity premium puzzle or the disposition effect. Indeed, Villas-Boas (2024) applies a setup similar to ours to risky choice. At the same time, a full treatment of such problems, potentially applying comparative information in both the prize and probability domain, lies beyond the scope of this paper.

A second application could be to intertemporal choice, where boosts from ordinal comparisons may give new interpretations to familiar phenomena. When asked how much \$100 in 6

months is worth today, the DM may be uncertain (Enke et al., 2025; Chakraborty, 2021), yet they will know it is less than \$100, giving a downward boost in the evaluation of delayed payments. Ordinal information could therefore make a Bayesian decision maker discount more steeply between today and tomorrow than between today and a year from now. It would also produce reference dependence in time delays, as has been documented empirically (Li and Zhong, 2023; Erakhtina et al., 2026).

Third, our approach may be applied to social reference points. While introspection and evidence suggest that social reference points matter partly for preference-based reasons (e.g., envy), it seems similarly intuitive that social reference points also contain information. For example, if I learn that a co-worker very similar to me makes twice as much as I do, this suggests I had a bad deal, or that my marginal product is higher than I used to believe.

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Online Appendix

A Evidence on Comparative and Ordinal Reasoning

A core component of our model is that, when people are uncertain how to judge or value objects, they rely on comparators as a source of information. This section reviews evidence that suggests that people indeed often rely on comparators (in a partially ordinal way) when they are uncertain what the right decision is.

The broad idea that decision making is often comparative in nature is one of the central and earliest insights of behavioral economics, as evidenced in prospect theory. A classic example is that warm water feels warmer after we stick our hand into cold rather than hot water. However, the comparative nature of cognition and decision making is not limited to reference-dependent preferences. Kahneman and Tversky (1979) already note that “the perceptual apparatus is attuned to the evaluation of changes or differences rather than to the evaluation of absolute magnitudes”.

Indeed, decades of psychophysical research suggest that performance on absolute judgment tasks – evaluating the magnitude of a stimulus – is substantially poorer than on tasks that require comparisons between two stimuli – evaluating which one is brighter, longer, heavier etc. This supports the view that human perception is optimized for detecting differences rather than encoding absolute magnitudes (Laming, 2009, 1984; Braida and Durlach, 1972; Garner, 1953). Empirically, the precision of the ordering of two stimuli implied by cardinal estimates is about two orders of magnitude lower than the precision of direct comparisons between the same stimuli (Laming, 1997). Moreover, responses in the classic psychological absolute judgment tasks (magnitude estimation and absolute identification) are more strongly shaped by contextual factors than in the comparative task of ordinal stimulus discrimination (Luce and Green, 1974; Parducci, 1965; Laming, 1997), suggesting that comparisons are more stable.

Much research suggests that such comparative thinking is partly ordinal in nature. It is generally easier to assess whether one stimulus is greater than, less than, or equal to another one than it is to make cardinal judgments. The importance of ordinal judgments has been most prominent in perceptual tasks but also heavily influenced work on decision-making. For example, it is intuitively easier to realize that six apples are better than four apples, compared to assessing how much better they are. One clear illustration of this in the perceptual domain is Hollands and Dyre (2000) who show that in magnitude estimation tasks with an exogenously-provided reference point of known value, subjects correctly recognize whether a stimulus is above or below the reference point but are insufficiently sensitive to how much, the two hallmark characteristics

of partial ordinal reasoning.

Importantly, much evidence suggests that people partly implicitly rely on ordinal comparisons even when they produce cardinal estimates. A classic reference is Braida and Durlach (1972), who showed that magnitude estimation (assigning a number proportional to intensity) and absolute identification (judging which stimulus is larger) yield essentially the same pattern of responses. Stewart et al. (2005) review this and related empirical observations, and propose a psychological model of absolute identification through relative judgment. Relatedly, the influential line of work on decision-by-sampling essentially asserts that people make decisions by ordinally comparing options with comparators retrieved from memory, and counting the number of advantages (Stewart et al., 2006, and references therein).

More recently, some contributions in economics suggested that relying on comparative thinking may partly reflect a simplification strategy and / or incomplete information about how to solve a problem. In studying belief updating, Augenblick et al. (2025) model a DM who knows in which direction to update (ordinally) but not by how much. Graeber et al. (2025) and Graeber & Enke (2026) show empirically that higher processing difficulty or uncertainty lead to stronger reliance on comparison points.

In summary, when people exhibit uncertainty about how to estimate the value or magnitude of a stimulus, they partially rely on comparisons, be they explicitly provided or retrieved from memory. Given its low cost and simplicity, these comparisons are partly ordinal in nature. Unlike some of the more extreme hypotheses and models in psychology, we do not posit that people only rely on comparative or ordinal information. Instead, our model features a standard absolute, cardinal signal, augmented with a comparative, ordinal one.²⁵

B Additional Results

This section includes several additional results. All proofs appear in Appendix E.

B.1 Cardinal Comparative Signals

A seemingly strong assumption of our model is that the comparative signal is only ordinal in nature—the DM may observe that u_x is better than u_r but not by how much. Yet in some settings, the DM may also have a sense of *how much* better or worse u_x is. We now provide two results. First, we show that the setting we study in the paper—a cardinal signal about u_x and

²⁵A notable caveat is that while humans generally perform better at, and tend to rely on, ordinal judgments, these judgments are not perfect, either. In particular, they break down when stimulus differences fall below the so-called “just noticeable difference” threshold. We capture these imperfections through the ordinal noise term in our model.

an ordinal signal about $(u_x - u_r)$ —is, in fact, informationally *equivalent* to a setting in which the DM also receives a cardinal signal about $u_x - u_r$, simply by adapting the parameters, under mild assumptions on the correlation matrix. Second, we provide an extension of our basic model that allows for more general cardinal comparative signals, showing how these are easy to incorporate.

An equivalence result. In the *Original Setting* studied in the main body of the paper, parametrized by $(\sigma_{\varepsilon_x}^2, \sigma_{\varepsilon_r}^2, \sigma_o^2)$, the DM holds independent priors $u_x \sim \mathcal{N}(\tilde{u}_x, \sigma_x^2)$, $u_r \sim \mathcal{N}(\tilde{u}_r, \sigma_r^2)$ and observes cardinal signals $s_x = u_x + \varepsilon_x$ and $s_r = u_r + \varepsilon_r$, with $\varepsilon_x \sim \mathcal{N}(0, \sigma_{\varepsilon_x}^2)$ and $\varepsilon_r \sim \mathcal{N}(0, \sigma_{\varepsilon_r}^2)$, and an ordinal signal $o = \text{sign}(u_x - u_r - v)$ where $v \sim \mathcal{N}(0, \sigma_o^2)$.²⁶ The utilities u_x, u_r and primitive errors $\varepsilon_x, \varepsilon_r, v$ are mutually independent.

Consider instead the following *Alternative Setting*, parametrized by $(\sigma_{e_x}^2, \sigma_{e_d}^2, \sigma_{e_r}^2, \sigma_{o'}^2)$, in which the DM holds the same priors and receives cardinal signals $s'_x = u_x + e_x$ and $s'_r = u_r + e_r$, an ordinal signal $o' = \text{sign}(u_x - u_r - v')$ where $v' \sim \mathcal{N}(0, \sigma_{o'}^2)$ (independent of everything else), and a *cardinal comparative signal* $s_d = u_x - u_r + e_d$. Assume that the error vector (e_x, e_d, e_r) is independent of (u_x, u_r) and distributed as $\mathcal{N}(\mathbf{0}, \Sigma)$, where

$$\Sigma = \begin{pmatrix} \sigma_{e_x}^2 & \sigma_{e_x}^2 & 0 \\ \sigma_{e_x}^2 & \sigma_{e_d}^2 & \sigma_{e_r}^2 \\ 0 & \sigma_{e_r}^2 & \sigma_{e_r}^2 \end{pmatrix},$$

with $\sigma_{e_d}^2 > \sigma_{e_x}^2 + \sigma_{e_r}^2$. This covariance structure implies that we can write $e_d = e_x + e_r + t$, where $t \sim \mathcal{N}(0, \sigma_{e_d}^2 - \sigma_{e_x}^2 - \sigma_{e_r}^2)$ is independent of e_x and e_r : intuitively, all this requires is that the comparative signal s_d is subject to the noise from the individual components plus additional independent noise.

Despite their different formulations, the Original and Alternative settings are informationally *equivalent*: for every Alternative Setting, there exists a corresponding Original Setting, and vice-versa, that generates the same distribution of posterior beliefs about (u_x, u_r) . This equivalence is formalized in the following Observation (proved in Appendix E for completeness).

Observation 3 (Equivalence of Settings). *The Alternative and Original Settings are informationally equivalent: they generate the same distribution of posterior beliefs over (u_x, u_r) under the following reparametrizations.*

²⁶The DM's knowledge of u_r can be interpreted as arising from a prior together with a cardinal signal about u_r , left implicit above.

1. Any Alternative Setting with $\sigma_{e_r}^2 > 0$ is equivalent to the Original Setting satisfying

$$\sigma_{\varepsilon_x}^2 = \sigma_{e_x}^2, \quad \frac{1}{\sigma_{\varepsilon_r}^2} = \frac{1}{\sigma_{e_r}^2} + \frac{4}{\sigma_{e_d}^2 - \sigma_{e_x}^2 - \sigma_{e_r}^2}, \quad \sigma_o^2 = \sigma_{o'}^2.$$

2. Conversely, any Original Setting with $\sigma_{e_r}^2 > 0$ is equivalent to any Alternative Setting satisfying $\sigma_{e_x}^2 = \sigma_{\varepsilon_x}^2$, $\sigma_{o'}^2 = \sigma_o^2$, any $\sigma_{e_r}^2 > \sigma_{\varepsilon_r}^2$, and

$$\sigma_{e_d}^2 = \sigma_{\varepsilon_x}^2 + \sigma_{e_r}^2 + 4 \left(\frac{1}{\sigma_{\varepsilon_r}^2} - \frac{1}{\sigma_{e_r}^2} \right)^{-1}.$$

If the reference point is perfectly known, so that $\sigma_r^2 = 0$, the two settings are equivalent whenever $\sigma_{\varepsilon_x}^2 = \sigma_{e_x}^2$ and $\sigma_o^2 = \sigma_{o'}^2$.

Generic comparative cardinal signals. Next, we extend our core characterization to the case in which the agent also receives a cardinal comparative signal.

Proposition 10. Consider the setup of Proposition 1, but suppose the DM additionally receives a cardinal comparative signal $d = u_x - u_r + \varepsilon_d$, where $\varepsilon_d \sim \mathcal{N}(0, \sigma_d^2)$ is independent of the utility vector and all other primitive signal errors. Let $K := \sigma_x^2 + \sigma_r^2 + \sigma_d^2$ and define

$$\tilde{u}_x^{s,d} := \left(1 - \frac{\sigma_x^2}{K}\right) \tilde{u}_x^s + \frac{\sigma_x^2}{K} (d + \tilde{u}_r), \quad \tilde{\Delta}^{s,d} := \frac{\sigma_d^2}{K} (\tilde{u}_x^s - \tilde{u}_r) + \frac{\sigma_x^2 + \sigma_r^2}{K} d, \quad \sigma_\Delta^2 := \frac{\sigma_d^2 (\sigma_x^2 + \sigma_r^2)}{K},$$

so that $\tilde{u}_x^{s,d} = \mathbb{E}[u_x \mid s, d]$, $\tilde{\Delta}^{s,d} = \mathbb{E}[u_x - u_r \mid s, d]$ and $\sigma_\Delta^2 = \mathbb{V}(u_x - u_r \mid s, d)$. If $\sigma_\Delta^2 + \sigma_o^2 > 0$ and $\sigma_x^2 + \sigma_r^2 > 0$. Then

$$\mathbb{E}[u_x \mid s, d, o] = \tilde{u}_x^{s,d} + \frac{\sigma_x^2}{\sigma_x^2 + \sigma_r^2} \frac{\sigma_\Delta^2}{\sqrt{\sigma_\Delta^2 + \sigma_o^2}} \psi \left(\frac{\tilde{\Delta}^{s,d}}{\sqrt{\sigma_\Delta^2 + \sigma_o^2}}, o \right).$$

B.2 Posterior Variances After Ordinal Signals and Average Posterior Means

We now present the characterization of the posterior variance after the ordinal signal, using the notation $\mathbb{V}(x)$ for the variance of a random variable x ; as well as the characterization of the average posterior mean that the DM will have, averaging over the realizations of the (noisy) cardinal and ordinal signals.

Figure 6 plots average posterior variances, while Figure 7 presents a heatmap of the full distribution of the mean posterior for a given value of $u_x - u_r$.

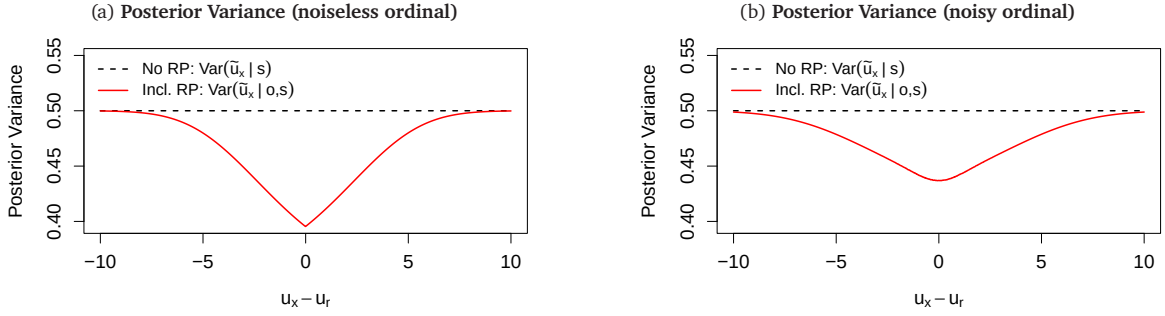


Figure 6: Posterior variance with and without ordinal information. Both panels show the reference-free posterior variance (black) for comparison. Parameters are held fixed across panels with $\sigma_p, \sigma_\varepsilon, \sigma_r = 1$; the left panel uses $\sigma_o = 0$ (noiseless), while the right panel uses $\sigma_o = 1$ (noisy ordinal).

Proposition 11 (Posterior Variances). *Suppose $\sigma_x^2 + \sigma_r^2 + \sigma_o^2 > 0$. Let $t_s := \frac{\tilde{u}_x^s - \tilde{u}_r}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}$. Then, for $o \in \{+, -\}$,*

$$\mathbb{V}(u_x | s, o) = \sigma_x^2 - \frac{\sigma_x^4}{\sigma_x^2 + \sigma_r^2 + \sigma_o^2} (t_s \psi(t_s, o) + \psi(t_s, o)^2).$$

Moreover, averaging over ordinal signals using the DM's predictive probabilities $\Pr(o = + | s) = \Phi(t_s)$ and $\Pr(o = - | s) = \Phi(-t_s)$ gives

$$\mathbb{E}_o[\mathbb{V}(u_x | s, o) | s] = \sigma_x^2 - \frac{\sigma_x^4}{\sigma_x^2 + \sigma_r^2 + \sigma_o^2} \frac{\phi(t_s)^2}{\Phi(t_s)\Phi(-t_s)}.$$

Holding variances fixed, this expectation is strictly increasing in $|t_s|$ when $\sigma_x^2 > 0$.

Proposition 12 (Average Posterior Mean). *Suppose $\sigma_x^2 + \sigma_r^2 + \sigma_o^2 > 0$. Fix the true values (u_x, u_r) , and let expectations be over signal realizations conditional on these values. For each s , let $t_s := \frac{\tilde{u}_x^s - \tilde{u}_r}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}$. Then,*

$$\begin{aligned} \mathbb{E}_{s,o}[\tilde{u}_x^{s,o}] &= \lambda u_x + (1 - \lambda) \tilde{u}_x \\ &+ \frac{\sigma_x^2}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}} \mathbb{E}_s \left[(\Pr(o = + | u_x, u_r) - \Phi(t_s)) \frac{\phi(t_s)}{\Phi(t_s)(1 - \Phi(t_s))} \right], \end{aligned} \quad (14)$$

where

$$\Pr(o = + | u_x, u_r) = \begin{cases} \Phi\left(\frac{u_x - u_r}{\sigma_o}\right), & \sigma_o > 0, \\ \mathbf{1}\{u_x \geq u_r\}, & \sigma_o = 0. \end{cases}$$

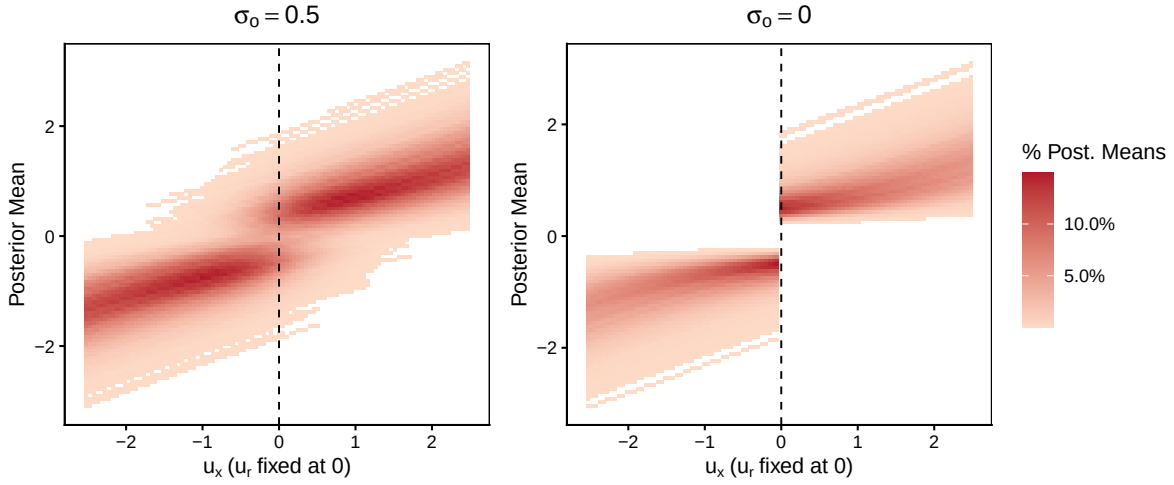


Figure 7: Distribution of posterior means $\mathbb{E}[u_x | s, o]$ for $\tilde{u}_x = u_r = 0$, and $\sigma_\varepsilon = \sigma_p = 1$. The left panel shows the case in which the ordinal signal is noisy, $\sigma_o = 0.5$. The right panel shows the case of noiseless ordinal signal ($\sigma_o = 0$). Each panel shows, for 100 steps of $u_x \in [-2.5, 2.5]$, the average over 10,000 draws of $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2)$ and $v \sim \mathcal{N}(0, \sigma_o^2)$.

B.3 Belief Updating with Correlated Signals

For some of our results, it is useful to prove a more general version of Proposition 1 that allows for correlation between the two variables.

Proposition 13. *Let (u_x, u_r) be jointly normal with means $(\tilde{u}_x, \tilde{u}_r)$, variances (σ_x^2, σ_r^2) , and covariance σ_{xr} . Let $v \sim \mathcal{N}(0, \sigma_o^2)$ be independent of (u_x, u_r) , and let $o = +$ if $u_x - u_r \geq v$ and $o = -$ otherwise. If*

$$\tau^2 := \sigma_x^2 + \sigma_r^2 - 2\sigma_{xr} + \sigma_o^2 > 0,$$

then, for $o \in \{+, -\}$,

$$\mathbb{E}[u_x | o] = \tilde{u}_x + \frac{\sigma_x^2 - \sigma_{xr}}{\tau} \psi\left(\frac{\tilde{u}_x - \tilde{u}_r}{\tau}, o\right).$$

B.4 Comparative Statics of Reference Dependence

We analyze comparative statics both for specific signal realizations and in expectation. Let $B(s, +)$ denote the average effect of the ordinal signal $+$ given cardinal signal s , and $C(u_x, u_r)$ the average effect of the ordinal signal given true values of u_x and u_r aggregating across possible ordinal and cardinal signals. That is

$$B(s, +) := \mathbb{E}[u_x | +, s] - \tilde{u}_x^s \quad \text{and} \quad C(u_x, u_r) := \mathbb{E}_{o,s} [\mathbb{E}[u_x | o, s] | u_x, u_r] - \mathbb{E}_s [\mathbb{E}[u_x | s] | u_x].$$

From Proposition 1, let $t_s := \frac{\tilde{u}_x^s - \tilde{u}_r}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}$, then

$$B(s, +) = \frac{\sigma_x^2}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}} \frac{\phi(t_s)}{\Phi(t_s)}, \quad B(s, -) = -\frac{\sigma_x^2}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}} \frac{\phi(-t_s)}{\Phi(-t_s)}.$$

The following proposition describes its comparative statics.

Proposition 14 (Boost from the ordinal signal). *Suppose $\sigma_x^2 > 0$. Fix a realized cardinal signal s . Then, $B(s, +) > 0$ and $B(s, -) < 0$. Moreover:*

(a) *Holding $\tilde{u}_x^s$ fixed, $B(s, +)$ is strictly increasing in σ_x^2 .*

(b) *For each $v \in \{\sigma_r^2, \sigma_o^2\}$,*

$$\frac{\partial B(s, +)}{\partial v} < 0 \quad \text{if } z_s \leq 0.$$

If $z_s > 0$, there exists a unique cutoff (numerically, $z^ \approx 0.84$) such that*

$$\frac{\partial B(s, +)}{\partial v} < 0 \quad \text{for } 0 < z_s < z^*, \quad \text{and} \quad \frac{\partial B(s, +)}{\partial v} > 0 \quad \text{for } z_s > z^*.$$

(c) *$B(s, +)$ is strictly decreasing in $\tilde{u}_x^s - \tilde{u}_r$.*

(d) *(Primitive variances.) The effect of $(\sigma_p^2, \sigma_\varepsilon^2)$ on $B(s, +)$ decomposes as*

$$\frac{\partial B(s, +)}{\partial \sigma_p^2} = B_{\sigma_x^2} \frac{\partial \sigma_x^2}{\partial \sigma_p^2} + B_{\tilde{u}_x^s} \frac{\partial \tilde{u}_x^s}{\partial \sigma_p^2}, \quad \frac{\partial B(s, +)}{\partial \sigma_\varepsilon^2} = B_{\sigma_x^2} \frac{\partial \sigma_x^2}{\partial \sigma_\varepsilon^2} + B_{\tilde{u}_x^s} \frac{\partial \tilde{u}_x^s}{\partial \sigma_\varepsilon^2},$$

where $B_{\sigma_x^2} > 0$ (by part (a)) and $B_{\tilde{u}_x^s} < 0$ (by part (c)). In particular,

$$\frac{\partial B(s, +)}{\partial \sigma_p^2} > 0 \quad \text{whenever } s \leq \tilde{u}_x, \quad \text{and} \quad \frac{\partial B(s, +)}{\partial \sigma_\varepsilon^2} > 0 \quad \text{whenever } s \geq \tilde{u}_x.$$

Parts (a) and (c) show how the boost is strictly *increasing* in σ_x^2 and *decreasing* in $\tilde{u}_x^s - \tilde{u}_r$. Both are intuitive: The more uncertain the DM is about u_x , captured by σ_x^2 , the more he will rely on the ordinal information. Similarly, the larger $\tilde{u}_x^s - \tilde{u}_r$, the smaller the surprise of a + ordinal signal, thus the smaller its effect.

More subtle are the effects of σ_r^2 and σ_o^2 , discussed in Part (b). Intuitively, one may expect the two noises to have a *negative* effect on the boost—at the end of the day, both make the ordinal information less precise. Indeed, this is the case when $\tilde{u}_x^s \leq \tilde{u}_r$ or when $\tilde{u}_x^s - \tilde{u}_r$ is positive but small enough. However, if $\tilde{u}_x^s - \tilde{u}_r$ is very large, an increase in σ_r^2 or σ_o^2 may *increase* the boost.

To see why, suppose that $\tilde{u}_x^s - \tilde{u}_r$ is extremely large and suppose first that there is no noise in the ordinal signal. Then, a + signal is not very informative—the DM expected this. However, it will be at least a bit more informative when σ_r^2 is higher—because even if $\tilde{u}_r$ is much below $\tilde{u}_x^s$, a high σ_r^2 increases the chance that u_r is above $\tilde{u}_r$ and therefore makes the signal more relevant. Similarly, suppose that $\tilde{u}_x^s - \tilde{u}_r$ is extremely large and $\sigma_r^2 = 0$. Again, adding a bit of noise to the ordinal signal may (paradoxically) make it *more* informative because in our model, the chances of a correct ordinal signal depend on the distance between the realized u_x and u_r ; therefore, noise in the ordinal signal makes the + signal more meaningful—because, with noise, a – signal could have been possible, but it didn't realize, increasing the chances that u_x is well above u_r .

We now turn to the average boost C , for which similar comparative statics also hold. (In the proposition, derivatives at zero variances are understood as right derivatives.)

Proposition 15 (Average boost). *Suppose $u_x > u_r = \tilde{u}_r = \tilde{u}_x$ and $\sigma_p^2, \sigma_\varepsilon^2 > 0$. Let $\delta := u_x - u_r$. Then $C(u_x, u_r) > 0$. Moreover:*

(a) *Holding λ fixed, $C(u_x, u_r)$ is strictly increasing in σ_x^2 .*

(b) *For sufficiently small $\delta > 0$, $\partial C / \partial \sigma_o^2 < 0$. Also $\partial C / \partial \sigma_r^2 < 0$ for sufficiently small $\delta > 0$ whenever*

$$2\lambda\sigma_o \leq \sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}.$$

This holds if $\lambda \leq 1/2$ or, for any $\lambda \in (0, 1)$, whenever σ_o^2 is sufficiently small.

(c) *If $\sigma_o = 0$, C is strictly decreasing in $\delta > 0$. If $\sigma_o > 0$, C tends to zero as $\delta \downarrow 0$ and as $\delta \rightarrow \infty$, is strictly increasing sufficiently close to zero, and is strictly decreasing for all sufficiently large δ . It therefore has an interior maximizer. In addition, there exists $\bar{\sigma}_o > 0$ such that, whenever $0 < \sigma_o < \bar{\sigma}_o$, there is a unique $\Delta^* > 0$ with*

$$C_\delta > 0 \quad (0 < \delta < \Delta^*), \quad C_\delta < 0 \quad (\delta > \Delta^*).$$

(d) *The primitive-variance effects decompose as*

$$\frac{\partial C}{\partial \sigma_p^2} = C_{\sigma_x^2} \frac{\partial \sigma_x^2}{\partial \sigma_p^2} + C_\lambda \frac{\partial \lambda}{\partial \sigma_p^2}, \quad \frac{\partial C}{\partial \sigma_\varepsilon^2} = C_{\sigma_x^2} \frac{\partial \sigma_x^2}{\partial \sigma_\varepsilon^2} + C_\lambda \frac{\partial \lambda}{\partial \sigma_\varepsilon^2}.$$

Here $C_{\sigma_x^2} > 0$, $\partial \sigma_x^2 / \partial \sigma_p^2 > 0$, $\partial \sigma_x^2 / \partial \sigma_\varepsilon^2 > 0$, $\partial \lambda / \partial \sigma_p^2 > 0$, and $\partial \lambda / \partial \sigma_\varepsilon^2 < 0$. Thus both primitive variances raise C through the posterior-variance channel, but move λ in opposite directions.

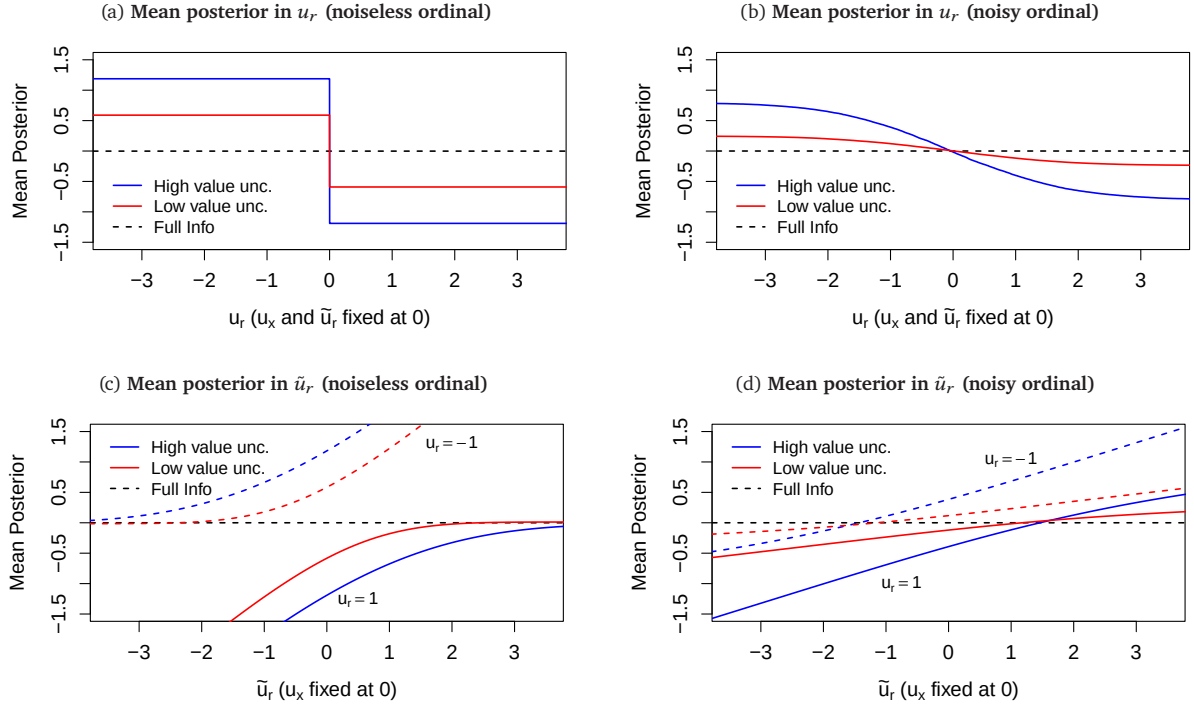


Figure 8: Illustration of comparative statics with respect to reference quantities. The top panels show the mean posterior as a function of the reference point u_r , holding constant $u_x = \tilde{u}_r = 0$. The bottom panels show the mean posterior as a function of the reference evaluation $\tilde{u}_r$, fixing $u_x = 0$ and $u_r \in \{-1, 1\}$ (solid lines: $u_r = 1$; dashed lines: $u_r = -1$). The left panels (a,c) illustrate the case with a *noisy reference, noiseless ordinal* signal ($\sigma_r = 0.25$, $\sigma_o = 0$). The right panels (b,d) illustrate the case with a *noisy reference, noisy ordinal* signal ($\sigma_r = 0.25$, $\sigma_o = 1.5$). All panels compare $\tilde{u}_x^{s,o}$ for a 'low value uncertainty' DM ($\sigma_\varepsilon = 0.75$, red lines) with a 'high value uncertainty' DM ($\sigma_\varepsilon = 2$, blue lines), where we hold constant $\sigma_p = 2$ and prior $\tilde{u}_x = 0$ and only vary σ_ε^2 . Black dashed lines indicate the full information benchmark. Simulation conducted over 10,000 draws.

Notice that the average boost C may be *negative* even if $u_x > u_r$ when $\tilde{u}_x$ is much above $\tilde{u}_r$.²⁷ Moreover, increasing σ_r^2 or σ_o^2 may actually increase the average boost C locally when $u_x - u_r$ is sufficiently large (the proof provides an explicit asymptotic example).

B.5 Comparative Statics With Respect to the Reference Point: Contrast and Assimilation

The mean posterior in Eq. (4) has novel comparative statics implications regarding the role of the reference point. As with the option itself, two distinct—but interrelated—reference quantities enter the mean posterior: the belief about the reference point, $\tilde{u}_r$, which affects the size of the boost, and the true utility of the reference point, u_r , which affects the probability of which ordinal signal will be received.

²⁷For example, if $u_r = \tilde{u}_r = 0$ and $\tilde{u}_x = 2$, $u_x = 0.1$, $\sigma_o = 1$, $\sigma_r^2 = 0$, $\sigma_p^2 = 0.04$, $\sigma_\varepsilon^2 = 4$. Then $C < 0$ (numerically ≈ -0.0402).

Fixing the ordinal signal, an increase in $\tilde{u}_r$ increases the evaluation of the option at hand, through the size of the boost. This is intuitive. Suppose the ordinal signal is positive: then, the belief about the utility of an option should be higher when you are told that it is better than an object with utility of one million than when told it is better than one with utility of zero. This produces an “assimilation effect:” a higher reference belief pushes evaluations of the option up. This is illustrated in the bottom panels of Figure 8.

On the other hand, fixing beliefs about the reference point $\tilde{u}_r$, the average posterior mean about u_x decreases in the true utility of the reference point u_r . This is because u_r determines which ordinal signal is observed, and a higher reference point makes it more likely that the comparative signal is negative, lowering beliefs on average. The top panels of Figure 8 illustrates how, varying u_r while fixing $\tilde{u}_r$, a “contrast effect” exists locally around the reference point.²⁸

The overall effect of a change in the utility of the reference point u_r is determined by the relative importance of these two effects. This, in turn, is affected by how beliefs $\tilde{u}_r$ vary with u_r . Depending on the parameterization of the problem, either channel could dominate. For example, if ordinal signals are noisy and $\tilde{u}_r$ is insensitive to u_r (e.g., with a strong prior about the value of the reference point), the contrast effect dominates, and an increase in u_r decreases the evaluation of u_x . On the other hand, if ordinal signals are noiseless and $\tilde{u}_r$ is very sensitive to u_r , then an increase in u_r that does not change the sign of $u_x - u_r$ will increase the evaluation of x as the assimilation effect will dominate.

The following proposition formalizes this discussion.

Proposition 16. Suppose $\sigma_p^2, \sigma_\epsilon^2, \sigma_o^2 > 0$.²⁹ Then,

1. (Contrast effect.) Holding $\tilde{u}_r$ fixed, $\partial \mathbb{E}[\tilde{u}_x^{s,o}] / \partial u_r < 0$.
2. (Assimilation effect.) Holding u_r fixed, $\partial \mathbb{E}[\tilde{u}_x^{s,o}] / \partial \tilde{u}_r > 0$.
3. Suppose that, at some u_r , $\tilde{u}_r$ varies differentially with respect to u_r . Let $\eta := d\tilde{u}_r / du_r$ and define, at the point being considered,

$$\eta^* := -\frac{\partial \mathbb{E}[\tilde{u}_x^{s,o}] / \partial u_r}{\partial \mathbb{E}[\tilde{u}_x^{s,o}] / \partial \tilde{u}_r} > 0.$$

²⁸If we add to the model a cardinal comparative signal (as discussed in Section 3.2), a global contrast effect obtains even in the absence of ordinal noise; see Appendix B.1 for a discussion of cardinal comparative signals.

²⁹When $\sigma_o = 0$, Part 2 remains true as is (and strict). Away from $u_r = u_x$, the partial derivative with respect to u_r is zero and the total derivative in Part 3 is $\eta \partial \mathbb{E}[\tilde{u}_x^{s,o}] / \partial \tilde{u}_r$; the relevant threshold is then zero. At $u_r = u_x$, the evaluation jumps downward as u_r increases past u_x .

Then

$$\frac{d}{du_r} \mathbb{E}[\tilde{u}_x^{s,o}] \begin{cases} > 0, & \eta > \eta^*, \\ = 0, & \eta = \eta^*, \\ < 0, & \eta < \eta^*. \end{cases}$$

Parts (1)–(2) say that the posterior mean exhibits a contrast effect with respect to the actual value of the reference point and an assimilation effect with respect to the beliefs about the reference point. Part (3) clarifies that, if only the reference point (or its underlying attributes) is observable by the analyst, its observed net effect depends on how much the changes in the reference point are internalized by the decision-maker: the assimilation effect dominates if and only if changes are sufficiently assimilated in the beliefs, while the contrast effect dominates otherwise.

In Prospect Theory and its variants, a higher reference point always *decreases* the evaluation of an option. This captures the compelling intuition that if I make \$100 today, I am happier if I made \$10 rather than \$80 yesterday. This contrast effect with respect to the true attribute values is present also in our model. Yet, as can be seen in Figure 8, our model adds the prediction that evaluations increase in subjective beliefs about the utility of the reference point. For example, our model also predicts that if I make \$100 today, I am happier if I believe that my utility from making \$80 yesterday was 300 rather than 200.

In many cases, we would expect the true value of the reference point and its perceived utility to move together. Indeed, in most datasets, only one of these two is observable. However, a recent body of work both in economics and in perceptual psychology measures both and provides evidence that is consistent with our model’s predictions.

Graeber & Enke (2026) propose that comparison points produce assimilation effects when they consist of decisions / evaluations, yet contrast effects when they are given by exogenous problem attributes. They implement experiments that induce both a reference point and a reference evaluation. For instance, subjects are asked to state their willingness-to-accept (WTA) for completing 10 real effort tasks, while being told that a different subject stated a WTA of \$8 for completing 20 tasks. Through the lens of our model, the reference point u_r is (some function of) the number of effort tasks the other subject was offered, while the evaluation of the reference point, $\tilde{u}_r$, is given by (some function of) the other subject’s WTA. Graeber & Enke (2026) find that stated WTA increases in the other subject’s WTA and simultaneously decreases in the other subject’s offered workload, in line with the predictions of our model.

There is related evidence also in the perceptual domain in psychology. In experiments in which subjects sequentially estimate the magnitude of visual stimuli, a recent result is that sub-

jects' estimates decrease in the reference stimulus (e.g., the previous stimulus) but increase in the subject's estimate of the reference stimulus (e.g., Moon and Kwon, 2022; Sadil et al., 2024). Thus, again, beliefs exhibit a contrast effect with respect to the true reference quantity (u_r) but increase in the evaluation of the reference quantity ($\tilde{u}_r$).

B.6 Beliefs with Two Uncertain Comparison Points

The following result on updating with two comparison points appears in Balakrishnan and Dean (2025); we reproduce it here for convenience and refer to that paper for a proof.

Proposition 17 (Balakrishnan and Dean 2025). *Let x , y and z be three choice alternatives with beliefs about utilities along a given dimension given by*

$$\begin{aligned} u_x &\sim N(\tilde{u}_x, \sigma_x^2) \\ u_y &\sim N(\tilde{u}_y, \sigma_y^2) \\ u_z &\sim N(\tilde{u}_z, \sigma_z^2) \end{aligned}$$

distributed independently. Then, the the expected value of x having learned that it is better than y which is in turn better than z is

$$\mathbb{E}(u_x | u_x > u_y > u_z) = \tilde{u}_x + T((\tilde{u}_x, \tilde{u}_y, \tilde{u}_z, \sigma_x, \sigma_y, \sigma_z))$$

with

$$\begin{aligned} T((\tilde{u}_x, \tilde{u}_y, \tilde{u}_z, \sigma_x, \sigma_y, \sigma_z)) &: = \frac{\sigma_x^2}{\sqrt{(\sigma_x^2 + \sigma_y^2)}} \phi\left(\frac{\tilde{u}_x - \tilde{u}_y}{\sqrt{(\sigma_x^2 + \sigma_y^2)}}\right) \\ &\cdot \Phi\left(\left(\frac{(\tilde{u}_y - \tilde{u}_x) \sigma_x}{\sigma_y \sqrt{\sigma_x^2 + \sigma_y^2}} - \left(\frac{\tilde{u}_z - \tilde{u}_x}{\sigma_x}\right) \left(\frac{\sqrt{\sigma_x^2 + \sigma_y^2}}{\sigma_y}\right)\right) \frac{1}{\sqrt{1 + \left(\frac{1}{\sigma_y^2} + \frac{1}{\sigma_x^2}\right) \sigma_z^2}}\right) \\ &\cdot \frac{1}{\Phi_2\left(\frac{\tilde{u}_x - \tilde{u}_y}{\sqrt{\sigma_x^2 + \sigma_y^2}}, \frac{\tilde{u}_y - \tilde{u}_z}{\sqrt{\sigma_y^2 + \sigma_z^2}}; g(\sigma_x, \sigma_y, \sigma_z)\right)} \end{aligned}$$

where Φ_2 is the CDF of the bivariate standard normal with correlation g given by

$$g(\sigma_x, \sigma_y, \sigma_z) = \frac{-1}{\sqrt{\left(1 + \frac{\sigma_x^2}{\sigma_y^2}\right) \left(1 + \frac{\sigma_z^2}{\sigma_y^2}\right)}}.$$

The expected utility of z is given by

$$\mathbb{E}(u_z | u_x > u_y > u_z) = \tilde{u}_z - B((\tilde{u}_x, \tilde{u}_y, \tilde{u}_z, \sigma_x, \sigma_y, \sigma_z))$$

with

$$\begin{aligned} B((\tilde{u}_x, \tilde{u}_y, \tilde{u}_z, \sigma_x, \sigma_y, \sigma_z)) &= \frac{\sigma_z^2}{\sqrt{(\sigma_y^2 + \sigma_z^2)}} \phi\left(\frac{\tilde{u}_z - \tilde{u}_y}{\sqrt{\sigma_z^2 + \sigma_y^2}}\right) \\ &\cdot \Phi\left(\left(\frac{(\tilde{u}_z - \tilde{u}_y) \sigma_z}{\sigma_y \sqrt{\sigma_z^2 + \sigma_y^2}} - \left(\frac{(\tilde{u}_z - \tilde{u}_x)}{\sigma_z}\right) \left(\frac{\sqrt{\sigma_y^2 + \sigma_z^2}}{\sigma_y}\right)\right) \frac{1}{\sqrt{1 + \left(\frac{1}{\sigma_y^2} + \frac{1}{\sigma_z^2}\right) \sigma_x^2}}\right) \\ &\cdot \frac{1}{\Phi_2\left(\frac{\tilde{u}_x - \tilde{u}_y}{\sqrt{\sigma_x^2 + \sigma_y^2}}, \frac{\tilde{u}_y - \tilde{u}_z}{\sqrt{\sigma_y^2 + \sigma_z^2}}; g(\sigma_x, \sigma_y, \sigma_z)\right)}. \end{aligned}$$

While for y it is

$$\mathbb{E}(u_y | u_x > u_y > u_z) = \tilde{u}_y + \frac{\sigma_y^2}{\sigma_z^2} B(\tilde{u}_x, \tilde{u}_y, \tilde{u}_z, \sigma_x, \sigma_y, \sigma_z) - \frac{\sigma_y^2}{\sigma_x^2} T(\tilde{u}_x, \tilde{u}_y, \tilde{u}_z, \sigma_x, \sigma_y, \sigma_z).$$

B.7 Avoiding Losses with Multiple Boosts

We show how the result of Section 4.1 extends to the case in which the DM takes into account all ordinal comparisons, not only those involving the reference point. First we show that the result holds in the absence of ordinal noise.

Corollary 2. *Let the random variables $u_{x,1}$, $u_{x,2}$, $u_{y,1}$, $u_{y,2}$ have nondegenerate, independent normal distributions; and reference utilities $u_{r,1}$, $u_{r,2}$ also have normal distributions (possibly degenerate), independent of each other and all other random variables. Subjects receive cardinal signals $s_{a,i} = u_{a,1} + \epsilon_{a,1}$ for $a \in \{x, y\}$ and $i \in \{1, 2\}$, with each $\epsilon_{a,1}$ independent of all other random variables. Assume subjects receive noiseless ordinal signals about each pairwise comparison. Then,*

$$\mathbb{E} \left[\mathbb{E}[U(x) - U(y) \mid s] \mid u_{x,1} > u_{y,1} > u_{r,1}, u_{y,2} > u_{x,2} > u_{r,2} \right] \\ < \mathbb{E} \left[\mathbb{E}[U(x) - U(y) \mid s] \mid u_{x,1} > u_{r',1} > u_{y,1}, u_{y,2} > u_{x,2} > u_{r,2} \right].$$

This result is a direct implication of Proposition 3. Moreover, note that since interim beliefs have full support, the probability that x is chosen (when $\mathbb{E}[U(x) - U(y) \mid \text{all observed signals}] > 0$) is also increasing.

The following proposition uses continuity to extend the result of Proposition 3 to the case where there are too small levels of ordinal noise. For any two random variables x and y , denote by o_{xy} the ordinal signal between them with normal noise (as in the rest of the paper).

Proposition 18. *Let u_x, u_y, u_z be independent normally distributed random variables with strictly positive variances, and suppose the DM observes the three pairwise ordinal signals o_{xy}, o_{yz}, o_{xz} , with independent noise of common variance σ_o^2 . Write $o := (o_{xy}, o_{yz}, o_{xz})$. Then there exists $\bar{\sigma}_o^2 > 0$ such that, for every $\sigma_o^2 < \bar{\sigma}_o^2$,*

$$\mathbb{E} \left[\mathbb{E}[u_x - u_y \mid o] \mid u_x > u_z > u_y \right] > \mathbb{E} \left[\mathbb{E}[u_x - u_y \mid o] \mid u_x > u_y > u_z \right].$$

A direct corollary of Proposition 18 is that Corollary 2 also extends to the case of small ordinal noise.

Corollary 3. *Modify the set up of Proposition 18 to allow for each ordinal comparison to be observed with an independent normally distributed noise term with common variance σ_o^2 . Then there exists $\bar{\sigma}_o^2 > 0$ such that, for every $\sigma_o^2 < \bar{\sigma}_o^2$,*

$$\mathbb{E} \left[\mathbb{E}[U(x) - U(y) \mid s, o] \mid u_{x,1} > u_{y,1} > u_{r,1}, u_{y,2} > u_{x,2} > u_{r,2} \right] \\ < \mathbb{E} \left[\mathbb{E}[U(x) - U(y) \mid s, o] \mid u_{x,1} > u_{r',1} > u_{y,1}, u_{y,2} > u_{x,2} > u_{r,2} \right].$$

B.8 Decoy Effects

B.8.1 Asymmetric Dominance

We can use our model to study a classic empirical pattern in choice: the asymmetric dominance, or decoy effect, a classic violation of WARP; Soltani et al. (2012) includes a review. In it, there are two dimensions, and subjects are first asked to choose between x and y , where x is better on dimension 2 and y is better on dimension 1. In a second choice set, subjects are asked to

choose between x , y , and a third option z , which is dominated by x in both dimensions but better than y in dimension 2. The empirical observation is that adding z increases the fraction of subjects choosing x . In what follows, assume that DMs receive cardinal signals $s_{i,a}$ as well as noiseless ordinal signals among all available options; let s and o denote the vectors of all such signals, respectively. Prior variances and cardinal-signal-error variances are strictly positive in every dimension for each option (and, as usual, all utility components are independent). Our model naturally gives rise to the attraction effect, as shown by the following result, a corollary of Proposition 3.

Corollary 4. *Suppose $n = 2$ and all ordinal signals are noiseless ($\sigma_{o,i}^2 = 0 \forall i$). Under the assumptions on x , y , and z of the paragraph above, then,*

$$\mathbb{E}[U(x) - U(y) | u_{y,1} > u_{x,1}, u_{x,2} > u_{y,2}, s] < \mathbb{E}[U(x) - U(y) | u_{y,1} > u_{x,1} > u_{z,1}, u_{x,2} > u_{z,2} > u_{y,2}, s].$$

Intuitively, the addition of z boosts u_x relative to u_y in both dimensions: In dimension 1, u_z boosts both u_x and u_y upwards, but as u_x is closer to u_z than u_y , it receives the bigger boost. In dimension 2, u_z boosts u_x upwards and u_y downwards, once again increasing their difference. Thus, for every signal realization, the introduction of z strictly increases the assessment of x relative to y . Since interim beliefs have full support, the probability that x is chosen (when $\mathbb{E}[U(x) - U(y) | s, o] > 0$) strictly increases, in line with the evidence. (Also in line with the evidence, our model predicts that, because x dominates z , the latter is never chosen.)

Moreover, because our effect is based on information, it does not rely on z being choosable: the exact same effects would be at play if z is observable but cannot be chosen. This is in line with the experimental evidence of the “phantom decoy” of Soltani et al. (2012).

B.8.2 The Compromise Effect

As noted, our model unambiguously predicts the asymmetric dominance effect because adding the decoy produces a negative boost for y but not for x . Notably, this asymmetry is absent in a second classic violation of the independence of irrelevant alternatives: the compromise effect. In it, introducing a decoy produces one additional positive and one additional negative comparison for both target options, x and y . Rather than asymmetrically adding advantages and disadvantages, the compromise-effect decoy makes x the “middle option” in both dimensions. Consistent with the intuition that our model explains decoy effects that reflect asymmetric advantages/disadvantages, our model does not generally give rise to the compromise effect. Instead, our model can generate either the compromise effect or the opposite, depending on where the decoy is placed.

As a concrete example, consider a case in which interim beliefs are such that the means of x are $\{2,2\}$ on dimensions 1 and 2, respectively, while the means of y are $\{3,1\}$. Assume that standard deviations are 1 for each alternative on each dimension, and ordinal rankings consistent with the stated means. If we add a phantom decoy z with means $\{1,3\}$ (and standard deviations also equal to 1) then this will have no effect on the average evaluation of x and y due to the symmetry of the problem.

If, instead, z has mean $\{1,4\}$ then our model predicts the compromise effect. This is because moving z further away in dimension 2 has a bigger effect on the closer option. In dimension 1, x receives 0 boost, as the positive boost from z and the negative boost from y cancel each other out. Using Proposition 17, y receives a positive boost of about 0.36. In dimension 2, x receives a positive boost of 0.17: the positive boost from y is larger than the negative boost from z because the former is closer. y receives a negative boost of 0.31, meaning that x is favored overall.

Finally, note that if we place the decoy at $\{0,3\}$, the pattern is reversed. Now x receives a negative boost in dimension 1 of about -0.17 and a 0 boost in dimension 2, while y receives a positive boost of 0.31 in dimension 1 and a negative boost of about 0.36 in dimension 2, meaning that overall the addition of z favors y .

B.9 Bracketing

B.9.1 Bayesian Narrow benchmark

Maintain the reference-centered homoskedastic assumptions of Section 6, so that $\tilde{u}_{x,i} = u_{r,i}$, $\tau_1 = \tau_2 = \tau > 0$, and $\rho \in (-1, 1)$. The reference utility components are known and no additional cardinal signals are received. Under Bayesian Narrow Bracketing, the agent uses the true joint distribution to report $\mathbb{E}[U(x) \mid o_1^N, o_2^N]$ rather than the sum of the two marginal posterior means. All ordinal-noise variances below are finite and nonnegative.

Proposition 19 (Bayesian Narrow versus Broad Bracketing). *Bayesian Narrow Bracketing has strictly lower expected MSPE than Broad Bracketing if and only if*

$$\sigma_B^2 > 2(\tau^2 + \sigma_N^2) \left[1 + \frac{2}{\pi} \arcsin \left(\frac{\tau^2}{\tau^2 + \sigma_N^2} \rho \right) \right] - 2\tau^2(1 + \rho).$$

The two rules have equal expected MSPE at any admissible equality. If $\rho = 0$, the cutoff is $2\sigma_N^2$.

Figure 9 depicts the comparison between Bayesian Narrow and Broad Bracketing under the same reference-centered homoskedastic assumptions.

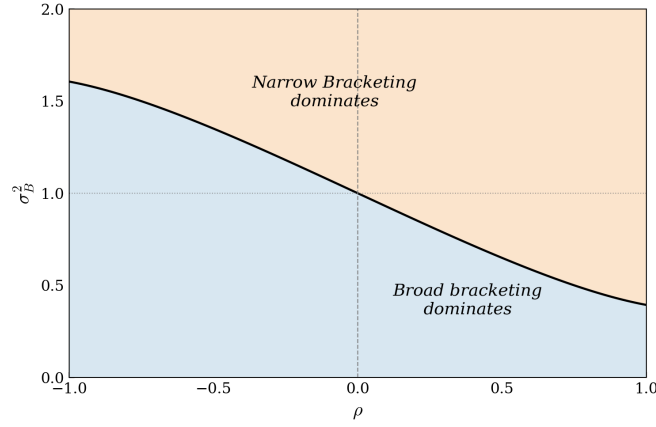


Figure 9: Dominance regions in the (ρ, σ_B^2) space under $\tilde{u}_{x,i} = u_{r,i}$, $\tau_1 = \tau_2 = \tau$, and common narrow ordinal-noise variance σ_N^2 . The figure sets $\tau = 1$ and $\sigma_N^2 = 1/2$ and varies σ_B^2 from 0 to $4\sigma_N^2$. The boundary is the Bayesian Narrow threshold in Proposition 19. Above the boundary, Bayesian Narrow Bracketing has lower expected MSPE than Broad Bracketing; below it, Broad Bracketing has lower expected MSPE. The two rules have equal expected MSPE on the boundary.

The next result separates the information effect from the naivete effect under the matched-noise normalization used in panel (b) of Figure 5.

Corollary 5 (Information and naivete effects). *Suppose $\sigma_B^2 = 2\sigma_N^2$. Let MSPE_B , MSPE_N , and MSPE_{BN} denote the true expected MSPEs under Broad, naive Narrow, and Bayesian Narrow Bracketing, respectively. Define*

$$\kappa := \frac{\tau^2}{\tau^2 + \sigma_N^2}, \quad q(\rho) := \frac{2}{\pi} \arcsin(\kappa\rho).$$

Then

$$\text{MSPE}_B - \text{MSPE}_{BN} = \frac{4\tau^4}{\pi(\tau^2 + \sigma_N^2)} \frac{(1 + \rho)^2 [\kappa\rho - q(\rho)]}{[1 + \kappa\rho][1 + q(\rho)]}, \quad (15)$$

$$\text{MSPE}_N - \text{MSPE}_{BN} = \frac{4\tau^4}{\pi(\tau^2 + \sigma_N^2)} \frac{[\rho - q(\rho)]^2}{1 + q(\rho)}. \quad (16)$$

The first expression has the sign of ρ , whereas the second is nonnegative and equals zero if and only if $\rho = 0$. Consequently,

$$\rho < 0 \implies \text{MSPE}_B < \text{MSPE}_{BN} < \text{MSPE}_N,$$

whereas

$$\rho > 0 \implies \text{MSPE}_{BN} < \min\{\text{MSPE}_B, \text{MSPE}_N\}.$$

For positive correlation, the ranking of Broad and naive Narrow Bracketing can depend on the parameters. It satisfies

$$\text{MSPE}_N < \text{MSPE}_B \quad \text{for every } \rho \in (0, 1) \quad \Longleftrightarrow \quad \sigma_N^2 \leq \tau^2.$$

If $\sigma_N^2 > \tau^2$, Broad Bracketing has lower expected MSPE than naive Narrow Bracketing for all ρ sufficiently close to 1. At $\rho = 0$, all three rules have the same expected MSPE.

B.9.2 General Gaussian case

We now extend the analysis in Section 6 to arbitrary component means, reference points, and unequal component and signal-noise variances. Let

$$U(x) = u_{x,1} + u_{x,2}, \quad U(r) = u_{r,1} + u_{r,2},$$

where the reference utility components are fixed and known, and suppose

$$\begin{pmatrix} u_{x,1} \\ u_{x,2} \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \tilde{u}_{x,1} \\ \tilde{u}_{x,2} \end{pmatrix}, \begin{pmatrix} \tau_1^2 & \rho\tau_1\tau_2 \\ \rho\tau_1\tau_2 & \tau_2^2 \end{pmatrix} \right), \quad \tau_1, \tau_2 > 0, \quad \rho \in (-1, 1).$$

Identifying the ordinal realizations $+$ and $-$ with $+1$ and -1 , the broad signal is

$$o^B = \text{sgn}(U(x) - U(r) - v_B), \quad v_B \sim \mathcal{N}(0, \sigma_B^2),$$

and the narrow signals are

$$o_i^N = \text{sgn}(u_{x,i} - u_{r,i} - v_i), \quad v_i \sim \mathcal{N}(0, \sigma_{N,i}^2), \quad i = 1, 2.$$

All noise terms are mutually independent and independent of $(u_{x,1}, u_{x,2})$, with finite nonnegative variances. No additional cardinal signals are received. The main-text model is recovered by imposing

$$\tilde{u}_{x,i} = u_{r,i}, \quad \tau_1 = \tau_2 = \tau, \quad \sigma_{N,1}^2 = \sigma_{N,2}^2 = \sigma_N^2.$$

Let $\Phi_2(\cdot, \cdot; \eta)$ denote the standard bivariate normal distribution function with correlation η . Define the two Gaussian truncation functions

$$\mathcal{T}(z) := \frac{\phi(z)^2}{\Phi(z)\Phi(-z)}$$

and

$$\mathcal{T}_{12}(z_1, z_2; \eta) := \frac{\phi(z_1)\phi(z_2)[\Phi_2(z_1, z_2; \eta) - \Phi(z_1)\Phi(z_2)]}{\Phi(z_1)\Phi(-z_1)\Phi(z_2)\Phi(-z_2)}.$$

Finally, write

$$V := \mathbb{V}(U(x)) = \tau_1^2 + \tau_2^2 + 2\rho\tau_1\tau_2, \quad s_i := \sqrt{\tau_i^2 + \sigma_{N,i}^2}, \quad i = 1, 2.$$

Define

$$\text{MSPE}_B := \mathbb{E} \left[(U(x) - \mathbb{E}[U(x) \mid o^B])^2 \right]$$

and

$$\text{MSPE}_N := \mathbb{E} \left[\left(U(x) - \{ \mathbb{E}[u_{x,1} \mid o_1^N] + \mathbb{E}[u_{x,2} \mid o_2^N] \} \right)^2 \right].$$

All expectations are taken under the true joint distribution.

Proposition 20. *In the general Gaussian model,*

$$\text{MSPE}_B = V - \frac{V^2}{V + \sigma_B^2} \mathcal{T} \left(\frac{\tilde{u}_{x,1} + \tilde{u}_{x,2} - U(r)}{\sqrt{V + \sigma_B^2}} \right), \quad (17)$$

while

$$\begin{aligned} \text{MSPE}_N = V - & \left[\sum_{i=1}^2 \frac{\tau_i^2(\tau_i^2 + 2\rho\tau_1\tau_2)}{s_i^2} \mathcal{T} \left(\frac{\tilde{u}_{x,i} - u_{r,i}}{s_i} \right) \right. \\ & \left. - \frac{2\tau_1^2\tau_2^2}{s_1s_2} \mathcal{T}_{12} \left(\frac{\tilde{u}_{x,1} - u_{r,1}}{s_1}, \frac{\tilde{u}_{x,2} - u_{r,2}}{s_2}; \frac{\rho\tau_1\tau_2}{s_1s_2} \right) \right]. \end{aligned} \quad (18)$$

Consequently, naive Narrow Bracketing has strictly lower MSPE than Broad Bracketing if and only if the bracketed reduction in (18) exceeds the reduction in (17).

Proposition 21. *Suppose $\tilde{u}_{x,i} = u_{r,i}$ for $i = 1, 2$, while allowing different component variances and different narrow-signal noise variances. Then*

$$\text{MSPE}_B = V - \frac{2}{\pi} \frac{V^2}{V + \sigma_B^2}, \quad (19)$$

and

$$\text{MSPE}_N = V - \frac{2}{\pi} \left[\sum_{i=1}^2 \frac{\tau_i^2(\tau_i^2 + 2\rho\tau_1\tau_2)}{s_i^2} - \frac{4\tau_1^2\tau_2^2}{\pi s_1s_2} \arcsin \left(\frac{\rho\tau_1\tau_2}{s_1s_2} \right) \right]. \quad (20)$$

If the bracketed expression in (20) is positive, naive Narrow Bracketing has strictly lower MSPE than

Broad Bracketing if and only if

$$\sigma_B^2 > \frac{V^2}{\sum_{i=1}^2 \frac{\tau_i^2(\tau_i^2 + 2\rho\tau_1\tau_2)}{s_i^2} - \frac{4\tau_1^2\tau_2^2}{\pi s_1 s_2} \arcsin\left(\frac{\rho\tau_1\tau_2}{s_1 s_2}\right)} - V.$$

If the right-hand side is nonnegative, the two rules have equal MSPE when σ_B^2 equals it. If the right-hand side is negative, naive Narrow Bracketing has strictly lower MSPE for every $\sigma_B^2 \geq 0$. If the bracketed expression is nonpositive, Broad Bracketing has strictly lower MSPE for every finite σ_B^2 .

Under the homoskedastic specialization $\tau_1 = \tau_2 = \tau$ and $\sigma_{N,1}^2 = \sigma_{N,2}^2 = \sigma_N^2$, we have $V = 2\tau^2(1 + \rho)$, and the bracketed expression becomes

$$\frac{2\tau^4}{\tau^2 + \sigma_N^2} \left[1 + 2\rho - \frac{2}{\pi} \arcsin\left(\frac{\rho\tau^2}{\tau^2 + \sigma_N^2}\right) \right] = \frac{2\tau^4}{\tau^2 + \sigma_N^2} \Gamma_N(\rho).$$

Thus the result reduces to Proposition 9.

B.10 Caution

In this section we expand on the cautious extension of our model discussed in Section 7.

As mentioned above, we model aversion to uncertainty using the classical approach: adding a concave transformation, which yields aversion to uncertainty exactly like the concavity of a utility function leads to risk aversion in standard Expected Utility; or like a concave transformation leads to ambiguity aversion in the Smooth model of ambiguity aversion (Klibanoff et al., 2005). In particular, let $\zeta : \mathbb{R} \rightarrow \mathbb{R}$, strictly increasing, continuous, and absolutely integrable under any distribution considered here. Then, if x is the item of interest with $U(x) = \sum u_{x,i}$ its utility distributed according to $p_{U(x)}$ (under the assumptions of the paper) and $\mathcal{I}$ a (Borel) information set, denote by M the set of all combinations of $x|\mathcal{I}$ that we study.³⁰ Then, the agent evaluates each $x|\mathcal{I} \in M$ by

$$V(x|\mathcal{I}) = \int \zeta(U(x)) dp_{U(x)|\mathcal{I}}. \quad (21)$$

When preferences are represented as in (21) for all combinations, we say that it admits a *cautious representation*.³¹ Note that this model is very general and requires very few assumptions: it is

³⁰Recall that we assume that $U(x) = \sum u_{x,i}$ where each $u_{x,i}$ are independent, and that we only consider information sets $\mathcal{I}$ that can be written independently across dimensions.

³¹A preference relation is a complete, reflexive, and transitive binary relation. As noted above, this representation is related to the one in Cerreia-Vioglio et al. (2024). Indeed, notice how the uncertainty is about the utilities in each dimension, and the min form of the representation in that paper can be seen as a limit case of aversion, just like the MaxMin Expected Utility model of (Gilboa and Schmeidler, 1989) can be seen as a limit case of the Smooth

a routine exercise to show that preferences can be represented this way if and only if they are complete preferences that satisfy monotonicity (higher utility is better), (topological) continuity, and standard vNM Independence (in the appropriate mixture space).

For the purposes of our paper, particularly natural is the case in which ζ is of the CARA form, that is $\zeta(x) := -e^{-\alpha x}/\alpha$ for $\alpha \neq 0$, and $\zeta(x) = x$ for $\alpha = 0$; α captures the attitude towards this uncertainty— $\alpha > 0$ captures aversion, $\alpha = 0$ neutrality, and $\alpha < 0$ an uncertainty seeking attitude.³² Importantly, this gives a very tractable functional form in our setup, as the following proposition illustrates.

Proposition 22. *Let $\succsim$ be a preference relation over M that admits a cautious representation with $\zeta(x) = -e^{-\alpha x}/\alpha$, where $\alpha \neq 0$. For every information set $\mathcal{I}$ under which utility components are conditionally independent across dimensions, $\succsim$ is represented by*

$$\bar{U}(x \mid \mathcal{I}) := \sum_{i=1}^n \text{CE}(u_{x,i} \mid \mathcal{I}), \quad \text{CE}(u_{x,i} \mid \mathcal{I}) := -\frac{1}{\alpha} \ln \mathbb{E}[e^{-\alpha u_{x,i}} \mid \mathcal{I}].$$

If $u_x \mid s \sim \mathcal{N}(\tilde{u}_x^s, \sigma_x^2)$, then

$$\text{CE}(u_x \mid s) = \tilde{u}_x^s - \frac{\alpha \sigma_x^2}{2}.$$

Moreover, suppose $u_r \sim \mathcal{N}(\tilde{u}_r, \sigma_r^2)$ and $o = +$ if $u_x - u_r \geq v$ and $o = -$ otherwise, with $v \sim \mathcal{N}(0, \sigma_o^2)$; assume $\sigma_x^2 > 0$, u_r is independent of anything else, and v is independent of (u_x, u_r) . Then

$$\text{CE}(u_x \mid s, o) = \tilde{u}_x^s - \frac{\alpha \sigma_x^2}{2} + \bar{\psi}_\alpha \left(\frac{\tilde{u}_x^s - \tilde{u}_r}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}, \frac{\alpha \sigma_x^2}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}, o \right)$$

where

$$\bar{\psi}_\alpha(m, q, o) := \begin{cases} \frac{1}{\alpha} [\ln \Phi(m) - \ln \Phi(m - q)], & o = +, \\ -\frac{1}{\alpha} [\ln \Phi(-m + q) - \ln \Phi(-m)], & o = -. \end{cases}$$

This proposition shows that preferences admit a very tractable representation, even if we add caution. First, it shows that we can represent aggregate preferences as the sum of the dimension-by-dimension *certainly equivalents* $\text{CE}(u_{x,i} \mid \mathcal{I})$.³³ Moreover, each dimension's component admits

ambiguity preferences of (Klibanoff et al., 2005). However, as noted earlier, unlike that paper, here we do not consider options relative to a reference point, but in absolute terms, and study the impact of information.

³²The CARA version of the model is characterized by the following behavioral property. For any $x \in M$ and any $m \in \mathbb{R}$, denote m_1x the object that returns (known) utility m in dimension 1 and x otherwise. Then, we say that a preference is *separable* if $m_1x \succsim m_1y \Leftrightarrow n_1x \succsim n_1y$ for all $x, y \in M$ and $m, n \in \mathbb{R}$. Assuming sufficiently rich domain, it is a routine exercise to show that if $\succsim$ admits a cautious representation, then $\succsim$ is separable if and only if $\exists \alpha \in \mathbb{R}$ such that $\succsim$ admits a cautious representation with index ζ with either $\zeta(x) = -e^{-\alpha x}/\alpha$, or ζ linear.

³³Indeed, note that $\text{CE}(u_{x,i} \mid \mathcal{I})$ is the certainty equivalent of $u_{x,i} \mid \mathcal{I}$ according to the agent's preferences. Note

a simple functional form. To understand it, let us compare with the mean beliefs after ordinal signals obtained in Proposition 1.

Suppose that $o = +$ and recall from Proposition 1 that

$$\mathbb{E}[u_x \mid +, s] = \underbrace{\tilde{u}_x^s}_{\text{Ref.-indep. component}} + \underbrace{\frac{\sigma_x^2}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}} \cdot \frac{\phi\left(\frac{\tilde{u}_x^s - \tilde{u}_r}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}\right)}{\Phi\left(\frac{\tilde{u}_x^s - \tilde{u}_r}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}\right)}}_{\text{Boost} > 0}.$$

With caution, we replace this with

$$\text{CE}(x \mid s, o) = \underbrace{\tilde{u}_x^s}_{\text{Ref.-indep.}} - \underbrace{\alpha \frac{\sigma_x^2}{2}}_{\text{Variance adjustment}} + \underbrace{\frac{1}{\alpha} \left(\ln \Phi\left(\frac{\tilde{u}_x^s - \tilde{u}_r}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}\right) - \ln \Phi\left(\frac{\tilde{u}_x^s - \tilde{u}_r - \alpha \sigma_x^2}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}\right) \right)}_{\alpha\text{-Adjusted boost} > 0}.$$

That is, adding caution implies *i*) adding a variance punishment, using the variance *before* the ordinal signal (which is simple to compute); and *ii*) changing the boost to an α -dependent formula. Note that the α -adjusted boost maintains the expected signs, and, in particular, it is strictly positive if $o = +$ and strictly negative otherwise.³⁴ Indeed, it is also easily verified that this modified boost coincides with that identified in Proposition 1 if $\alpha = 0$, that is, $\lim_{\alpha \downarrow 0} \bar{\psi}_\alpha(m, \alpha k, +) = k \frac{\phi(m)}{\Phi(m)}$.³⁵ Moreover, the boost maintains the core properties that we identified for mean beliefs. In addition, unless the posterior is degenerate, the CE is strictly decreasing in α . Holding the cardinal posterior and the ordinal-signal structure fixed, the signed ordinal correction increases with α : the positive correction becomes larger, whereas the negative correction becomes less negative; however, this increase is more than offset by the change in the term $-\alpha \sigma_x^2/2$.³⁶ Thus, greater aversion to value uncertainty lowers each nondegenerate component's

that using certainty equivalents for doing dimension-by-dimension aggregation not only is the result given by the proposition, but it is also the natural step—if we wanted to aggregate across dimensions, we need to make sure items are normalized to the same unit of measure, and certainty equivalents guarantee this is the case.

³⁴Suppose $\alpha > 0$ and $o = +$. Then, the boost is obviously positive since $\alpha \sigma_x^2 > 0$ and both Φ and $\ln$ are strictly increasing. When $\alpha < 0$, then $\alpha \sigma_x^2 < 0$ but also $\frac{1}{\alpha} < 0$. The case $o = -$ is symmetric.

³⁵To see why, suppose $o = +$ and let $g(z) := \ln \Phi(z)$, so $g'(z) = \phi(z)/\Phi(z)$ and g is C^1 on $\mathbb{R}$. By the mean value theorem, for each $\alpha > 0$ there exists $\xi_\alpha \in (m - \alpha k, m)$ such that

$$\bar{\psi}_\alpha(m, \alpha k, +) = \frac{g(m) - g(m - \alpha k)}{\alpha} = k g'(\xi_\alpha) = k \frac{\phi(\xi_\alpha)}{\Phi(\xi_\alpha)}.$$

Since $\xi_\alpha \rightarrow m$ as $\alpha \downarrow 0$ and ϕ/Φ is continuous, $\lim_{\alpha \downarrow 0} \bar{\psi}_\alpha(m, \alpha k, +) = k \frac{\phi(m)}{\Phi(m)}$.

³⁶To verify monotonicity in α , let P denote the conditional law of X given $\mathcal{I}$, set $F(\alpha) := \log \mathbb{E}_P[e^{-\alpha X}]$, and define $\frac{dQ_\alpha}{dP} := \exp\{-\alpha X - F(\alpha)\}$. The Gaussian and truncated-Gaussian laws used here have finite exponential

certainty equivalent and hence the overall evaluation $\bar{U}$ whenever at least one component is nondegenerate.

We should also highlight how, despite aversion to uncertainty, preferences that admit a cautious representation respect first-order stochastic dominance: an option whose posterior utility distribution first-order stochastically dominates another's is weakly preferred, and strictly preferred whenever the two distributions differ. This follows from the strict monotonicity of ζ and the assumed absolute integrability. It is also intuitive, since this model is akin to Expected Utility with a strictly increasing index, which are known to satisfy FOSD.

We are now ready to study the implications of caution in our context.

The Endowment Effect and Status Quo Bias. Consider the classic evidence of the endowment effect; here, we review and expand on the setup discussed in the main body. For an object x of true utility $u_x \in \mathbb{R}_{++}$, let $p_{\text{WTP}} > 0$ be the maximum price the DM is willing to pay to acquire it, $p_{\text{WTA}} > 0$ the minimum price they are willing to accept to sell it, and $p_{\text{Ch}} > 0$ the amount that makes them indifferent, starting from the reference bundle of no object and zero additional money, between receiving the object and receiving p_{Ch} . We assume that priors are correctly centered, ordinal comparisons are noiseless, and the DM receives an additional cardinal signal about each component of the reference bundle, with error variance σ_e^2 .³⁷ For transparency, we evaluate all cardinal signals at their expected values, that is, at realizations with zero signal error. The true utility of no object is $u_n = 0$. The object and no-object outcomes have the same prior variance and ordinary signal precision. Let $\sigma_x^2 > 0$ denote their posterior variance after the ordinary cardinal signal and, for $\sigma_e^2 > 0$, let $\hat{\sigma}_x^2 := \left(\frac{1}{\sigma_x^2} + \frac{1}{\sigma_e^2}\right)^{-1} < \sigma_x^2$ denote their posterior variance when they are the reference outcome. Write

$$\Delta_x := \sigma_x^2 - \hat{\sigma}_x^2, \quad \tau_x := \sqrt{\sigma_x^2 + \hat{\sigma}_x^2}.$$

Beliefs about the utility of monetary gains/losses have the following structure. Let u_y denote the utility of a monetary transfer $y \in \mathbb{R}$, and let u_0 denote the utility of no transfer. True monetary utility is the identity, so that the true values are $u_y = y$ and $u_0 = 0$. Conditional on

moments of every order. Differentiation gives

$$\frac{d}{d\alpha} \text{CE}_\alpha(X \mid \mathcal{I}) = \frac{F(\alpha) - \alpha F'(\alpha)}{\alpha^2} = -\frac{D(Q_\alpha \| P)}{\alpha^2} < 0 \quad (\alpha \neq 0)$$

whenever X is not almost surely constant. At $\alpha = 0$, the continuous extension satisfies $\text{CE}_0(X \mid \mathcal{I}) = \mathbb{E}[X \mid \mathcal{I}]$ and $\partial_\alpha \text{CE}_\alpha|_{\alpha=0} = -\text{Var}(X \mid \mathcal{I})/2$.

³⁷Correct calibration rules out WTP–WTA gaps generated solely by asymmetric Bayesian shrinkage, for example because the prior mean for the object lies below its true value.

the ordinary cardinal information, let

$$u_y = u_0 + d_y, \quad u_0 \sim \mathcal{N}(0, A_m), \quad d_y \sim \mathcal{N}(y, \vartheta y^2), \quad u_0 \perp d_y, \quad (22)$$

where $A_m \geq 0$ and $\vartheta \geq 0$. (As in the rest of the paper, these random variables are independent of the object-side utilities.) Equivalently,

$$\begin{pmatrix} u_0 \\ u_y \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ y \end{pmatrix}, \begin{pmatrix} A_m & A_m \\ A_m & A_m + \vartheta y^2 \end{pmatrix} \right).$$

Thus,

$$\mathbb{V}(u_y) = A_m + \vartheta y^2, \quad \text{Cov}(u_0, u_y) = A_m, \quad \mathbb{V}(u_y - u_0) = \vartheta y^2.$$

The variance A_m captures uncertainty about a common monetary level (and may or may not equal that of having no object). An additional signal about the reference outcome u_0 may reduce A_m , but it leaves the distribution of $d_y = u_y - u_0$ unchanged. When $\vartheta > 0$, uncertainty about a monetary difference increases with the absolute amount. When $\vartheta = 0$, the common level u_0 may remain uncertain, but the DM knows every monetary utility difference exactly ($u_y - u_0 = y$ is known).

We can now apply Proposition 22. What is the advantage of object vs. no object? Suppose first that the endowment is having no object. Before ordinal signals, DM's posteriors about object and no object are, respectively, $\mathcal{N}(u_x, \sigma_x^2)$ and $\mathcal{N}(0, \hat{\sigma}_x^2)$. Then, the agent receives the (noiseless) ordinal signal that the object is strictly better than no object. Applying Proposition 22, the utility gain becomes

$$\Gamma_x := u_x - \frac{\alpha}{2} \Delta_x + \frac{1}{\alpha} \ln \frac{\Phi((u_x + \alpha \hat{\sigma}_x^2)/\tau_x)}{\Phi((u_x - \alpha \sigma_x^2)/\tau_x)}. \quad (23)$$

When the endowment is having the object, the posteriors before ordinal signals about object and no object are now $\mathcal{N}(u_x, \hat{\sigma}_x^2)$ and $\mathcal{N}(0, \sigma_x^2)$. Again applying the proposition, the utility gain becomes

$$\Lambda_x := u_x + \frac{\alpha}{2} \Delta_x + \frac{1}{\alpha} \ln \frac{\Phi((u_x + \alpha \sigma_x^2)/\tau_x)}{\Phi((u_x - \alpha \hat{\sigma}_x^2)/\tau_x)}. \quad (24)$$

We now derive the values of receiving or paying p , denoted $\mathcal{R}(p)$ and $\mathcal{P}(p)$. For $\vartheta = 0$, these are simply $\mathcal{R}(p) = \mathcal{P}(p) = p$ as there is no uncertainty about the monetary gains/losses (the common component u_0 cancels from all comparisons). For $\vartheta > 0$, let Z have the distribution of a $\mathcal{N}(1, \vartheta)$ random variable conditional on being positive; when $\vartheta = 0$, let $Z = 1$. Because the

common component u_0 is independent of the relevant monetary difference, it cancels from the corresponding certainty-equivalent differences:

$$\begin{aligned}\text{CE}(u_p \mid u_p > u_0) - \text{CE}(u_0 \mid u_p > u_0) &= -\frac{1}{\alpha} \ln \mathbb{E}[e^{-\alpha p Z}], \\ \text{CE}(u_0 \mid u_0 > u_{-p}) - \text{CE}(u_{-p} \mid u_0 > u_{-p}) &= \frac{1}{\alpha} \ln \mathbb{E}[e^{\alpha p Z}].\end{aligned}$$

For $\vartheta > 0$, the cautious value of receiving p and the cautious cost of paying p can be written as

$$\mathcal{R}(p) := -\frac{1}{\alpha} \ln \mathbb{E}[e^{-\alpha p Z}] = p - \frac{\alpha \vartheta p^2}{2} + \frac{1}{\alpha} \ln \frac{\Phi(1/\sqrt{\vartheta})}{\Phi((1 - \alpha \vartheta p)/\sqrt{\vartheta})}, \quad (25)$$

$$\mathcal{P}(p) := \frac{1}{\alpha} \ln \mathbb{E}[e^{\alpha p Z}] = p + \frac{\alpha \vartheta p^2}{2} + \frac{1}{\alpha} \ln \frac{\Phi((1 + \alpha \vartheta p)/\sqrt{\vartheta})}{\Phi(1/\sqrt{\vartheta})}. \quad (26)$$

Since object-side and monetary uncertainty are independent, the value of a bundle is the sum of its object-side and monetary certainty equivalents. The three indifference conditions are therefore

$$\mathcal{P}(p_{\text{WTP}}) = \Gamma_x, \quad (27)$$

$$\mathcal{R}(p_{\text{WTA}}) = \Lambda_x, \quad (28)$$

$$\mathcal{R}(p_{\text{Ch}}) = \Gamma_x. \quad (29)$$

Proposition 23. *Suppose $u_x, \alpha, \sigma_x^2, \sigma_e^2 > 0$ and $\vartheta \geq 0$. Then Equations (27)–(29) admit unique strictly positive solutions, and $p_{\text{WTA}} > p_{\text{Ch}} \geq p_{\text{WTP}} > 0$. The second inequality is strict if and only if $\vartheta > 0$. Hence, $p_{\text{WTA}} > p_{\text{Ch}} > p_{\text{WTP}}$ when the utility differences induced by monetary gains/losses are uncertain, whereas $p_{\text{WTA}} > p_{\text{Ch}} = p_{\text{WTP}}$ when they are known.*

Figure 10 illustrates how the endowment effect varies with object-side value uncertainty in the benchmark in which monetary utility differences are known ($\vartheta = 0$). Lastly, as already mentioned in the main body, an identical line of argument can be adapted to show how the model generates status quo bias.

C Approximations

C.1 Approximation of the Boost

In this appendix, we derive the approximation of the boost $\psi^a(t, o)$ and we show its performance.

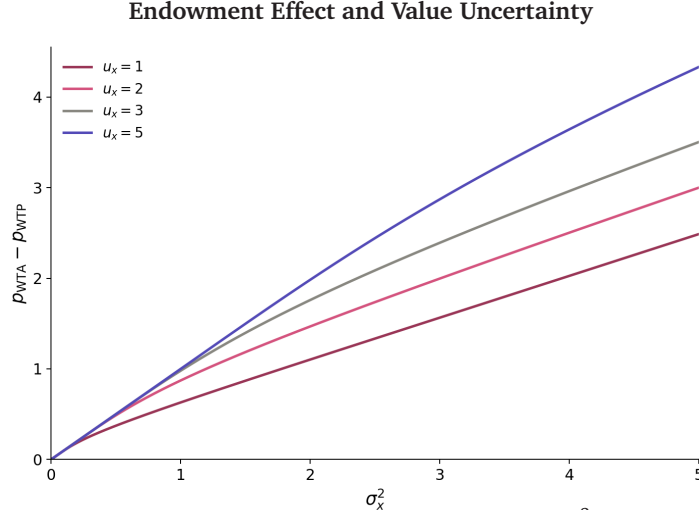


Figure 10: Comparative statics of the endowment effect ($p_{WTA} - p_{WTP}$) as σ_x^2 varies, assuming known monetary utility differences ($\vartheta = 0$), the limiting case of a perfectly known reference outcome ($\hat{\sigma}_x^2 = 0$), and $\alpha = 1$.

Derivation. We obtain our approximation by first taking a first-order Taylor approximation of $\psi(t, o)$ around $t = 0$. Note that

$$\psi(0, +) = \frac{\phi(0)}{\Phi(0)} = \frac{(1/\sqrt{2\pi})}{1/2} = \sqrt{\frac{2}{\pi}}, \quad \psi(0, -) = -\frac{\phi(-0)}{\Phi(-0)} = -\frac{\phi(0)}{\Phi(0)} = -\sqrt{\frac{2}{\pi}}.$$

Next, differentiate $\psi(t, +)$ with respect to t . Since $\phi'(t) = -t\phi(t)$ and $\Phi'(t) = \phi(t)$,

$$\psi_t(t, +) = \frac{\phi'(t)\Phi(t) - \phi(t)\Phi'(t)}{\Phi(t)^2} = \frac{(-t\phi(t))\Phi(t) - \phi(t)\phi(t)}{\Phi(t)^2} = -\frac{\phi(t)}{\Phi(t)} \left(t + \frac{\phi(t)}{\Phi(t)} \right) = -\psi(t, +)(t + \psi(t, +)).$$

Evaluating this at $t = 0$ gives $\psi_t(0, +) = -\frac{2}{\pi}$. A similar argument shows that, for $o = -$, we have $\psi_t(0, -) = -\frac{2}{\pi}$. Therefore, we can approximate

$$\begin{aligned} \psi(t, o) &\approx \psi(0, o) + \psi_t(0, o) t =: \hat{\psi}^a(t, o) \\ &= \text{sgn}(o) \sqrt{\frac{2}{\pi}} - \frac{2}{\pi} t. \end{aligned}$$

Note, however, that this approximation will result in $\hat{\psi}^a(t, +) < 0$ for sufficiently large positive t , and $\hat{\psi}^a(t, -) > 0$ for sufficiently large negative t . Since we know that $\psi(t, +) \geq 0$ and $\psi(t, -) \leq 0$, we can restrict the approximation to also have these signs. We can then define our desired approximation pointwise as the function that is closest to $\hat{\psi}^a(t, \cdot)$ while respecting these

sign restrictions, i.e.,

$$\psi^a(t, +) = \arg \min_{a \in [0, \infty)} (a - \hat{\psi}^a(t, +))^2, \quad \psi^a(t, -) = \arg \min_{a \in (-\infty, 0]} (a - \hat{\psi}^a(t, -))^2.$$

This gives our final expression

$$\psi^a(t, +) = \max \left\{ \sqrt{\frac{2}{\pi}} - \frac{2}{\pi}t, 0 \right\} \quad \psi^a(t, -) = -\max \left\{ \sqrt{\frac{2}{\pi}} + \frac{2}{\pi}t, 0 \right\}.$$

Note that $\psi^a(t, o)$ is a better approximation for $\psi(t, o)$ than $\hat{\psi}^a(t, o)$, in the sense that, for every t and o

$$|\psi^a(t, o) - \psi(t, o)| \leq |\hat{\psi}^a(t, o) - \psi(t, o)|$$

with the inequality strict for some values.

Performance. We compare our approximation $\psi^a(\cdot)$ with the true inverse Mills ratio (IMR) function $\psi(\cdot)$. In order to do so, we simulate the posterior mean $\mathbb{E}[\tilde{u}_x^{s,o}]$ (averaged over signal realizations) when the agent receives an ordinal signal under both models. Formally, we draw cardinal signal (s_i) and ordinal signals (o_i) with errors drawn independently across N simulations $i = 1, 2, \dots, N$, and we compute

$$\text{Full Model:} \quad \frac{1}{N} \sum_{i=1}^N \left[\lambda s_i + (1 - \lambda) \tilde{u}_x + \frac{\sigma_x^2}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}} \psi \left(\frac{\lambda s_i + (1 - \lambda) \tilde{u}_x - \tilde{u}_r}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}, o_i \right) \right]$$

$$\text{Approximation:} \quad \frac{1}{N} \sum_{i=1}^N \left[\lambda s_i + (1 - \lambda) \tilde{u}_x + \frac{\sigma_x^2}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}} \psi^a \left(\frac{\lambda s_i + (1 - \lambda) \tilde{u}_x - \tilde{u}_r}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}, o_i \right) \right]$$

We let $N = 10,000$ and fix parameters $\tilde{u}_x = \tilde{u}_r = u_r = 0$ and $\sigma_p = \sigma_r = 1$. We compare the simulation average across $u_x \in [-2, 2]$ for all combinations of $\sigma_\epsilon \in \{\frac{1}{2}, 1, 2\}$ and $\sigma_o \in \{0, \frac{1}{4}, \frac{1}{2}\}$. Figure 11 displays the simulated means under the full and approximated model. The approximation $\psi^a(\cdot)$ performs remarkably well, and is virtually indistinguishable from the full model.

C.2 Approximation of the Average Boost and Average Posterior Mean

Next, we present a useful approximation of the average boost and average posterior mean, integrating over signal realizations. Let $t := \frac{\lambda u_x + (1 - \lambda) \tilde{u}_x - \tilde{u}_r}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}$. Then, the average posterior mean belief

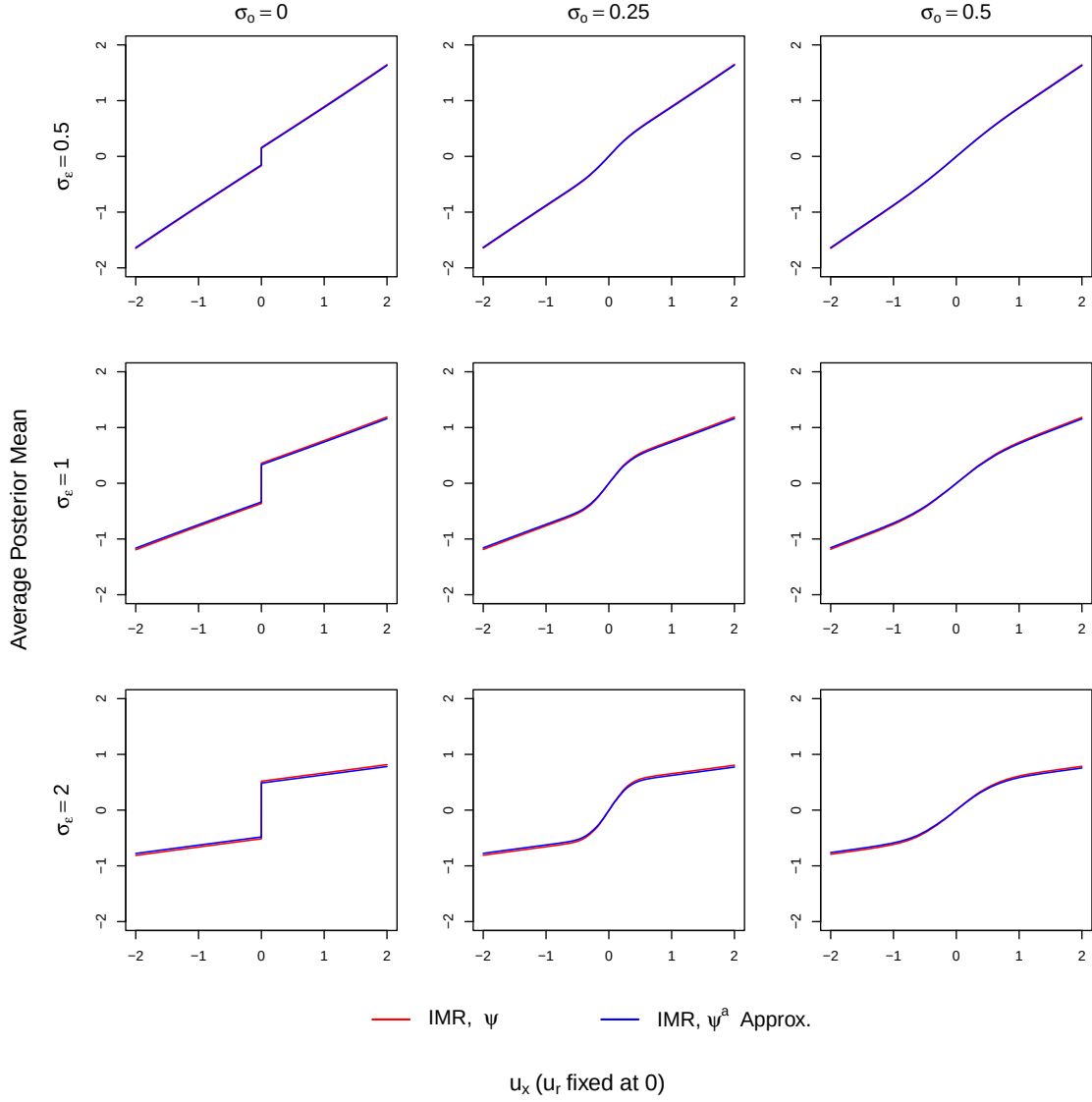


Figure 11: Simulation of posterior mean using IMR and its approximation $\psi^a(\cdot)$, averaged in both cases over 10,000 error draws. Prior and reference point parameters are set to $\tilde{u}_x = \tilde{u}_r = u_r = 0$ and utility is given by $u_x = x$, which we vary between $[-2, 2]$.

can be approximated by

$$\begin{aligned}
 \mathbb{E}[\tilde{u}_x^{s,o}] &\approx \underbrace{\lambda u_x + (1 - \lambda) \tilde{u}_x}_{\text{Bayesian shrinkage to prior; Attenuation}} + \underbrace{\frac{\sigma_x^2}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}}}_{\text{Weight of ref.-dep.}} \underbrace{\bar{\psi}^a(t, u_x, u_r, \sigma_o)}_{\text{Avg. Boost Approx.}} \\
 \text{where } \bar{\psi}^a(t, u_x, u_r, \sigma_o) &:= \underbrace{\left[\Phi\left(\frac{u_x - u_r}{\sigma_o}\right) \psi^a(t, +) + \left(1 - \Phi\left(\frac{u_x - u_r}{\sigma_o}\right)\right) \psi^a(t, -) \right]}_{\text{Gain / loss boosts weighted by prob. of correct ordinal}}. \quad (30)
 \end{aligned}$$

When the ordinal signal is noiseless, the last part simplifies to the gain-loss component in the illustrative example in Section 2.

Derivation. Write $t_s := (\lambda s + (1 - \lambda)\tilde{u}_x - \tilde{u}_r) / \sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}$, and $p := \Phi((u_x - u_r) / \sigma_o)$, with the appropriate indicator when $\sigma_o = 0$. Conditional on the true utilities, the cardinal and ordinal signals are independent, so

$$\mathbb{E}_{s,o} [\psi(t_s, o)] = p \mathbb{E}_s [\psi(t_s, +)] + (1 - p) \mathbb{E}_s [\psi(t_s, -)] .$$

Since $\mathbb{E}_s [t_s] = t$, replacing each expected boost by its clipped linear approximation evaluated at t gives (30).

Performance. We simulate the average posterior mean $\mathbb{E}[\tilde{u}_x^{s,o}]$ for the full model as we did in Appendix C.1 by averaging across 10,000 realizations of cardinal and ordinal signals. We compare this simulation average to the approximation of (30) (computed once per u_x) for all combinations of $\sigma_\varepsilon \in \{\frac{1}{2}, 1, 2\}$ and $\sigma_o \in \{0, \frac{1}{4}, \frac{1}{2}\}$. In all simulations, we set $\tilde{u}_x = \tilde{u}_r = u_r = 0$ and $\sigma_p = \sigma_r = 1$. Figure 12 illustrates the simulation results. We find that the approximation computed using $\bar{\psi}^a(\cdot)$ closely tracks the true average.

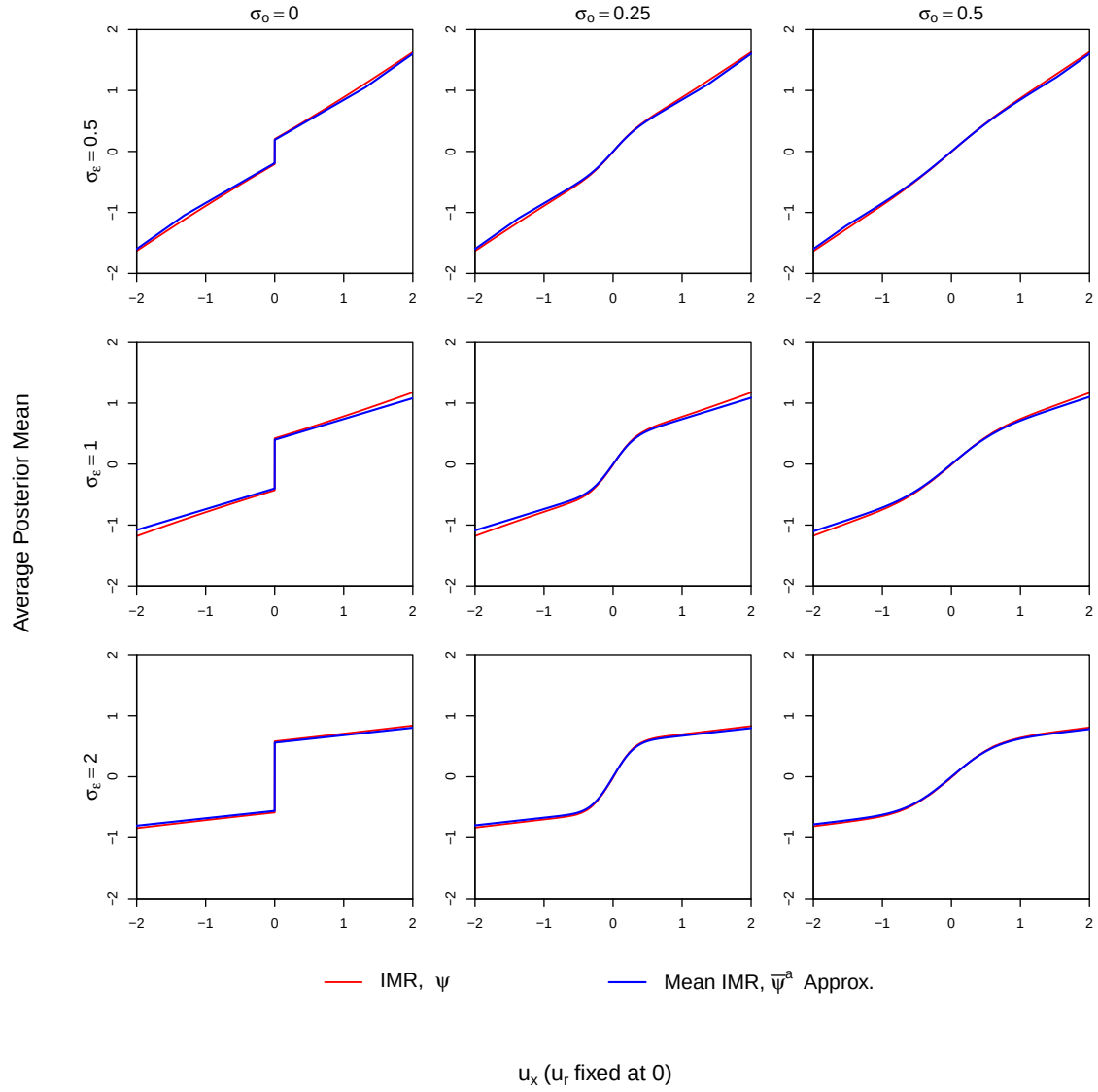


Figure 12: Simulation of Posterior Mean using IMR (average over 10,000 error draws) and its approximation $\bar{\psi}^a(\cdot)$. Prior and reference point parameters are set to $\tilde{u}_x = \tilde{u}_r = u_r = 0$ and utility is given by $u_x = x$, which we vary between $[-2, 2]$.

C.3 Approximation of Bayesian Continuous Action Choice Model

In what follows, we use simulations to show how our quasi-Bayesian, simpler model discussed in Section 4.2—in which signals are perfectly correlated but the agent ignores this—is a very good approximation, on a finite grid of values, of our original model—in which signals are independent, the agent receives ordinal signals across actions, and follows Bayes’ rule.

Recall that the DM evaluates the utilities of different levels of an action, a . Given the additive separability of our model, we here focus on a single dimension.

In our baseline Bayesian model with ordinal signals—which we refer to simply as *Bayesian model*—the DM has a prior belief about the dimension-specific utility of action a , $u_a | a \sim \mathcal{N}(\tilde{u}_a, \sigma_p^2)$, and receives two types of signals: for each action, a cardinal signal with a noise term independent across a , $s_{\text{Bayesian}}^a \sim \mathcal{N}(u_a, \sigma_{\varepsilon, \text{Bayesian}}^2)$; second, noiseless ordinal signals that tell him, for each action and associated utility outcome u_a , whether $u_a \geq u_{a'}$ for all a, a' . In other words, the DM understands monotonicity not just with respect to the reference point but also with respect to the outcomes induced by his actions. These ordinal signals ensure that the posterior mean about u_a is monotonic in the action. However, the interdependent truncations arising from the many ordinal signals make the Bayesian posterior mean intractable.

In our alternative, simpler formulation, which we refer to as *Quasi-Bayesian*, there is instead a single error draw $\varepsilon_0 \sim \mathcal{N}(0, \sigma_{\varepsilon, \text{quasi-Bayesian}}^2)$, and the DM receives signals $s_{\text{quasi-Bayesian}}^a = u_a + \varepsilon_0$. Crucially, however, the DM updates beliefs from these cardinal signals, but does so ignoring the correlation among them. Then, the (quasi-Bayesian) posterior mean is given by

$$\mathbb{E}[u_a | s_{\text{quasi-Bayesian}}^a] = \lambda s_{\text{quasi-Bayesian}}^a + (1 - \lambda) \tilde{u}_a \quad (31)$$

where λ is the usual shrinkage weight.

The thought behind the approximation is that, by picking a noise variance, the quasi-Bayesian posterior mean from the model with perfectly correlated noise terms approximates the Bayesian posterior of the model with independent noise and ordinal signals. We thus pick the variance $\sigma_{\varepsilon, \text{quasi-Bayesian}}^2$ that minimizes the distance between the two posterior means through nested interval optimization. Specifically, given the prior mean, prior variance, and the Bayesian DM’s signal variance, we identify the optimal quasi-Bayesian variance that minimizes the Mean Squared Error between the posterior means of the quasi-Bayesian and constrained Bayesian models.

Simulation. We simulate both models over a range of parameter combinations using Python’s PyMC package with the Hamiltonian Monte Carlo No-U-Turn Sampler (NUTS). Figure 13 displays the average posterior mean as a function of a for both the Bayesian and quasi-Bayesian DMs for the case in which there is no reference point (or the outcome is very far from the reference point).

The Bayesian posterior mean is computed by explicitly enforcing monotonicity via an ordered transformation, whereas the quasi-Bayesian posterior mean utilizes the analytical formula given by equation (31). The results indicate that the quasi-Bayesian approximation closely replicates the average posterior means from the fully Bayesian model, with deviations concentrated near the boundaries of the simulated range.

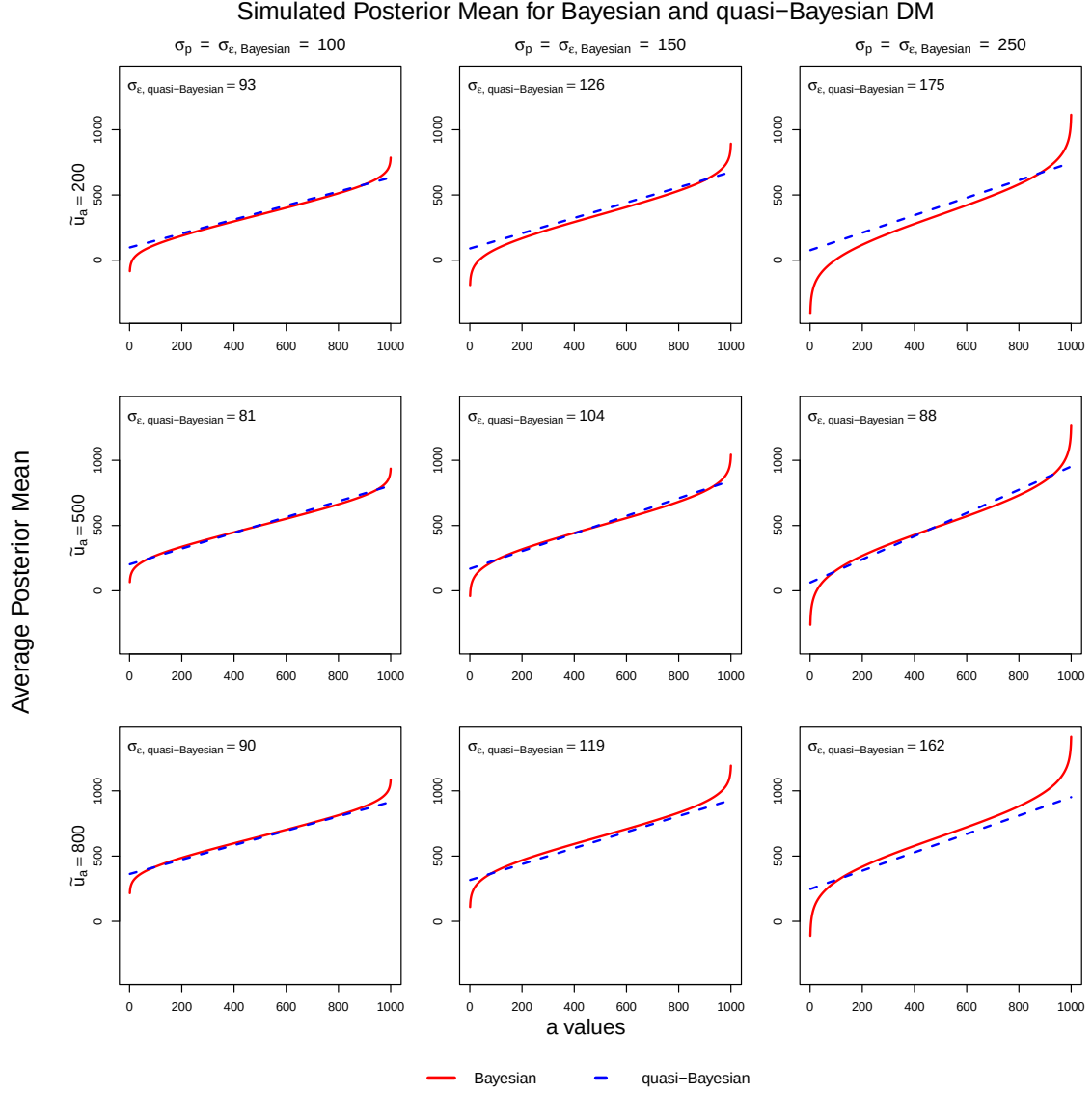


Figure 13: Simulated posterior mean utilities for a fully Bayesian agent enforcing monotonicity (solid line) and a quasi-Bayesian agent using the single-error approximation (dashed line). Each panel reports averages across 200 simulation iterations. Rows differ by prior mean $\bar{u}_a$ (left label), and columns differ by the prior standard deviation, set equal to the Bayesian agent's signal standard deviation (top label). For each panel, the quasi-Bayesian signal standard deviation minimizing the MSE relative to the Bayesian posterior mean is indicated in the top-left corner. The continuous action variable a ranges from 1 to 1000 with unit increments for simulation purposes.

Quality of approximation with a reference point. The simulations above suggest that the posterior means of the two models are close to each other when there is no reference point. We now extend the analysis to a reference point, r . Specifically, the DM now additionally receives ordinal signals o_a that indicate whether $u_a \geq u_r$. Because this adds another layer of truncation on top of the monotonicity constraints among actions, simulating the fully Bayesian model for many actions becomes computationally prohibitive. As a proof-of-concept, we instead conduct a brute-force simulation over ten randomly sampled values of $u_a \in (0, 1000)$, but constrain them to include at least two values close to the reference point. For each iteration, we first draw a noisy signal about each u_a , from which the DM creates a posterior distribution $u_a | s^a \sim \mathcal{N}(\lambda s^a + (1 - \lambda)\tilde{u}_a, \sigma_x^2)$ (where $\sigma_\varepsilon = \sigma_p = \tilde{u}_a = 0$). We truncate the posterior distributions on $u_r = 500 \geq u_a$. We then sample a candidate utility path from the sequence of truncated posterior distributions and accept this path only if it is strictly monotonic across all outcomes. We repeat this process until we collect a sufficient number of valid monotonic paths; their average forms our “fully Bayesian” benchmark. Figure 14 plots this path along with our quasi-Bayesian model. Once again, it shows that the quasi-Bayesian version provides a good approximation, even to the boost around the reference point. Notably, both the Bayesian model and its approximation generate a jump in perceived utility around the reference point.

Two forces combine to generate the jump in the Bayesian model. First, the value of the reference point is known more precisely than the value of the choice alternatives, meaning that comparative information relating the reference point is stronger. Second, the boost from the reference point is stronger for choice alternatives closer to it. Thus nearby alternatives are affected more than those further away, creating the jump.

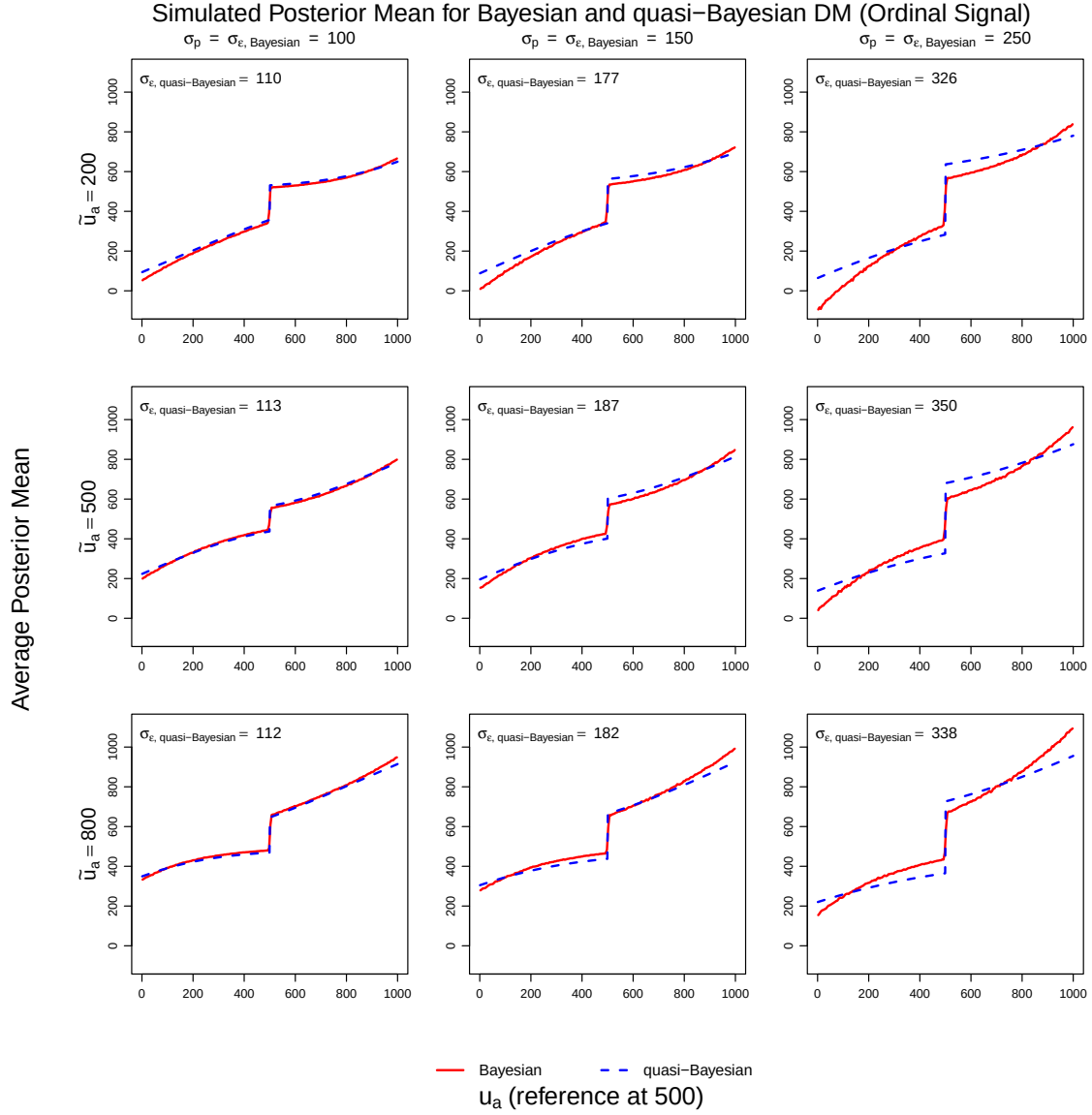


Figure 14: Comparison of posterior means for a reference-dependent DM. The solid line represents the Bayesian benchmark, while the dashed line shows the quasi-Bayesian model. Each panel reports the average of 50,000 accepted monotonic sample paths, generated via a brute-force acceptance-rejection method with posteriors truncated at the reference point ($u_r = 500$ across all panels). The quasi-Bayesian approximation is computed as the average over 1,000 draws of signal error $\epsilon_0 \sim \mathcal{N}(0, \sigma_{\epsilon}^2)$, where the quasi-Bayesian signal standard deviation $\sigma_{\epsilon, \text{quasi-Bayesian}}$ that minimizes the MSE relative to the Bayesian posterior mean is indicated in the top-left corner. Rows differ by prior mean $\tilde{u}_a$ (left label), and columns differ by the prior standard deviation, set equal to the Bayesian agent's signal standard deviation.

D Proofs of the Results in the Text

D.1 Normal identities used below

We now include several known results pertaining to the Normal distribution that will be used repeatedly below. Let $\rho(z) := \phi(z)/\Phi(z)$. Then:

$$\rho'(z) = -\rho(z)(z + \rho(z)), \quad -1 < \rho'(z) < 0, \quad \rho''(z) > 0, \quad \rho(z) \leq \rho(0) + |z|. \quad (32)$$

Note also that $z + \rho(z)$ is the strictly positive mean of a $N(z, 1)$ variable conditional on being positive, while its variance is $1 + \rho'(z) > 0$.

Moreover, ρ has several useful convexity properties. Let $h_0(z) := z + \rho(z)$ and $q(z) := \frac{\rho''(z)}{\rho(z)} = 2h_0(z)^2 - zh_0(z) - 1$. Integration by parts in the normal tail gives $\rho(-x) = x + x^{-1} - 2x^{-3} + 10x^{-5} + O(x^{-7})$ as $x \rightarrow \infty$; hence $q(-x) \sim 2x^{-4} > 0$. At any zero of q , differentiation gives $q' = 2h_0^3 > 0$. Thus q has no zero, proving $\rho'' > 0$. In fact, q is strictly increasing. To verify this, use $h'_0 = 1 + zh_0 - h_0^2$ to obtain $q' = (4h_0 - z)h'_0 - h_0$. At any zero of q' , setting $k = z/h_0 < 1$ gives

$$h_0^2 = \frac{3 - k}{(4 - k)(1 - k)}, \quad q'' = \frac{4(2k - 5)}{(k - 4)^2(k - 1)} > 0.$$

Every hypothetical zero of q' would therefore be an upward crossing. If q' were nonpositive anywhere, it would be negative on a left half-line, contradicting $q > 0$ and $q(-\infty) = 0$. Consequently $q' > 0$.

Gaussian smoothing. For $v \geq 0$, let $f_v(m) := \mathbb{E}[\rho(m + \sqrt{v}Z)]$, where $Z \sim N(0, 1)$, and let $Q_v := f_v''/f_v$. For $v > 0$, $Q_v(m)$ is the expectation of $q(z)$ under the density proportional to $\rho(z)e^{-(z-m)^2/(2v)}$. Differentiation in m gives $Q'_v(m) = \text{Cov}(q(z), z)/v > 0$ under that density. For $v = 0$, the same conclusion follows from $q' > 0$. The heat identity $\partial_v f_v = f_v''/2$ now yields, for $m > 0$,

$$\partial_v \log \frac{f_v(-m)}{f_v(m)} = \frac{1}{2}[Q_v(-m) - Q_v(m)] < 0.$$

It follows that

$$\frac{f_v(-m)}{f_v(m)} \leq \frac{\rho(-m)}{\rho(m)} = \frac{\Phi(m)}{\Phi(-m)}, \quad (33)$$

strictly when $v > 0$. In particular, if $m > 0$ and $p > \Phi(m)$, then

$$pf_v(m) - (1 - p)f_v(-m) > 0, \quad pf_v''(m) - (1 - p)f_v''(-m) > 0. \quad (34)$$

The second inequality uses the first and $Q_v(m) > Q_v(-m) > 0$. All differentiations here are justified by the normal tails and the polynomial bounds on the displayed functions and their derivatives.

Tail estimates. For fixed $v \geq 0$, as $m \rightarrow \infty$,

$$\begin{aligned} f_v(m) &\sim \frac{1}{\sqrt{1+v}} \phi\left(\frac{m}{\sqrt{1+v}}\right), \\ f'_v(m) &\sim -\frac{m}{1+v} f_v(m), \quad f''_v(m) \sim \frac{m^2}{(1+v)^2} f_v(m). \end{aligned} \tag{35}$$

These estimates are locally uniform in v . To see this, on the positive half-line use $\rho(z) = \phi(z)[1 + O(\Phi(-z))]$ and complete the square in the product of the two normal densities. The negative half-line and the error term have strictly faster Gaussian decay. For the derivative estimates, apply the same calculation to $\rho'(z) = -z\phi(z)$ and $\rho''(z) = (z^2 - 1)\phi(z)$ up to errors with faster Gaussian decay on the positive half-line. Completing the square gives leading factors $-m/(1+v)$ and $m^2/(1+v)^2 - 1/(1+v)$, respectively. The case $v = 0$ follows directly from $\rho = \phi/\Phi$.

The information kernel. For $h(z) := \phi(z)^2/[\Phi(z)\Phi(-z)] = \rho(z)\rho(-z)$,

$$(\log h)''(z) = -2 - \rho'(z) - \rho'(-z) < 0.$$

Thus h is even, strictly decreasing on $(0, \infty)$, and tends to zero at infinity. Also $h(0) = 2/\pi$ and $h''(0) = 8/\pi^2 - 4/\pi < 0$. Its derivatives used below are bounded.

Normal orthants. If (Z_1, Z_2) is standard bivariate normal with correlation γ , rotational symmetry of two independent standard normals gives

$$\Pr(Z_1 > 0, Z_2 > 0) = \frac{1}{4} + \frac{\arcsin \gamma}{2\pi}, \quad \mathbb{E}[\operatorname{sgn} Z_1 \operatorname{sgn} Z_2] = \frac{2}{\pi} \arcsin \gamma. \tag{36}$$

Indeed, writing $Z_1 = R \cos \theta$ and $Z_2 = R \cos(\theta - \arccos \gamma)$, the intersection of the two positive half-planes has angular length $\pi - \arccos \gamma$.

D.2 Proof of Proposition 1

Conditional on s , let $D := u_x - u_r - v$, $\mu_D := \tilde{u}_x^s - \tilde{u}_r$, and $\tau^2 := \sigma_x^2 + \sigma_r^2 + \sigma_o^2 > 0$. The jointly normal pair (u_x, D) has covariance σ_x^2 . Consequently it admits the regression decomposition

$$u_x = \tilde{u}_x^s + \frac{\sigma_x^2}{\tau^2}(D - \mu_D) + \xi, \quad \mathbb{E}[\xi \mid s] = 0,$$

where ξ is independent of D conditional on s . For $z := \mu_D/\tau$, standardization and integration of $t\phi(t) = -\phi'(t)$ give

$$\begin{aligned} \mathbb{E}[D - \mu_D \mid s, D \geq 0] &= \tau \frac{\int_{-z}^{\infty} t\phi(t) dt}{\Phi(z)} = \tau \frac{\phi(z)}{\Phi(z)}, \\ \mathbb{E}[D - \mu_D \mid s, D < 0] &= \tau \frac{\int_{-\infty}^{-z} t\phi(t) dt}{\Phi(-z)} = -\tau \frac{\phi(z)}{\Phi(-z)}. \end{aligned}$$

Taking expectations in the regression decomposition conditional on each sign of D gives

$$\mathbb{E}[u_x \mid s, o] = \tilde{u}_x^s + \frac{\sigma_x^2}{\tau} \psi(z, o),$$

as claimed. □

D.3 Proof of Proposition 2

Write $\delta := u_x - u_r$, $a := \sigma_x^2 > 0$, $e := \sigma_\varepsilon^2 > 0$, $t := \sigma_o$, $\tau_t^2 := a + \sigma_r^2 + t^2$, and $k_t := \lambda/\tau_t$. For $t > 0$ let $p_t(\delta) := \Phi(\delta/t)$ and $Z_t := k_t(\delta + \varepsilon)$, with $\varepsilon \sim N(0, e)$. The average posterior mean is

$$M_t(\delta) = u_r + \lambda\delta + \frac{a}{\tau_t} \mathbb{E}[H_{p_t(\delta)}(Z_t)], \quad H_p(z) := p\rho(z) - (1-p)\rho(-z). \quad (37)$$

At $t = 0$, use $p_0(\delta) = 1$ for $\delta \geq 0$ and zero otherwise. All primitive means and variances not being varied are held fixed. The normal bounds proved above justify differentiation under the expectations below.

Monotonicity and sensitivity. For $t > 0$, put $d(z) := \rho(z) + \rho(-z)$. Differentiation gives

$$M'_t(\delta) = \lambda + \frac{a}{\tau_t} \left[p'_t(\delta) \mathbb{E}d(Z_t) + k_t \mathbb{E}H'_{p_t(\delta)}(Z_t) \right]. \quad (38)$$

Since $-1 < H'_p < 0$,

$$M'_t(\delta) > \lambda - \frac{a\lambda}{\tau_t^2} \geq 0.$$

At zero, symmetry gives

$$M'_t(0) = \lambda + \frac{a}{\tau_t} \left[\frac{\sqrt{2/\pi}}{t} \mathbb{E}\rho(k_t\varepsilon) + k_t \mathbb{E}\rho'(k_t\varepsilon) \right] \longrightarrow \infty \quad (t \downarrow 0),$$

because $\mathbb{E}\rho(k_t\varepsilon) \geq \rho(0) > 0$ and $|\rho'| < 1$. Continuity gives oversensitivity near zero for all sufficiently small positive t . The tail estimates (35), together with the faster decay of $1 - p_t$ and p'_t , give $M'_t(\delta) \rightarrow \lambda \in (0, 1)$ as $|\delta| \rightarrow \infty$. For large t , the linear bound on d and boundedness of $\phi(x)(1 + |x|)$ imply, uniformly in δ ,

$$p'_t \mathbb{E}d(Z_t) = O(t^{-1}), \quad k_t |\mathbb{E}H'_{p_t}(Z_t)| = O(t^{-1}).$$

Thus $\sup_\delta |M'_t(\delta) - \lambda| = O(t^{-2})$, proving the uniform attenuation result. At $t = 0$, M_0 has an upward jump at zero, of size $2a\mathbb{E}\rho(k_0\varepsilon)/\tau_0$. Away from zero,

$$M'_0(\delta) = \lambda + \frac{a\lambda}{\tau_0^2} \begin{cases} \mathbb{E}\rho'(Z_0), & \delta > 0, \\ \mathbb{E}\rho'(-Z_0), & \delta < 0, \end{cases}$$

which lies strictly between zero and λ . This proves Part 1.

A uniform argument near zero. We record a simple observation used twice below. Suppose a family $A_t(\delta)$ is continuously differentiable near $(t, \delta) = (0, 0)$, with $A_0(0) > 0$. On a sufficiently small neighborhood, there are $c, L > 0$ such that

$$p_t(\delta)A_t(\delta) - (1 - p_t(\delta))A_t(-\delta) \geq c\{2\Phi(\delta/t) - 1\} - L\delta.$$

For $0 < \delta \leq \eta$, the ratio $\{2\Phi(\delta/t) - 1\}/\delta$ is bounded below by $\{2\Phi(\eta/t) - 1\}/\eta$, because ϕ decreases on $[0, \infty)$. Choosing η small and then t small shows that the expression in the display exceeds any prescribed constant times δ , uniformly on $(0, \eta]$.

The bias crossing. Apply this observation to $A_t(\delta) = \mathbb{E}\rho(k_t(\delta + \varepsilon))$. It follows that the bias $\beta_t(\delta) := M_t(\delta) - u_r - \delta$ is positive on $(0, \eta]$ for small $t > 0$. From (38),

$$\beta'_t(\delta) \leq -(1 - \lambda) + \frac{a}{\tau_t} p'_t(\delta) \mathbb{E}d(Z_t).$$

Uniformly for $\delta \geq \eta$, the last term tends to zero as $t \downarrow 0$, since it is bounded by a constant times $(1 + \delta)t^{-1}\phi(\delta/t)$. Hence $\beta'_t < -(1 - \lambda)/2$ on $[\eta, \infty)$ for small t . Since $\beta_t(\delta) \rightarrow -\infty$, there is exactly one positive zero Δ_t . Oddness of $M_t - u_r$, and hence of β_t , gives all the signs in Part 2. For large t , $\beta_t(0) = 0$ and $\beta'_t = M'_t - 1 < 0$, which gives the remaining assertions.

The reference-variance crossing. Let $\ell(z) := \rho(z) + z\rho'(z)$ and $F_k(\delta) := \mathbb{E}\ell(k(\delta + \varepsilon))$. Differentiating (37) gives

$$\frac{\partial M_t(\delta)}{\partial \sigma_r^2} = -\frac{a}{2\tau_t^3}G_t(\delta), \quad G_t(\delta) := p_t(\delta)F_{k_t}(\delta) - (1 - p_t(\delta))F_{k_t}(-\delta). \quad (39)$$

The function ℓ is positive for $z \leq 0$. For $z > 0$, its sign is the opposite of $z^2 + z\rho(z) - 1$, whose derivative is $2z + \rho(z) + z\rho'(z) > z + \rho(z) > 0$. Thus ℓ has exactly one zero $z^* > 0$, crossing from positive to negative.

Each F_k has at most one zero and crosses from positive to negative. Indeed, for $\delta_2 > \delta_1$, the ratio of the densities of $k(\delta_2 + \varepsilon)$ and $k(\delta_1 + \varepsilon)$ is strictly increasing in z . Comparing this ratio with its value at z^* gives $F_k(\delta_2) < r(z^*)F_k(\delta_1)$ whenever $F_k(\delta_1) \leq 0$. Also

$$F_k(0) = \frac{1}{2}\mathbb{E}[d(k\varepsilon) + k\varepsilon d'(k\varepsilon)] > 0.$$

For $m = k\delta$ and $v = k^2e$, Gaussian integration by parts gives

$$F_k(\delta) = f_v(m) + mf'_v(m) + vf''_v(m) \sim -\frac{m^2}{(1+v)^2}f_v(m) < 0 \quad (\delta \rightarrow \infty),$$

by (35). Thus there is a unique positive zero d_k . At that zero, differentiating the normal density yields

$$F'_k(d_k) = \frac{1}{ke}\mathbb{E}[(Z - z^*)\ell(Z)] < 0, \quad Z \sim N(kd_k, k^2e).$$

The crossing is strict and d_k is continuous in k .

At $t = 0$, $G_0 = F_{k_0}$ for $\delta > 0$, proving the required pattern. For small $t > 0$, the preceding near-zero argument, applied to $A_t = F_{k_t}$, gives $G_t > 0$ on $(0, \eta]$. On compact subsets of $(0, \infty)$, $G_t \rightarrow F_{k_0}$ in C^1 . This preserves the signs away from d_{k_0} and a strictly negative derivative in a neighborhood of that zero. Finally, the zeros d_{k_t} remain in a fixed compact interval; beyond their largest value, $F_{k_t}(\delta) < 0 < F_{k_t}(-\delta)$, so $G_t < 0$. These three observations exclude any additional zeros and prove Part 3(a), including the right derivative at $\sigma_r^2 = 0$. Choose the common $\underline{\sigma}$ below all the small-noise thresholds just used, and $\bar{\sigma} > \underline{\sigma}$ large enough for uniform attenuation.

Prior and cardinal-signal uncertainty. First hold e fixed and vary σ_p^2 , equivalently λ . Then $a = \lambda e$. Put $c := \sigma_r^2 + t^2$ and $\tau^2 := a + c$. For $y := \delta + \varepsilon$,

$$\frac{\partial}{\partial \lambda} \left[\lambda y + \frac{a}{\tau} H_p(\lambda y / \tau) \right] = \frac{\tau}{\lambda} q_{\alpha,p}(z), \quad q_{\alpha,p}(z) := z + \alpha \{H_p(z) + z H_p'(z)\},$$

where $z = \lambda y / \tau$ and $\alpha = a(c + a/2) / \tau^4 = \frac{1}{2}(1 - c^2 / \tau^4) \in (0, 1/2]$. The function $q_{\alpha,1/2}$ is odd and strictly positive for $z > 0$: $H_{1/2}(z) > -z$ and $H'_{1/2}(z) > -1$ give $q_{\alpha,1/2}(z) > (1 - 2\alpha)z \geq 0$. Moreover,

$$\partial_p q_{\alpha,p}(z) = \alpha [d(z) + z d'(z)] > 0.$$

Since z is normal with positive mean and $p \geq 1/2$, reflection of its density about zero gives $\mathbb{E} q_{\alpha,p}(z) > 0$. As $\partial \lambda / \partial \sigma_p^2 > 0$, this proves $\partial M_t / \partial \sigma_p^2 > 0$.

Finally hold σ_p^2 fixed and vary e . In (37), the derivative of $\lambda \delta$ is $\delta \lambda_e < 0$. To differentiate the boost while accounting for the changing cardinal-noise distribution, write its expectation as $p f_v(m) - (1 - p) f_v(-m)$, with $m = \lambda \delta / \tau$ and $v = \lambda^2 e / \tau^2$. Here $m_e = O(\delta)$, $v_e = O(1)$, and $\partial_v f_v = f_v'' / 2$. The estimates (35) show that all positive-argument terms in this derivative are $o(\delta)$; the negative-argument terms have at most polynomial growth and are multiplied by $1 - p$, which has Gaussian decay (or is zero when $t = 0$). Therefore

$$\frac{\partial M_t(\delta)}{\partial e} = \delta \lambda_e + o(\delta) < 0$$

for all sufficiently large δ . The signs below the reference point reverse by symmetry. This proves Part 3(b) and completes the proof. $\square$

D.4 Proof of Proposition 3

Write $X = u_x$, $Y = u_y$, and $Z = u_z$, and let $D := X - Y$, $S := \sigma_X^2 + \sigma_Y^2$, and $k := \sigma_Y^2 / S \in (0, 1)$. Conditional on $D = d$,

$$Y \sim N\left(\mu_Y - k[d - (\mu_X - \mu_Y)], \frac{\sigma_X^2 \sigma_Y^2}{S}\right), \quad X = Y + d.$$

Hence, conditional on $D = d > 0$, the probabilities of the events $Y > Z$ and $Z > X$ are both strictly decreasing in d .

For the remaining event, let $q^2 := \sigma_Z^2 + \sigma_X^2 \sigma_Y^2 / S$ and $a := \mu_Z - \mu_Y - k(\mu_X - \mu_Y)$. Since

$$Z - Y \mid D = d \sim N(a + kd, q^2),$$

$$\Pr(Y < Z < X \mid D = d) = \Phi\left(\frac{(1-k)d - a}{q}\right) - \Phi\left(\frac{-kd - a}{q}\right),$$

whose derivative with respect to d is strictly positive.

Thus, relative to the distribution of $D \mid D > 0$, conditioning on $X > Y > Z$ or on $Z > X > Y$ reweights D by a strictly decreasing function, whereas conditioning on $X > Z > Y$ reweights it by a strictly increasing function. Since

$$\frac{\mathbb{E}[Dw(D) \mid D > 0]}{\mathbb{E}[w(D) \mid D > 0]} - \mathbb{E}[D \mid D > 0] = \frac{\text{Cov}(D, w(D) \mid D > 0)}{\mathbb{E}[w(D) \mid D > 0]},$$

the first two reweightings strictly lower the conditional mean of D and the third strictly raises it. Strictness follows from $2\text{Cov}(D, w(D)) = \mathbb{E}[(D - D')(w(D) - w(D'))]$, where D' is an independent copy under the conditional law $D \mid D > 0$. This proves all three inequalities. $\square$

D.5 Proof of Proposition 4

For $q \in \{r, r'\}$, let $P_q(t)$ denote the probability in the statement when the ordinal-signal variance is t . We first establish that $P_r(0) < P_{r'}(0)$.

Suppose $t = 0$, and fix a realization of the cardinal signals. In dimension 1, x receives a positive ordinal signal under both reference points. Because the reference-point priors have the same variance and are correctly centered, Proposition 1 and the strict decrease of $\psi(\cdot, +)$ imply

$$\mathbb{E}[u_{x,1} \mid s, o_{x,r',1}] > \mathbb{E}[u_{x,1} \mid s, o_{x,r,1}],$$

since $u_{r',1} > u_{r,1}$. Option y , instead, receives a positive signal under r and a negative signal under r' , and therefore

$$\mathbb{E}[u_{y,1} \mid s, o_{y,r',1}] < \mathbb{E}[u_{y,1} \mid s] < \mathbb{E}[u_{y,1} \mid s, o_{y,r,1}].$$

In dimension 2, the reference-point priors and the realized ordinal signals are identical under r and r' , because $u_{r',2} = u_{r,2} < u_{x,2} < u_{y,2}$. Hence, defining

$$D_q(s) := \mathbb{E}[U(x) \mid s, o_{x,q}] - \mathbb{E}[U(y) \mid s, o_{y,q}],$$

we have

$$D_{r'}(s) > D_r(s) \quad \text{for every } s.$$

This pointwise inequality increases the choice probability strictly. Indeed, fix the dimension-1 cardinal signals. The positive gap $D_{r'}(s) - D_r(s)$ is then fixed, while the common dimension-2 contribution is continuous in $s_{x,2}$ and $s_{y,2}$ and can be made arbitrarily positive or negative. Hence there is a nonempty open set of cardinal-signal realizations on which

$$D_r(s) < 0 < D_{r'}(s).$$

Because the cardinal signals have an everywhere-positive Gaussian density, $P_r(0) < P_{r'}(0)$.

It remains to extend the result to small positive ordinal noise. There are only finitely many ordinal-signal realizations. For each such realization, the posterior expected-utility difference given by Proposition 1 is continuous in (s, t) . Moreover, posterior ties occur with probability zero: holding the other signals fixed, the posterior expected-utility difference is strictly increasing in $s_{x,1}$: its derivative is $\lambda_{x,1} [1 + (\sigma_{x,1}^2 / \tau_{x,1}^2) \psi'(z, o)] > 0$, where $\tau_{x,1}^2 := \sigma_{x,1}^2 + \sigma_{r,1}^2 + \sigma_o^2$. Thus, bounded convergence implies that the choice probability conditional on each ordinal-signal realization is continuous at $t = 0$.

Because all true ordinal comparisons are strict, as $t \downarrow 0$ the probability of the true ordinal-signal realization converges to one, while the probabilities of all other realizations converge to zero. Consequently,

$$P_q(t) \longrightarrow P_q(0) \quad \text{for } q \in \{r, r'\}.$$

Since $P_r(0) < P_{r'}(0)$, there exists $\bar{\sigma}_o^2 > 0$ such that $P_r(t) < P_{r'}(t)$ whenever $0 \leq t < \bar{\sigma}_o^2$. $\square$

D.6 Proof of Corollary 1

Let q be an auxiliary reference point with $u_{q,1} = u_{r,1}$ and $u_{q,2} = u_{r,2}$, a correctly centered prior, and the common dimension-specific reference variances. Let $P_z(x)$ denote the probability that x is chosen with reference point z . Moving from q to r raises the first reference component from below $u_{y,1}$ to a point between $u_{y,1}$ and $u_{x,1}$, and leaves the second component unchanged below both options. Proposition 4 gives $P_r(x) > P_q(x)$ for all sufficiently small ordinal noise. Moving from q to r' instead raises the second reference component from below $u_{x,2}$ to a point between $u_{x,2}$ and $u_{y,2}$. Apply the same proposition with the options and dimensions interchanged to obtain $P_{r'}(y) > P_q(y)$.

The two posterior valuations tie with probability zero: conditional on each ordinal-signal profile, their difference is strictly increasing in x 's first cardinal signal, because its derivative is $\lambda_{x,1} [1 + (\sigma_{x,1}^2 / \tau_{x,1}^2) \psi'(z, o)] > 0$, where $\tau_{x,1}^2 := \sigma_{x,1}^2 + \sigma_{r,1}^2 + \sigma_o^2$. Thus $P_{r'}(x) < P_q(x)$. Taking the smaller of the two positive noise thresholds yields $P_r(x) > P_q(x) > P_{r'}(x)$, as required. $\square$

D.7 Proof of Observation 1

Fix a dimension i , and write

$$v_i := \sigma_{\varepsilon_x, i}^2 = \sigma_{\varepsilon_r, i}^2, \quad \widehat{\sigma}_i^2 := \frac{\sigma_{p, i}^2 v_i}{\sigma_{p, i}^2 + v_i}.$$

Because x and r have the same prior mean and variance in dimension i , their posterior variances after the cardinal signals are both $\widehat{\sigma}_i^2$, while their posterior-mean difference is

$$d_i(s) := \tilde{u}_{x, i}^s - \tilde{u}_{r, i}^s = \frac{\sigma_{p, i}^2}{\sigma_{p, i}^2 + v_i} (s_{x, i} - s_{r, i}).$$

Thus, conditional on s ,

$$u_{x, i} - u_{r, i} \sim N(d_i(s), 2\widehat{\sigma}_i^2).$$

Gaussian truncation therefore gives

$$\mathbb{E}[u_{x, i} - u_{r, i} \mid s, o_i] = d_i(s) + \sqrt{2\widehat{\sigma}_i^2} \psi\left(\frac{d_i(s)}{\sqrt{2\widehat{\sigma}_i^2}}, o_i\right),$$

where $o_i = +$ for $i \in D_x$ and $o_i = -$ for $i \in D_r$.

Holding the true utilities fixed,

$$\mathbb{E}_s[d_i(s)^2] = \left(\frac{\sigma_{p, i}^2}{\sigma_{p, i}^2 + v_i}\right)^2 [(u_{x, i} - u_{r, i})^2 + 2v_i] \longrightarrow 0$$

as $v_i \rightarrow \infty$, while $\widehat{\sigma}_i \rightarrow \sigma_{p, i}$. Moreover, the expression above, viewed as a function of $d_i(s)$, has derivative $1 + \psi'(\cdot, o_i) \in (0, 1)$. It follows that

$$\mathbb{E}_s[\mathbb{E}[u_{x, i} - u_{r, i} \mid s, o_i]] \longrightarrow \begin{cases} \frac{2\sigma_{p, i}}{\sqrt{\pi}}, & i \in D_x, \\ -\frac{2\sigma_{p, i}}{\sqrt{\pi}}, & i \in D_r, \end{cases}$$

where we used $\psi(0, +) = \sqrt{2/\pi}$ and $\psi(0, -) = -\sqrt{2/\pi}$.

Summing across dimensions,

$$\mathbb{E}_s[\mathbb{E}[U(x) - U(r) \mid s, o]] \longrightarrow \frac{2}{\sqrt{\pi}} \left(\sum_{i \in D_x} \sigma_{p, i} - \sum_{i \in D_r} \sigma_{p, i} \right).$$

The limit is nonzero by assumption. Since there are finitely many dimensions, there therefore exists $\bar{\sigma}_\varepsilon^2$ such that, whenever $v_i > \bar{\sigma}_\varepsilon^2$ for every i , the expected posterior-utility difference has the same sign as this limit. This proves the stated equivalence. The same Lipschitz bound gives convergence of each posterior gap in L^2 , not only convergence of its expectation. Their sum therefore converges in probability to the displayed nonzero constant, so the choice probability converges to one for the option with the larger weighted tally.

Finally, if the prior variances are identical across dimensions, then $\sigma_{p,i} = \sigma_p > 0$ for every i , so

$$\sum_{i \in D_x} \sigma_{p,i} - \sum_{i \in D_r} \sigma_{p,i} = \sigma_p(|D_x| - |D_r|).$$

Hence the condition on the right is equivalent to $|D_x| > |D_r|$. $\square$

D.8 Proof of Proposition 5

Fix ε , write $\bar{b} := \overline{\beta w}$, and let

$$\mu(w) := \lambda \beta w + (1 - \lambda) \bar{b}, \quad q(w) := \frac{r}{w}, \quad t(a, w) := \frac{\mu(w)a + \lambda \varepsilon - \beta r}{\sigma_x}.$$

For $o \in \{-, +\}$, write

$$F_o(a, w) := \mu(w)a - \frac{a^2}{2} + \sigma_x \psi(t(a, w), o) + \lambda \varepsilon.$$

The objective is $F_-(a, w)$ for $a < q(w)$ and $F_+(a, w)$ for $a \geq q(w)$. It is upper semicontinuous (the value at the jump is its upper branch) and tends to $-\infty$ as $a \rightarrow \infty$, since the boost grows at most linearly and the effort cost is quadratic. Thus a global maximizer exists for every $w > 0$. We use the standard properties

$$\psi_+(t) > 0 > \psi_-(t), \quad -1 < \psi'_\pm(t) < 0, \quad \psi''_-(t) < 0.$$

Because μ is strictly increasing and q strictly decreasing, there is a unique $w_2 > 0$ such that $\mu(w_2) = q(w_2)$. Fix $w < w_2$. The below-target branch is strictly concave, with a unique maximizer $a_-(w) \in (0, \mu(w))$ satisfying

$$a_-(w) = \mu(w) \left[1 + \psi'_-(t(a_-(w), w)) \right].$$

On the above-target branch, $F_{+,a}(a, w) < \mu(w) - a < 0$ for $a \geq q(w)$, so $q(w)$ uniquely maximizes that branch.

Let

$$\Delta(w) := F_+(q(w), w) - F_-(a_-(w), w).$$

The envelope theorem and $q'(w) = -q(w)/w$ give

$$\frac{d}{dw} F_-(a_-(w), w) = \mu'(w) \frac{a_-(w)^2}{\mu(w)}$$

and

$$\frac{d}{dw} F_+(q(w), w) = \frac{q(w)}{w} \{q(w) - (1 - \lambda)\bar{b}[1 + \psi'_+(t(q(w), w))]\}.$$

Since $q(w) > \mu(w) = w\mu'(w) + (1 - \lambda)\bar{b}$ and $1 + \psi'_+(\cdot) \in (0, 1)$, the second derivative above is strictly larger than $\mu'(w)q(w)$, and hence strictly larger than the first. Thus Δ is strictly increasing on $(0, w_2)$. Moreover, $\Delta(w) \rightarrow -\infty$ as $w \downarrow 0$, whereas $\Delta(w_2) > 0$. There is therefore a unique $w_1 \in (0, w_2)$ with $\Delta(w_1) = 0$. The unique optimum is $a_-(w)$ for $w < w_1$ and $q(w)$ for $w_1 < w \leq w_2$; at w_1 , both are optimal. The largest-maximizer convention selects $a^*(w_1) = q(w_1) = r/w_1$.

To establish the earnings comparative static below w_1 , let

$$\eta(w) := -\frac{\mu(w)^2}{\sigma_x} \psi''_-(t(a_-(w), w)) > 0.$$

Differentiating the first-order condition yields

$$\frac{d}{dw} [wa_-(w)] = \frac{1 + \psi'_-(t(a_-(w), w))}{1 + \eta(w)} [2w\mu'(w) + (1 - \lambda)\bar{b}(1 + \eta(w))] > 0.$$

Since $a_-(w_1) < \mu(w_1) < q(w_1)$, the left limit of earnings at w_1 is below r , while earnings equal r immediately to the right.

Finally, if $w > w_2$, every $a < q(w)$ is dominated by $q(w)$, and every $a > \mu(w)$ is dominated by $\mu(w)$. Since $F_{+,a}(\mu(w), w) < 0$, $\mu(w)$ is not optimal. Hence every optimum satisfies

$$q(w) \leq a^*(w) < \mu(w) = a^{\text{no rp}}(w),$$

which proves the proposition. □

D.9 Proof of Lemma 1

Because squared-error loss is minimized by the posterior mean, the law of total variance gives

$$V(r) = \mathbb{E}_s \left[\mathbb{V}_{o|s}(\mathbb{E}[u_x | s, o]) \right],$$

where $s = (s_x, s_r)$; including s_r in the first information set is harmless because u_x and s_r are independent conditional on s_x . Let

$$\Sigma := \sigma_x^2 + \sigma_r^2 + \sigma_o^2, \quad z_s := \frac{\tilde{u}_x^{s_x} - \tilde{u}_r^{s_r}}{\sqrt{\Sigma}}.$$

Conditional on s , the positive ordinal signal has probability $\Phi(z_s)$. Proposition 1 implies that the update to the posterior mean of u_x equals

$$\frac{\sigma_x^2}{\sqrt{\Sigma}} \frac{\phi(z_s)}{\Phi(z_s)} \quad \text{or} \quad -\frac{\sigma_x^2}{\sqrt{\Sigma}} \frac{\phi(z_s)}{1 - \Phi(z_s)},$$

with probabilities $\Phi(z_s)$ and $1 - \Phi(z_s)$, respectively. This binary update has mean zero and variance

$$\frac{\sigma_x^4}{\Sigma} \frac{\phi(z_s)^2}{\Phi(z_s)[1 - \Phi(z_s)]} = \frac{\sigma_x^4}{\Sigma} h(z_s).$$

Taking expectations over s proves the result. $\square$

D.10 Proof of Proposition 6

Let

$$\Sigma := \sigma_x^2 + \sigma_r^2 + \sigma_o^2, \quad b := \tilde{u}_r - \tilde{u}_x.$$

By Lemma 1,

$$V(r) = \frac{\sigma_x^4}{\Sigma} \mathbb{E}[h(Z_b)], \quad Z_b := \frac{\tilde{u}_r^{s_r} - \tilde{u}_x^{s_x}}{\sqrt{\Sigma}}.$$

The random variable Z_b is Normal with mean $b/\sqrt{\Sigma}$ and variance independent of b . For variance $v > 0$ and $b, y > 0$,

$$\partial_b \Pr(|N(b, v)| \leq y) = \frac{\phi((y+b)/\sqrt{v}) - \phi((y-b)/\sqrt{v})}{\sqrt{v}} < 0.$$

For zero variance the absolute value increases deterministically. Since h is even and strictly decreasing on $[0, \infty)$, while $|Z_b|$ strictly increases in the first-order stochastic order as $|b|$ increases,

$\mathbb{E}[h(Z_b)]$, and hence $V(r)$, is strictly decreasing in $|b|$.

Next, consider two reference-signal noise levels $0 \leq v_L < v_H$, and couple the corresponding signals as

$$s_r^H = s_r^L + \eta, \quad \eta \sim N(0, v_H - v_L),$$

where η is independent of all other variables. Let $\mathcal{I}_j := (s_x, s_r^j, o)$, $j \in \{L, H\}$. Then $\mathcal{I}_H$ is a Blackwell garbling of $\mathcal{I}_L$, so

$$\mathbb{E}[u_x \mid \mathcal{I}_H] = \mathbb{E}[\mathbb{E}[u_x \mid \mathcal{I}_L] \mid \mathcal{I}_H].$$

Using the law of total variance and cancelling the term that depends only on s_x ,

$$V_L - V_H = \mathbb{E}[\mathbb{V}(\mathbb{E}[u_x \mid \mathcal{I}_L] \mid \mathcal{I}_H)] \geq 0.$$

If $\sigma_{p,r}^2 > 0$, the posterior mean $\mathbb{E}[u_x \mid \mathcal{I}_L]$ depends strictly on s_r^L : the posterior mean of u_r is strictly increasing in s_r^L , and Proposition 1, together with $\psi'(\cdot, o) < 0$, implies that the posterior mean of u_x is strictly increasing in it. Since s_r^L remains nondegenerate conditional on the garbled signal $\mathcal{I}_H$, the last displayed expectation is strictly positive. Thus $V_L > V_H$. $\square$

D.11 Proof of Observation 2

Under the stated assumptions,

$$V(r_{\text{SQ}}) = \sigma_x^2 h\left(\frac{b_{\text{SQ}}}{\sigma_x}\right), \quad V(r_A) = \frac{\sigma_x^4}{\sigma_x^2 + \sigma_{r_A}^2} h(0).$$

Since $h(0) = 2/\pi$ and h is strictly decreasing on $[0, \infty)$,

$$V(r_{\text{SQ}}) > V(r_A) \iff b_{\text{SQ}} < \sigma_x h_+^{-1}\left(\frac{2}{\pi} \frac{\sigma_x^2}{\sigma_x^2 + \sigma_{r_A}^2}\right).$$

This is the stated exact threshold. $\square$

D.12 Proof of Proposition 7

Put $a := \sigma_x^2 > 0$, $b := \tilde{u}_r - \tilde{u}_x$, $\Sigma := a + \sigma_r^2 + \sigma_o^2$, and

$$W := (\sigma_{p,r}^2 - \sigma_r^2) + (\sigma_p^2 - a).$$

The posterior-mean difference before the ordinal signal is $N(b, W)$. Lemma 1 therefore gives

$$V(r) = a^2 F(\Sigma, W), \quad F(\Sigma, W) := \frac{1}{\Sigma} \mathbb{E} \left[h \left(\frac{b + \sqrt{W}Z}{\sqrt{\Sigma}} \right) \right], \quad Z \sim N(0, 1).$$

For $W > 0$, a change of variables and differentiation give

$$F_\Sigma = -\frac{1}{2W\sqrt{2\pi W\Sigma}} \int_{\mathbb{R}} h(z) e^{-(\sqrt{\Sigma}z-b)^2/(2W)} \left[\left(z - \frac{b}{2\sqrt{\Sigma}} \right)^2 + \frac{W - b^2/4}{\Sigma} \right] dz < 0$$

whenever $b^2 < 4W$. Also, for $y > 0$,

$$\partial_W \Pr(|N(b, W)| \leq y) = -\frac{(y-b)\phi((y-b)/\sqrt{W}) + (y+b)\phi((y+b)/\sqrt{W})}{2W^{3/2}} < 0$$

whenever $b^2 < W$. To check the only non-immediate case, assume $0 < y < b$ and note that

$$\log \frac{b+y}{b-y} = 2 \operatorname{arctanh}(y/b) > \frac{2y}{b} > \frac{2by}{W}.$$

Since h strictly decreases in absolute value, integration against these strictly decreasing central probabilities gives $F_W < 0$.

If the cardinal-signal variance for x is finite, then

$$w_0 := \sigma_p^2 - a = \frac{(\sigma_p^2)^2}{\sigma_p^2 + \sigma_{\varepsilon_x}^2} > 0.$$

Thus the explicit choice $\bar{\delta} = \sqrt{w_0} > 0$ ensures $b^2 < W$ for every reference variance and every ordinal-noise variance. This proves both required partial signs throughout the stated neighborhood.

We also verify existence of a positive neighborhood in the no-further-cardinal-information limit, where $w_0 = 0$ and the preceding explicit bound becomes zero. The following argument is uniform over $\Sigma \geq a$ and $W \geq 0$. If $W > a$ and $|b| < \sqrt{a}$, the preceding inequalities apply. For $W \leq a$, put $m = b/\sqrt{\Sigma}$ and $v = W/\Sigma \in [0, 1]$. Direct differentiation, including the right derivative at $W = 0$, gives

$$F_\Sigma = -\Sigma^{-2} \mathbb{E}[h(Y) + Yh'(Y)/2], \quad F_W = (2\Sigma^2)^{-1} \mathbb{E}h''(Y), \quad Y = m + \sqrt{v}Z.$$

At $m = 0$ and $v > 0$, Gaussian integration by parts gives

$$\mathbb{E}[h(Y) + Yh'(Y)/2] = \frac{1}{2}\mathbb{E}[(1 + Y^2/v)h(Y)] > 0, \quad \mathbb{E}h''(Y) = v^{-1}\mathbb{E}[Yh'(Y)] < 0.$$

At $v = 0$ these signs follow from $h(0) > 0$ and $h''(0) < 0$. Continuity and compactness of $v \in [0, 1]$ therefore give an $\epsilon \in (0, 1)$ for which both signs persist whenever $|m| < \epsilon$. Taking $\bar{\delta} = \epsilon\sqrt{a} > 0$ proves $F_\Sigma, F_W < 0$ uniformly also in this limiting case.

Increasing σ_o^2 raises Σ and leaves W fixed. Finally, if $P := \sigma_{p,r}^2$ and $E := \sigma_{\epsilon_r}^2$, then

$$\sigma_r^2 = \frac{PE}{P + E}, \quad P - \sigma_r^2 = \frac{P^2}{P + E}.$$

Both are nondecreasing in P , and their derivatives sum to one. The same statement holds under the natural definitions at $E = 0$ and in the no-signal limit $E = \infty$. Thus an increase in P increases at least one of Σ, W ; as both partial derivatives of F are strictly negative in the neighborhood just established, V strictly decreases in P as well. $\square$

D.13 Proof of Proposition 8

Conditional on c_o , the utility u_x is normal with mean $m_o := \mathbb{E}[\tilde{u}_x \mid c_o]$ and variance $\sigma_x^2 + \mathbb{V}(\tilde{u}_x \mid c_o)$. This variance is independent of the cue realization and of the reference point. Because all potential reference points have the same information quality and every reference-point mean is available, Proposition 6 implies that the selected reference point has mean equal to the DM's posterior mean of $\tilde{u}_x$. Hence

$$\tilde{u}_{r^*} = \mathbb{E}[\tilde{u}_x \mid c_o] = \hat{u}_x + \frac{\sigma_\mu^2}{\sigma_\mu^2 + \sigma_e^2} (c_o - \hat{u}_x).$$

Since $c = \tilde{u}_x$ and $c_o = c + e$, the pair $(c, \tilde{u}_{r^*})$ is jointly Gaussian and satisfies

$$\text{Cov}(c, \tilde{u}_{r^*}) = \mathbb{V}(\tilde{u}_{r^*}) = \frac{\sigma_\mu^4}{\sigma_\mu^2 + \sigma_e^2}.$$

Therefore,

$$c \mid \tilde{u}_{r^*} = m \sim N\left(m, \frac{\sigma_\mu^2 \sigma_e^2}{\sigma_\mu^2 + \sigma_e^2}\right).$$

It follows that

$$f(c \mid m) \propto \exp\left\{-\frac{\sigma_\mu^2 + \sigma_e^2}{2\sigma_\mu^2 \sigma_e^2} (c - m)^2\right\} = e^{-\tau(c-m)^2}.$$

Bayes' rule now gives

$$f(m | c) = \frac{e^{-\tau(c-m)^2} f(m)}{\int_{\mathbb{R}} e^{-\tau(c-m')^2} f(m') dm'},$$

as claimed. $\square$

D.14 Proof of Proposition 9

Identify the ordinal-signal realizations $+$ and $-$ with $+1$ and -1 . Write

$$D := U(x) - U(r), \quad v := \mathbb{V}(D) = 2\tau^2(1 + \rho) > 0, \quad w := \tau^2 + \sigma_N^2.$$

The reference-centered assumption gives $\mathbb{E}[D] = 0$. Let

$$Y_B := D - v_B, \quad Y_i := u_{x,i} - u_{r,i} - v_i, \quad i = 1, 2,$$

where $v_B \sim \mathcal{N}(0, \sigma_B^2)$ and $v_i \sim \mathcal{N}(0, \sigma_N^2)$ are mutually independent and independent of the utility components. Thus $o^B = \text{sgn}(Y_B)$ and $o_i^N = \text{sgn}(Y_i)$.

For centered jointly normal variables A, Y with $\mathbb{V}(Y) = s^2 > 0$, Gaussian regression gives $\mathbb{E}[A | Y] = \text{Cov}(A, Y)Y/s^2$. Since $\mathbb{E}|Y| = s\sqrt{2/\pi}$, symmetry then gives

$$\mathbb{E}[A | \text{sgn}(Y)] = \frac{\text{Cov}(A, Y)}{s} \sqrt{\frac{2}{\pi}} \text{sgn}(Y), \quad \mathbb{E}[A \text{sgn}(Y)] = \frac{\text{Cov}(A, Y)}{s} \sqrt{\frac{2}{\pi}}.$$

Broad Bracketing. Here $\text{Cov}(D, Y_B) = v$ and $\mathbb{V}(Y_B) = v + \sigma_B^2$, so

$$\widehat{U}_B - U(r) := \mathbb{E}[U(x) | o^B] - U(r) = \frac{v}{\sqrt{v + \sigma_B^2}} \sqrt{\frac{2}{\pi}} o^B.$$

The conditional-expectation projection identity therefore yields

$$\text{MSPE}_B = v - \mathbb{V}(\widehat{U}_B) = v - \frac{2}{\pi} \frac{v^2}{v + \sigma_B^2}.$$

Narrow Bracketing. Applying the same sign-update formula within each bracket gives

$$\widehat{U}_N - U(r) := \sum_{i=1}^2 \mathbb{E}[u_{x,i} | o_i^N] - U(r) = \frac{\tau^2}{\sqrt{w}} \sqrt{\frac{2}{\pi}} (o_1^N + o_2^N).$$

The centered normal pair (Y_1, Y_2) has common variance w and correlation $\eta := \tau^2 \rho / w$. Consequently,

$$\Pr(o_1^N = o_2^N = 1) = \int_0^\infty \phi(z) \Phi\left(\frac{\eta z}{\sqrt{1-\eta^2}}\right) dz = \frac{1}{4} + \frac{\arcsin(\eta)}{2\pi}.$$

For the last equality, the derivative of $\int_0^\infty \phi(z) \Phi(az) dz$ with respect to a is

$$\int_0^\infty z \phi(z) \phi(az) dz = \frac{1}{2\pi(1+a^2)}.$$

Integrating from $a = 0$, where the integral is $1/4$, and substituting $a = \eta/\sqrt{1-\eta^2}$ gives the stated expression. By sign symmetry,

$$q := \mathbb{E}[o_1^N o_2^N] = 4 \Pr(o_1^N = o_2^N = 1) - 1 = \frac{2}{\pi} \arcsin(\eta).$$

It follows that

$$\mathbb{V}(\widehat{U}_N) = \frac{4\tau^4}{\pi w} (1 + q).$$

Moreover, $\text{Cov}(D, Y_i) = \tau^2(1 + \rho)$, so the sign-update identity gives

$$\text{Cov}(D, o_i^N) = \frac{\tau^2(1 + \rho)}{\sqrt{w}} \sqrt{\frac{2}{\pi}}, \quad \text{Cov}(D, \widehat{U}_N) = \frac{4\tau^4(1 + \rho)}{\pi w}.$$

Both forecasts have mean $U(r)$. Computing the narrow forecast's MSPE under the true joint distribution therefore gives

$$\begin{aligned} \text{MSPE}_N &= v + \mathbb{V}(\widehat{U}_N) - 2\text{Cov}(D, \widehat{U}_N) \\ &= v - \frac{4\tau^4}{\pi w} (1 + 2\rho - q) = v - \frac{4\tau^4}{\pi w} \Gamma_N(\rho). \end{aligned}$$

Comparison. If $\Gamma_N(\rho) \leq 0$, then $\text{MSPE}_N \geq v$, whereas $\text{MSPE}_B < v$ for every finite σ_B^2 . If $\Gamma_N(\rho) > 0$, comparing the two MSPEs gives

$$\begin{aligned} \text{MSPE}_N \leq \text{MSPE}_B &\iff \frac{4\tau^4 \Gamma_N(\rho)}{\pi w} \geq \frac{2v^2}{\pi(v + \sigma_B^2)} \\ &\iff \sigma_B^2 \geq \frac{2(1 + \rho)^2(\tau^2 + \sigma_N^2)}{\Gamma_N(\rho)} - 2\tau^2(1 + \rho). \end{aligned}$$

At $\rho = 0$, the threshold is $2\sigma_N^2$.

Finally, suppose $\sigma_N^2 > 0$ and put $\kappa := \tau^2/w \in (0, 1)$. The continuous extension of Γ_N satisfies

$$\Gamma_N(-1) = -1 + \frac{2}{\pi} \arcsin(\kappa) < 0, \quad \Gamma_N(-1/2) = \frac{2}{\pi} \arcsin(\kappa/2) > 0.$$

Since

$$\Gamma_N''(\rho) = -\frac{2\kappa^3\rho}{\pi(1 - \kappa^2\rho^2)^{3/2}} > 0 \quad (-1 < \rho < 0),$$

strict convexity and these endpoint signs imply a unique zero $\rho_- \in (-1, -1/2)$. On $[-1/2, 0]$, $\Gamma_N \geq 1 + 2\rho$ with strict inequality at $-1/2$; on $[0, 1)$, $(2/\pi) \arcsin(\kappa\rho) \leq \rho$, by convexity of $\arcsin$ on $[0, 1]$. Thus Γ_N is positive on $[-1/2, 1)$, and $\Gamma_N(\rho) \leq 0$ if and only if $\rho \leq \rho_-$, as described in the figure caption. $\square$

E Proofs of the Results in the Appendix

E.1 Proof of Observation 3

When $\sigma_o^2 = \sigma_{o'}^2$, the ordinal signals have the same conditional distribution given (u_x, u_r) and are independent of the cardinal-signal errors. It therefore suffices to compare the cardinal signals. In the Alternative Setting, let $t := e_d - e_x - e_r$ and $\sigma_t^2 := \sigma_{e_d}^2 - \sigma_{e_x}^2 - \sigma_{e_r}^2 > 0$. The assumed covariance matrix implies that t is independent of (e_x, e_r) . The invertible transformation

$$(s'_x, s'_r, s_d) \longleftrightarrow \left(s'_x, s'_r, \bar{s}_r := \frac{s'_r + s'_x - s_d}{2} \right)$$

yields

$$s'_x = u_x + e_x, \quad s'_r = u_r + e_r, \quad \bar{s}_r = u_r - \frac{t}{2}.$$

Thus, in addition to the signal about u_x , the DM observes two independent Gaussian signals about u_r , with error variances $\sigma_{e_r}^2$ and $\sigma_t^2/4$. Completing the square in their Gaussian likelihood shows that their sufficient statistic is

$$s_r^* := \frac{s'_r/\sigma_{e_r}^2 + 4\bar{s}_r/\sigma_t^2}{1/\sigma_{e_r}^2 + 4/\sigma_t^2} = u_r + \varepsilon_r^*, \quad \mathbb{V}(\varepsilon_r^*) = \left(\frac{1}{\sigma_{e_r}^2} + \frac{4}{\sigma_t^2} \right)^{-1}.$$

The remaining linear combination of $(s'_r, \bar{s}_r)$ is ancillary. Hence the Alternative Setting is equivalent to an Original Setting with $\sigma_{\varepsilon_x}^2 = \sigma_{e_x}^2$ and

$$\frac{1}{\sigma_{\varepsilon_r}^2} = \frac{1}{\sigma_{e_r}^2} + \frac{4}{\sigma_{e_d}^2 - \sigma_{e_x}^2 - \sigma_{e_r}^2},$$

which proves the first part. Conversely, fix an Original Setting with $\sigma_{\varepsilon_r}^2 > 0$, choose any $\sigma_{e_r}^2 > \sigma_{\varepsilon_r}^2$, and set

$$\sigma_t^2 := 4 \left(\frac{1}{\sigma_{\varepsilon_r}^2} - \frac{1}{\sigma_{e_r}^2} \right)^{-1} > 0.$$

Take e_x, e_r, t to be independent centered Gaussian variables with variances $\sigma_{\varepsilon_x}^2, \sigma_{e_r}^2, \sigma_t^2$, respectively, and let $e_d = e_x + e_r + t$. Then (e_x, e_d, e_r) has the required covariance matrix and

$$\sigma_{e_d}^2 = \sigma_{\varepsilon_x}^2 + \sigma_{e_r}^2 + \sigma_t^2.$$

The first part therefore implies equivalence with the given Original Setting. Finally, suppose that $\sigma_r^2 = 0$, so that $u_r = \tilde{u}_r$ almost surely. Under the invertible transformation above, s'_r and $\bar{s}_r$ then depend only on the known value of u_r and on errors independent of (u_x, s'_x, o') , and are therefore ancillary. Likewise, s_r is ancillary in the Original Setting. Posterior beliefs thus depend only on (s'_x, o') in the Alternative Setting and on (s_x, o) in the Original Setting. These pairs have the same conditional distribution whenever $\sigma_{\varepsilon_x}^2 = \sigma_{e_x}^2$ and $\sigma_o^2 = \sigma_{o'}^2$, proving the final claim. $\square$

E.2 Proof of Proposition 10

Conditional on s , write $\Delta := u_x - u_r$ and $\mu_\Delta := \tilde{u}_x^s - \tilde{u}_r$. Since $\mathbb{V}(d | s) = K$, $\text{Cov}(u_x, d | s) = \sigma_x^2$, and $\text{Cov}(\Delta, d | s) = \sigma_x^2 + \sigma_r^2$, normal regression gives

$$\begin{aligned} \mathbb{E}[u_x | s, d] &= \tilde{u}_x^s + \frac{\sigma_x^2}{K}(d - \mu_\Delta), \\ \mathbb{E}[\Delta | s, d] &= \mu_\Delta + \frac{\sigma_x^2 + \sigma_r^2}{K}(d - \mu_\Delta), \\ \mathbb{V}(\Delta | s, d) &= \sigma_x^2 + \sigma_r^2 - \frac{(\sigma_x^2 + \sigma_r^2)^2}{K} = \sigma_\Delta^2, \\ \text{Cov}(u_x, \Delta | s, d) &= \sigma_x^2 - \frac{\sigma_x^2(\sigma_x^2 + \sigma_r^2)}{K} = \frac{\sigma_x^2 \sigma_d^2}{K}. \end{aligned}$$

The first two expressions are precisely the means in the statement. Conditional on (s, d) , apply the regression and sign-truncation calculation in Proposition 1 to the pair $(u_x, \Delta - \nu)$. Its second component has variance $\sigma_\Delta^2 + \sigma_o^2$ and covariance $\sigma_x^2 \sigma_\Delta^2 / (\sigma_x^2 + \sigma_r^2)$ with u_x . This gives the asserted formula. $\square$

E.3 Proof of Proposition 11

Write $D := u_x - u_r - v$, $\tau^2 := \sigma_x^2 + \sigma_r^2 + \sigma_o^2$, and $Z := (D - \mathbb{E}[D \mid s])/\tau$. The regression decomposition in the proof of Proposition 1 has residual variance $\sigma_x^2 - \sigma_x^4/\tau^2$. Integration by parts gives

$$\int_{-t_s}^{\infty} z^2 \phi(z) dz = \Phi(t_s) - t_s \phi(t_s), \quad \int_{-\infty}^{-t_s} z^2 \phi(z) dz = \Phi(-t_s) + t_s \phi(t_s).$$

Together with the conditional means already calculated, these imply

$$\mathbb{V}(Z \mid s, o) = 1 - t_s \psi(t_s, o) - \psi(t_s, o)^2.$$

The residual remains independent after conditioning on the sign. Therefore

$$\mathbb{V}(u_x \mid s, o) = \sigma_x^2 - \frac{\sigma_x^4}{\tau^2} + \frac{\sigma_x^4}{\tau^2} \mathbb{V}(Z \mid s, o),$$

which is the stated formula. Moreover, the DM's predictive probabilities of $o = +$ and $o = -$, conditional on s , are $\Phi(t_s)$ and $\Phi(-t_s)$, respectively. Averaging the displayed variance with these probabilities gives

$$\mathbb{E}_o[\mathbb{V}(u_x \mid s, o) \mid s] = \sigma_x^2 - \frac{\sigma_x^4}{\tau^2} h(t_s).$$

The information-kernel calculation in Appendix D.1 shows that, holding variances fixed, this expectation is strictly increasing in $|t_s|$ when $\sigma_x^2 > 0$. $\square$

E.4 Proof of Proposition 12

By Proposition 1,

$$\tilde{u}_x^{s,o} = \tilde{u}_x^s + \frac{\sigma_x^2}{\sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}} \psi(z_s, o).$$

Conditional on (u_x, u_r) , the cardinal and ordinal signals are independent. Writing $p := \Pr(o = + \mid u_x, u_r)$, we therefore have

$$\begin{aligned} \mathbb{E}_o[\psi(z_s, o)] &= p \frac{\phi(z_s)}{\Phi(z_s)} - (1-p) \frac{\phi(z_s)}{1 - \Phi(z_s)} \\ &= (p - \Phi(z_s)) \frac{\phi(z_s)}{\Phi(z_s)(1 - \Phi(z_s))}. \end{aligned}$$

Finally,

$$\mathbb{E}_s[\tilde{u}_x^s] = \mathbb{E}_s[\lambda s + (1 - \lambda)\tilde{u}_x] = \lambda u_x + (1 - \lambda)\tilde{u}_x.$$

Taking expectations over s gives (14). $\square$

E.5 Proof of Proposition 13

Let $D := u_x - u_r - v$, $\mu_D := \tilde{u}_x - \tilde{u}_r$, and $c := \sigma_x^2 - \sigma_{xr}$. The pair (u_x, D) is jointly normal, with $\mathbb{V}(D) = \tau^2$ and $\text{Cov}(u_x, D) = c$, so

$$u_x = \tilde{u}_x + \frac{c}{\tau^2}(D - \mu_D) + \xi,$$

where ξ is independent of D and has mean zero. The two truncated-normal integrals evaluated in the proof of Proposition 1 give

$$\mathbb{E}[D - \mu_D \mid o] = \tau\psi(\mu_D/\tau, o).$$

The sign signal depends only on D ; hence $\mathbb{E}[\xi \mid o] = 0$. Substitution proves the result, including the cases $c = 0$ and $c < 0$. $\square$

E.6 Proof of Proposition 14

Fix s , and write $a := \sigma_x^2$, $\tau := \sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}$, and $z := (\tilde{u}_x^s - \tilde{u}_r)/\tau$. Proposition 1 gives

$$B(s, +) = \frac{a}{\tau}\psi(z, +) > 0, \quad B(s, -) = \frac{a}{\tau}\psi(z, -) < 0.$$

Moreover, $\partial_1\psi(z, +) = -\psi(z, +)[z + \psi(z, +)]$.

Holding $\tilde{u}_x^s - \tilde{u}_r$ fixed, differentiation with respect to a gives

$$B_a(s, +) = \frac{\psi(z, +)}{2\tau^3} [a[1 + z^2 + z\psi(z, +)] + 2(\sigma_r^2 + \sigma_o^2)] > 0.$$

For $z \geq 0$ the bracket is immediate. If $z = -x < 0$, the standard Mills bound $\psi(-x, +) < x + 1/x$ implies $1 + z^2 + z\psi(z, +) = 1 + x^2 - x\psi(-x, +) > 0$. This proves (a).

For $v \in \{\sigma_r^2, \sigma_o^2\}$,

$$B_v(s, +) = -\frac{a\psi(z, +)}{2\tau^3} g(z), \quad g(z) := 1 - z^2 - z\psi(z, +).$$

If $z = -x \leq 0$, the Mills bound $\psi(-x, +) > x$ gives $g(z) > 1$. On $(0, \infty)$, $g(0) = 1$, $g(z) \rightarrow -\infty$,

and $g'(z) = -2z - \psi(z, +)g(z)$; hence every positive zero has derivative $-2z < 0$. Thus g has a unique positive zero z^* , with the sign pattern stated in (b).

Holding the variances fixed,

$$\frac{\partial B(s, +)}{\partial(\tilde{u}_x^s - \tilde{u}_r)} = \frac{a}{\tau^2} \partial_1 \psi(z, +) < 0,$$

because $z + \psi(z, +) > 0$ by the same Mills bound. This proves (c).

Finally, $\tilde{u}_x^s - \tilde{u}_r = \tilde{u}_x - \tilde{u}_r + \lambda(s - \tilde{u}_x)$, so the chain-rule identities in (d) follow. Both primitive variances strictly increase σ_x^2 ; increasing σ_p^2 weakly lowers $\tilde{u}_x^s - \tilde{u}_r$ when $s \leq \tilde{u}_x$, whereas increasing σ_ε^2 weakly lowers it when $s \geq \tilde{u}_x$. Parts (a) and (c) therefore give the two strict inequalities in (d). $\square$

E.7 Proof of Proposition 15

Put $a := \sigma_x^2$, $R := \sigma_r^2$, $t := \sigma_o$, $\tau^2 := a + R + t^2$, and let $Z \sim N(0, 1)$. Holding λ fixed means that $\sigma_\varepsilon^2 = a/\lambda$ and $\sigma_p^2 = a/(1 - \lambda)$. Consequently

$$m := \frac{\lambda\delta}{\tau}, \quad v := \frac{\lambda a}{\tau^2}, \quad p := \Phi(\delta/t), \quad C = \frac{a}{\tau} H(m, v; p),$$

where $p = 1$ when $t = 0$, and

$$H(m, v; p) := pf_v(m) - (1 - p)f_v(-m), \quad f_v(m) := \mathbb{E}[\rho(m + \sqrt{v}Z)].$$

For $t > 0$, $\delta/t > m > 0$, so $p > \Phi(m)$; the same inequality holds when $t = 0$. By (34),

$$H > 0, \quad H_m < 0, \quad H_{mm} > 0.$$

This proves positivity. With $K := R + t^2$, $H_v = H_{mm}/2$ and direct differentiation give

$$C_a = \frac{a + 2K}{2\tau^3} H - \frac{a\lambda\delta}{2\tau^4} H_m + \frac{\lambda a K}{2\tau^5} H_{mm} > 0, \quad (40)$$

proving (a).

Local effects of noise. First take $t > 0$. Set $c := \sqrt{2/\pi}$, $A(v) := f_v(0)$, and $L(v) := -f'_v(0)/f_v(0)$. The normal identities above imply

$$A'(v) > 0, \quad L'(v) = -\frac{1}{2}Q'_v(0) < 0, \quad 0 < L(v) < L(0) = c.$$

Since C is odd and smooth in δ ,

$$C(\delta) = S\delta + O(\delta^3), \quad S := C_\delta(0) = \frac{aA(v)}{\tau t} \left[c - \lambda \frac{t}{\tau} L(v) \right] > 0. \quad (41)$$

As t increases, v decreases. The positive prefactor $aA(v)/(\tau t)$ therefore strictly decreases, while $(t/\tau)L(v)$ strictly increases. The positive bracket in (41) strictly decreases as well. Thus $S_t < 0$, proving $C_{t^2} < 0$ for sufficiently small $\delta > 0$.

For reference variance, differentiation of the original expectation with the cardinal-noise draw fixed gives

$$C_R = -\frac{a}{2\tau^3} \mathbb{E}[H_p(Y) + YH'_p(Y)], \quad Y = m + \sqrt{v}Z, \quad H_p(y) := p\rho(y) - (1-p)\rho(-y). \quad (42)$$

The coefficient of δ in the expectation is

$$J := \frac{c}{t} \{A + v f_v''(0)\} + \frac{\lambda}{\tau} \{2f_v'(0) + v f_v'''(0)\}.$$

Because $Q'_v(0) > 0$, we have $f_v'''(0) > -L(v)f_v''(0)$. Hence

$$J > A \left(\frac{c}{t} - \frac{2\lambda L(v)}{\tau} \right) + v f_v''(0) \left(\frac{c}{t} - \frac{\lambda L(v)}{\tau} \right) > 0$$

whenever $2\lambda t \leq \tau$. This proves the claimed sufficient condition for $C_R < 0$ near zero. At $t = 0$, the derivative of p with respect to t^2 is zero for each $\delta > 0$, so $C_{t^2} = C_R$. Both are negative near zero: writing $\ell(y) := \rho(y) + y\rho'(y)$, symmetry gives

$$\mathbb{E}[\ell(\sqrt{v}Z)] = \frac{1}{2} \mathbb{E}[d(\sqrt{v}Z) + \sqrt{v}Z d'(\sqrt{v}Z)] > 0, \quad d(y) := \rho(y) + \rho(-y).$$

Here d is even and increasing in absolute value, by $\rho'' > 0$. This completes (b).

There is no unconditional local negative sign for C_R . Indeed, for fixed a, R, λ , the preceding expression for J gives

$$\lim_{\delta \downarrow 0} \frac{C_R(\delta)}{\delta} = \frac{a(2/\pi)(\lambda - 1/2)}{t^4} + O(t^{-6}) \quad (t \rightarrow \infty). \quad (43)$$

Thus the local effect is positive for sufficiently large t whenever $\lambda > 1/2$.

Shape in the utility difference. At $t = 0$, $p = 1$ and $C_\delta = (a\lambda/\tau^2)f'_v(m) < 0$. For $t > 0$, (41) gives $C(0) = 0$ and $C_\delta(0) > 0$. By (35),

$$C(\delta) \sim \frac{a}{\tau} f_v(\lambda\delta/\tau), \quad C_\delta(\delta) \sim \frac{a\lambda}{\tau^2} f'_v(\lambda\delta/\tau) < 0 \quad (\delta \rightarrow \infty).$$

The terms containing $1 - p$ or p' have faster Gaussian decay, since $\lambda^2/[\tau^2(1+v)] < 1/t^2$. Therefore C tends to zero at infinity and is eventually strictly decreasing. Positivity and continuity give an interior maximizer.

We next prove that the maximizer is unique for sufficiently small t . Here the primitive cardinal variances are fixed. Write

$$F_t(\delta) := \mathbb{E}[\rho(k_t(\delta + \varepsilon))], \quad k_t := \lambda/\tau_t, \quad A_t(\delta) := -F'_t(\delta) > 0,$$

where $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ and $\tau_t^2 = a + R + t^2$. Define

$$D_t(\delta) := F_t(\delta) + F_t(-\delta), \quad N_t(\delta) := p_t(\delta)A_t(\delta) + (1 - p_t(\delta))A_t(-\delta), \quad \mathcal{K}_t(\delta) := \frac{N_t(\delta)}{p'_t(\delta)D_t(\delta)}.$$

Then

$$C_\delta = \frac{a}{\tau_t} p'_t D_t (1 - \mathcal{K}_t), \quad \frac{d}{d\delta} \log \mathcal{K}_t = \frac{\delta}{t^2} + \frac{N'_t}{N_t} - \frac{D'_t}{D_t}.$$

Fix $\eta > 0$. On $[0, \eta]$, for all sufficiently small t , A_t, D_t, N_t are bounded above and away from zero, and $N'_t/N_t - D'_t/D_t$ is uniformly bounded. For the only potentially unbounded term, note that $p'_t\{A_t(\delta) - A_t(-\delta)\} = O(\delta p'_t) = O(1)$. On $[0, t]$, $p'_t \geq \phi(1)/t$, so $\mathcal{K}_t < 1$ for small t . On $[t, \eta]$, the logarithmic derivative is at least $1/t - O(1) > 0$.

Finally, uniformly for small t and $\delta \geq \eta$, there are constants $c_1, c_2, c_3 > 0$ such that

$$A_t(\delta) \geq c_1 e^{-c_2 \delta^2}, \quad D_t(\delta) \leq c_3 (1 + \delta).$$

The first bound follows by restricting the normal integral defining $-F'_t$ to $k_t(\delta + \varepsilon) \in [0, 1]$; the second follows from the linear-growth bound on ρ . Since $N_t \geq A_t/2$, these bounds imply

$$\inf_{\delta \geq \eta} \mathcal{K}_t(\delta) \geq \inf_{\delta \geq \eta} \frac{c_1 t}{2c_3(1 + \delta)\phi(0)} \exp\left[\left(\frac{1}{2t^2} - c_2\right)\delta^2\right] \rightarrow \infty.$$

Thus $\mathcal{K}_t$ crosses one exactly once, on (t, η) , and stays above one thereafter. This proves the small-noise uniqueness and the strict derivative signs in (c).

Primitive variances. Part (d) follows from the chain rule and

$$\frac{\partial a}{\partial \sigma_p^2} = (1 - \lambda)^2, \quad \frac{\partial a}{\partial \sigma_\varepsilon^2} = \lambda^2, \quad \frac{\partial \lambda}{\partial \sigma_p^2} = \frac{\sigma_\varepsilon^2}{(\sigma_p^2 + \sigma_\varepsilon^2)^2} > 0, \quad \frac{\partial \lambda}{\partial \sigma_\varepsilon^2} = -\frac{\sigma_p^2}{(\sigma_p^2 + \sigma_\varepsilon^2)^2} < 0.$$

The sign of C_a was proved in (a). □

E.8 Proof of Proposition 16

Write $a := \sigma_x^2 > 0$, $t := \sigma_o > 0$, $\tau^2 := a + \sigma_r^2 + t^2$, and

$$p := \Phi((u_x - u_r)/t), \quad z_s := \frac{\lambda s + (1 - \lambda)\tilde{u}_x - \tilde{u}_r}{\tau}, \quad H_p(z) := p\rho(z) - (1 - p)\rho(-z).$$

Conditional on the fixed true utilities, the cardinal-signal distribution is independent of both reference quantities. Hence

$$M := \mathbb{E}[\tilde{u}_x^{s,o}] = \lambda u_x + (1 - \lambda)\tilde{u}_x + \frac{a}{\tau} \mathbb{E}_s H_p(z_s).$$

Set

$$D := \mathbb{E}_s[\rho(z_s) + \rho(-z_s)] > 0, \quad N := -\mathbb{E}_s[p\rho'(z_s) + (1 - p)\rho'(-z_s)] > 0.$$

The normal bounds established above justify differentiation, and give

$$M_{u_r} = -\frac{a}{\tau t} \phi((u_x - u_r)/t) D < 0, \quad M_{\tilde{u}_r} = \frac{a}{\tau^2} N > 0.$$

These are Parts 1 and 2. If all other primitives are fixed and $d\tilde{u}_r/du_r = \eta$, the chain rule gives

$$\frac{dM}{du_r} = M_{\tilde{u}_r}(\eta - \eta^*), \quad \eta^* = \frac{\tau}{t} \phi((u_x - u_r)/t) \frac{D}{N} > 0,$$

which proves Part 3. When $t = 0$, p is constant on either side of $u_r = u_x$, so $M_{u_r} = 0$ there, while $M_{\tilde{u}_r} > 0$ remains true. At $u_r = u_x$ there is a downward jump as u_r increases past u_x . Away from that point, $dM/du_r = \eta M_{\tilde{u}_r}$. □

E.9 Proof of Corollary 2

Apply Proposition 3, conditional on s , in dimension 1. It gives

$$\mathbb{E}[u_{x,1} - u_{y,1} \mid u_{x,1} > u_{y,1} > u_{r,1}, s] < \mathbb{E}[u_{x,1} - u_{y,1} \mid u_{x,1} > u_{y,1}, s] < \mathbb{E}[u_{x,1} - u_{y,1} \mid u_{x,1} > u_{r',1} > u_{y,1}, s].$$

The same argument allows a perfectly known reference component: in the proof of Proposition 3, the conditional comparison variance remains positive because $q^2 = \sigma_Z^2 + \sigma_X^2 \sigma_Y^2 / (\sigma_X^2 + \sigma_Y^2) > 0$ even when $\sigma_Z^2 = 0$. The conditioning event in dimension 2 is identical on the two sides, so its contribution to $\mathbb{E}[U(x) - U(y) \mid \cdot]$ is the same. Adding the two dimensions proves the claim. $\square$

From posterior gaps to choice probabilities. Under the strictly positive cardinal-signal variances in Proposition 4, the preceding pointwise inequality also gives a strict inequality between choice probabilities. Here is the required support argument. In dimension 2, let $G := u_{y,2} - u_{x,2}$. Conditional on its noiseless ordering, G has support $(0, \infty)$. Varying the two cardinal posterior means along $\mu_{y,2}(\theta) = \mu_{y,2}(0) + \theta v_{y,2}$ and $\mu_{x,2}(\theta) = \mu_{x,2}(0) - \theta v_{x,2}$ tilts the conditional density by $e^{\theta G}$, where $v_{x,2}, v_{y,2}$ are the posterior variances. Its mean has derivative $\mathbb{V}_\theta(G) > 0$ and ranges from zero to infinity. Indeed, exponential tilting concentrates above any fixed lower bound as $\theta \rightarrow \infty$ and below any fixed positive upper bound as $\theta \rightarrow -\infty$; the normal tails make every such moment finite. Thus this negative contribution to $U(x) - U(y)$ can be chosen strictly between the two positive dimension-1 gaps. Continuity then gives an open set of cardinal signals on which the choice switches, and the Gaussian signal density assigns that set positive probability. The same tilting argument and Fubini's theorem show that ties have probability zero.

For fixed true utilities with strict rankings, the probability of each correct ordinal profile tends to one as $\sigma_o \downarrow 0$. Conditional on that profile and any fixed cardinal signals, the posterior gap converges to its noiseless value: apply dominated convergence to the numerator and denominator of its posterior integral, whose limiting denominator is positive. Since the limiting gap is nonzero almost surely, bounded convergence for the choice indicators shows convergence of the unconditional choice probabilities. Their strict noiseless inequality therefore persists for all sufficiently small ordinal noise. This establishes the all-pairwise extension stated in the main text, not merely an inequality between average posterior gaps.

E.10 Proof of Proposition 18

Write $D := u_x - u_y$ and $g(o; \sigma_o^2) := \mathbb{E}[D \mid o]$ for the DM's posterior mean, and set

$$G(\sigma_o^2) := \mathbb{E}[g(o; \sigma_o^2) \mid u_x > u_z > u_y], \quad H(\sigma_o^2) := \mathbb{E}[g(o; \sigma_o^2) \mid u_x > u_y > u_z].$$

We show that $G(0) > H(0)$ and that G and H are continuous at $\sigma_o^2 = 0$.

Step 1: the noiseless benchmark. When $\sigma_o^2 = 0$ the three signals reveal the ordering of (u_x, u_y, u_z) exactly. Hence, conditional on $u_x > u_z > u_y$, the signal profile is degenerate at the profile o^* consistent with that ordering, and $g(o^*; 0) = \mathbb{E}[D \mid u_x > u_z > u_y]$; likewise $H(0) = \mathbb{E}[D \mid u_x >$

$u_y > u_z]$. The first and third inequalities of Proposition 3 give

$$G(0) = \mathbb{E}[D \mid u_x > u_z > u_y] > \mathbb{E}[D \mid u_x > u_y] > \mathbb{E}[D \mid u_x > u_y > u_z] = H(0).$$

Step 2: continuity at $\sigma_o^2 = 0$. Let f denote the joint density of (u_x, u_y, u_z) , which factors by independence. Since $o_{ab} = +$ if and only if $u_a - u_b \geq v_{ab}$ with $v_{ab} \sim \mathcal{N}(0, \sigma_o^2)$,

$$\Pr(o^\star \mid u_x, u_y, u_z) = \Phi\left(\frac{u_x - u_y}{\sigma_o}\right) \Phi\left(\frac{u_z - u_y}{\sigma_o}\right) \Phi\left(\frac{u_x - u_z}{\sigma_o}\right) \xrightarrow{\sigma_o \rightarrow 0} \mathbf{1}\{u_x > u_z > u_y\}$$

pointwise off the null set where two coordinates coincide, and analogously for every other profile. Write $G(\sigma_o^2) = \sum_o \Pr(o \mid u_x > u_z > u_y; \sigma_o^2) g(o; \sigma_o^2)$.

(a) Each $\Pr(o \mid u_x > u_z > u_y; \sigma_o^2)$ is continuous in σ_o^2 and converges to $\mathbf{1}\{o = o^\star\}$: numerator and denominator are integrals of products of Φ 's against f , and dominated convergence applies with dominating function f , the integrands being bounded by 1.

(b) For $o = o^\star$,

$$g(o^\star; \sigma_o^2) = \frac{\int (u_x - u_y) \Pr(o^\star \mid u_x, u_y, u_z) f d(u_x, u_y, u_z)}{\int \Pr(o^\star \mid u_x, u_y, u_z) f d(u_x, u_y, u_z)}.$$

Dominated convergence applies to the numerator with dominating function $|u_x - u_y|f$, which is integrable because u_x and u_y are Gaussian, and to the denominator with f ; the denominator converges to $\Pr(u_x > u_z > u_y) > 0$. Hence $g(o^\star; \cdot)$ is continuous at 0 with limit $\mathbb{E}[D \mid u_x > u_z > u_y]$.

(c) For $o \neq o^\star$ the contribution vanishes. If $\liminf_{\sigma_o^2 \rightarrow 0} \Pr(o) > 0$, then $g(o; \cdot)$ is continuous at 0 by the argument in (b), hence bounded near 0, while $\Pr(o \mid u_x > u_z > u_y; \sigma_o^2) \rightarrow 0$ by (a). If instead $\Pr(o) \rightarrow 0$ (the two cyclic profiles), Cauchy–Schwarz gives $|g(o; \sigma_o^2)| \leq \|D\|_2 \Pr(o)^{-1/2}$ and $\Pr(o \mid u_x > u_z > u_y) \leq \Pr(o)/\Pr(u_x > u_z > u_y)$, so the product is bounded by $\|D\|_2 \Pr(o)^{1/2}/\Pr(u_x > u_z > u_y) \rightarrow 0$.

Combining (a)–(c), $G(\sigma_o^2) \rightarrow G(0)$; the same argument gives $H(\sigma_o^2) \rightarrow H(0)$.

Step 3: conclusion. Since $G(0) > H(0)$ and both are continuous at $\sigma_o^2 = 0$, there exists $\bar{\sigma}_o^2 > 0$ such that $G(\sigma_o^2) > H(\sigma_o^2)$ for every $\sigma_o^2 < \bar{\sigma}_o^2$. $\square$

E.11 Proof of Corollary 3

This follows directly from Proposition 18. Note that the change in reference point does not affect the expected utility of x or y along dimension 2. Along dimension 1, Proposition 18 shows that

the expected value of x will increase relative to the expected value of y for each signal realization, and thus will increase once taking expectations over signal realizations.

E.12 Proof of Corollary 4

Conditional on s , apply the first inequality in Proposition 3 to $(u_{y,1}, u_{x,1}, u_{z,1})$ and multiply by -1 . This strictly raises the expected contribution of dimension 1. Apply the third inequality to $(u_{x,2}, u_{y,2}, u_{z,2})$; this strictly raises the expected contribution of dimension 2. Adding the two inequalities gives the result. For the strict choice-probability statement in the text, vary the two dimension-2 cardinal means by exponential tilting as in the proof of Corollary 2. The no-decoy total gap crosses zero, while the with-decoy gap is strictly larger there. Continuity and full support give a positive-probability set of choice switches. $\square$

E.13 Proof of Proposition 19

Write

$$D := U(x) - U(r), \quad v := 2\tau^2(1 + \rho), \quad w := \tau^2 + \sigma_N^2, \quad \eta := \frac{\tau^2 \rho}{w}, \quad q := \frac{2}{\pi} \arcsin(\eta).$$

Identify the ordinal realizations with $\{-1, 1\}$ and let $Y_i := u_{x,i} - u_{r,i} - v_i$, where $v_i \sim \mathcal{N}(0, \sigma_N^2)$ are independent of each other and of the utility components. Thus $o_i^N = \text{sgn}(Y_i)$. The sign-moment calculations in the proof of Proposition 9 give

$$\mathbb{E}[o_1^N o_2^N] = q, \quad \mathbb{E}[D o_i^N] = \frac{\tau^2(1 + \rho)}{\sqrt{w}} \sqrt{\frac{2}{\pi}}, \quad i = 1, 2.$$

The joint centered Gaussian law of (D, Y_1, Y_2) is invariant under $(D, Y_1, Y_2) \mapsto (-D, -Y_2, -Y_1)$, since $\mathbb{V}(Y_1) = \mathbb{V}(Y_2)$ and $\text{Cov}(D, Y_1) = \text{Cov}(D, Y_2)$. This transformation preserves each mixed-sign event and reverses D , so $\mathbb{E}[D \mid o_1^N, o_2^N] = 0$ on both mixed-sign events. Simultaneous sign reversal also implies that the two aligned-sign posterior means are opposites. Hence, for some coefficient b , the Bayesian Narrow forecast satisfies

$$\widehat{U}_{BN} - U(r) := \mathbb{E}[D \mid o_1^N, o_2^N] = b(o_1^N + o_2^N).$$

Multiplying by o_1^N and taking expectations gives

$$b(1 + q) = \mathbb{E}[D o_1^N], \quad b = \frac{\tau^2(1 + \rho)}{\sqrt{w}(1 + q)} \sqrt{\frac{2}{\pi}}.$$

Therefore the projection identity yields

$$\text{MSPE}_{BN} = v - \mathbb{V}(\widehat{U}_{BN}) = v - \frac{4\tau^4(1+\rho)^2}{\pi w(1+q)}.$$

The proof of Proposition 9 also gives

$$\text{MSPE}_B = v - \frac{2}{\pi} \frac{v^2}{v + \sigma_B^2}.$$

Since $v > 0$ and $1 + q > 0$, comparing the two reductions gives

$$\text{MSPE}_{BN} < \text{MSPE}_B \iff \sigma_B^2 > 2w(1+q) - v,$$

which is the stated cutoff. Admissible equality gives equal MSPEs; at $\rho = 0$, the cutoff is $2\sigma_N^2$. $\square$

E.14 Proof of Corollary 5

Write $w := \tau^2 + \sigma_N^2$ and $c_0 := 4\tau^4/(\pi w)$. Under $\sigma_B^2 = 2\sigma_N^2$,

$$2\tau^2(1+\rho) + \sigma_B^2 = 2w(1+\kappa\rho).$$

The reductions in MSPE relative to the prior mean are therefore $c_0(1+\rho)^2/(1+\kappa\rho)$ under Broad Bracketing and $c_0(1+\rho)^2/(1+q(\rho))$ under Bayesian Narrow Bracketing. Subtracting the former from the latter gives (15).

To obtain the naive effect, let $b_0 := \tau^2\sqrt{2/\pi}/\sqrt{w}$. The preceding proofs give

$$\widehat{U}_N - U(r) = b_0(o_1^N + o_2^N), \quad \widehat{U}_{BN} - U(r) = b_0 \frac{1+\rho}{1+q(\rho)}(o_1^N + o_2^N).$$

Since $\widehat{U}_{BN}$ is the conditional expectation of $U(x)$ given (o_1^N, o_2^N) , the projection identity implies

$$\text{MSPE}_N - \text{MSPE}_{BN} = \mathbb{E}[(\widehat{U}_N - \widehat{U}_{BN})^2].$$

Substituting the two forecasts and using $\mathbb{E}[(o_1^N + o_2^N)^2] = 2(1+q(\rho))$ gives (16).

For the sign conclusions, the function $f(t) := t - (2/\pi) \arcsin(t)$ is odd and strictly concave on $(0, 1)$, with $f(0) = f(1) = 0$. Thus $f(t)$ has the sign of t for $t \in (-1, 1) \setminus \{0\}$. Since $\kappa \in (0, 1]$ and $|\rho| < 1$, this shows that $\kappa\rho - q(\rho)$ has the sign of ρ and that $|q(\rho)| < |\rho|$ whenever $\rho \neq 0$. The displayed information and naive effects therefore imply all the stated

rankings and equality at $\rho = 0$.

It remains to compare Broad and naive Narrow Bracketing for positive correlation. Their difference is $c_0 d(\rho, \kappa)$, where

$$d(\rho, \kappa) := 1 + 2\rho - \frac{2}{\pi} \arcsin(\kappa\rho) - \frac{(1 + \rho)^2}{1 + \kappa\rho}.$$

For each fixed $\rho \in (0, 1)$,

$$\frac{\partial^2 d}{\partial \kappa^2} = -\frac{2\rho^2(1 + \rho)^2}{(1 + \kappa\rho)^3} - \frac{2\kappa\rho^3}{\pi(1 - \kappa^2\rho^2)^{3/2}} < 0.$$

At the endpoints of $[1/2, 1]$,

$$d(\rho, 1) = \rho - \frac{2}{\pi} \arcsin(\rho) > 0, \quad d(\rho, 1/2) = \frac{\rho}{2 + \rho} - \frac{2}{\pi} \arcsin(\rho/2) > 0.$$

Both inequalities follow because the displayed functions of ρ are strictly concave on $[0, 1]$ and vanish at its endpoints. Concavity in κ therefore gives $d(\rho, \kappa) > 0$ for every $\rho \in (0, 1)$ and $\kappa \in [1/2, 1]$. Since $\kappa \geq 1/2$ is equivalent to $\sigma_N^2 \leq \tau^2$, this proves sufficiency.

Conversely, the continuous extension to $\rho = 1$ is

$$d(1, \kappa) = 3 - \frac{2}{\pi} \arcsin(\kappa) - \frac{4}{1 + \kappa}.$$

It vanishes at $\kappa = 1/2$ and is strictly increasing on $[0, 1/2]$:

$$\frac{\partial d(1, \kappa)}{\partial \kappa} = \frac{4}{(1 + \kappa)^2} - \frac{2}{\pi\sqrt{1 - \kappa^2}} \geq \frac{16}{9} - \frac{4}{\pi\sqrt{3}} > 0.$$

Hence $d(1, \kappa) < 0$ for $\kappa < 1/2$. By continuity, $d(\rho, \kappa) < 0$ for all ρ sufficiently close to 1. This proves necessity and the remaining claim. $\square$

E.15 Proof of Proposition 20

Identify the ordinal realizations $+$ and $-$ with $+1$ and -1 . We first record the sign-update identity used below. If (A, Y) is jointly normal, $\mathbb{E}[Y] = m$, and $\mathbb{V}(Y) = s^2 > 0$, Gaussian regression and integration of the normal density give

$$\mathbb{E}[A \mid Y] = \mathbb{E}[A] + \frac{\text{Cov}(A, Y)}{s^2}(Y - m), \quad \mathbb{E}[(Y - m)\mathbf{1}\{Y \geq 0\}] = s\phi(m/s).$$

Since $\Pr(Y \geq 0) = \Phi(m/s)$ and $\mathbb{E}[(Y - m)\mathbf{1}\{Y < 0\}] = -s\phi(m/s)$, it follows that

$$\mathbb{E}[A \mid \text{sgn}(Y)] = \mathbb{E}[A] + \frac{\text{Cov}(A, Y)}{s} \psi(m/s, \text{sgn}(Y)).$$

Here $\psi(z, o) = o\phi(z)/\Phi(oz)$ is the boost function defined in the main text. For a sign O with $\Pr(O = o) = \Phi(oz)$,

$$\mathbb{E}[\psi(z, O)] = 0, \quad \mathbb{E}[\psi(z, O)^2] = \mathcal{T}(z), \quad \psi(z, O) = \frac{\phi(z)[\mathbf{1}\{O = 1\} - \Phi(z)]}{\Phi(z)\Phi(-z)}.$$

Consequently, if (Y_1, Y_2) is jointly normal with means z_1, z_2 , unit variances, and correlation η , then

$$\mathbb{E}[\psi(z_1, \text{sgn}(Y_1))\psi(z_2, \text{sgn}(Y_2))] = \mathcal{T}_{12}(z_1, z_2; \eta),$$

because $\Pr(Y_1 \geq 0, Y_2 \geq 0) = \Phi_2(z_1, z_2; \eta)$.

For Broad Bracketing, put

$$z_B := \frac{\tilde{u}_{x,1} + \tilde{u}_{x,2} - U(r)}{\sqrt{V + \sigma_B^2}}.$$

The covariance of $U(x)$ with $U(x) - U(r) - v_B$ is V . The sign-update identity therefore yields

$$\widehat{U}_B := \mathbb{E}[U(x) \mid o^B] = \tilde{u}_{x,1} + \tilde{u}_{x,2} + \frac{V}{\sqrt{V + \sigma_B^2}} \psi(z_B, o^B).$$

Hence

$$\mathbb{V}(\widehat{U}_B) = \frac{V^2}{V + \sigma_B^2} \mathcal{T}(z_B).$$

The law of total variance gives $\text{MSPE}_B = V - \mathbb{V}(\widehat{U}_B)$, proving (17).

For Narrow Bracketing, write

$$z_i := \frac{\tilde{u}_{x,i} - u_{r,i}}{s_i}, \quad i = 1, 2, \quad \eta := \frac{\rho\tau_1\tau_2}{s_1s_2}.$$

The standardized comparison variables $(u_{x,i} - u_{r,i} - v_i)/s_i$ have means z_i , unit variances, and correlation η . Their covariances with $u_{x,i}$ and $U(x)$ are τ_i^2/s_i and $(\tau_i^2 + \rho\tau_1\tau_2)/s_i$, respectively. Thus

$$\mathbb{E}[u_{x,i} \mid o_i^N] = \tilde{u}_{x,i} + \frac{\tau_i^2}{s_i} \psi(z_i, o_i^N)$$

and

$$\mathbb{E}[U(x) \mid o_i^N] = \tilde{u}_{x,1} + \tilde{u}_{x,2} + \frac{\tau_i^2 + \rho\tau_1\tau_2}{s_i} \psi(z_i, o_i^N).$$

Let $\widehat{U}_N := \mathbb{E}[u_{x,1} \mid o_1^N] + \mathbb{E}[u_{x,2} \mid o_2^N]$. The moment identities give

$$\mathbb{V}(\widehat{U}_N) = \sum_{i=1}^2 \frac{\tau_i^4}{s_i^2} \mathcal{T}(z_i) + \frac{2\tau_1^2\tau_2^2}{s_1s_2} \mathcal{T}_{12}(z_1, z_2; \eta),$$

while iterated expectations give

$$\text{Cov}(U(x), \widehat{U}_N) = \sum_{i=1}^2 \frac{\tau_i^2(\tau_i^2 + \rho\tau_1\tau_2)}{s_i^2} \mathcal{T}(z_i).$$

Since $\mathbb{E}[\widehat{U}_N] = \mathbb{E}[U(x)] = \tilde{u}_{x,1} + \tilde{u}_{x,2}$,

$$\text{MSPE}_N = V + \mathbb{V}(\widehat{U}_N) - 2\text{Cov}(U(x), \widehat{U}_N).$$

Substitution yields (18). Comparing the two reductions from V proves the final assertion. $\square$

E.16 Proof of Proposition 21

Under $\tilde{u}_{x,i} = u_{r,i}$, the standardized means z_B, z_1, z_2 in the preceding proof are zero. Directly,

$$\mathcal{T}(0) = \frac{2}{\pi}.$$

The quadrant-probability calculation in the proof of Proposition 9 gives

$$\Phi_2(0, 0; \eta) = \frac{1}{4} + \frac{\arcsin(\eta)}{2\pi}, \quad \mathcal{T}_{12}(0, 0; \eta) = \frac{4}{\pi^2} \arcsin(\eta).$$

Substituting into (17) and (18) proves (19) and (20).

For the comparison, let R_N denote the bracketed expression in (20). Then

$$\text{MSPE}_N < \text{MSPE}_B \iff R_N > \frac{V^2}{V + \sigma_B^2}.$$

If $R_N > 0$, rearranging gives $\sigma_B^2 > V^2/R_N - V$, the stated cutoff. Equality gives equal MSPEs whenever the cutoff is nonnegative; if it is negative, the strict inequality holds for every $\sigma_B^2 \geq 0$. If $R_N \leq 0$, then $\text{MSPE}_N \geq V > \text{MSPE}_B$ for every finite σ_B^2 , since $V > 0$.

Finally, if $\tau_1 = \tau_2 = \tau$ and $\sigma_{N,1}^2 = \sigma_{N,2}^2 = \sigma_N^2$, then $s_1^2 = s_2^2 = \tau^2 + \sigma_N^2$ and $V = 2\tau^2(1 + \rho)$, so

$$R_N = \frac{2\tau^4}{\tau^2 + \sigma_N^2} \left[1 + 2\rho - \frac{2}{\pi} \arcsin \left(\frac{\rho\tau^2}{\tau^2 + \sigma_N^2} \right) \right] = \frac{2\tau^4}{\tau^2 + \sigma_N^2} \Gamma_N(\rho).$$

Substituting into the cutoff and distinguishing $R_N > 0$ from $R_N \leq 0$ recovers Proposition 9. $\square$

E.17 Proof of Proposition 22

Conditional independence across dimensions gives

$$V(x \mid \mathcal{I}) = -\frac{1}{\alpha} \prod_{i=1}^n \mathbb{E}[e^{-\alpha u_{x,i}} \mid \mathcal{I}].$$

The map $h_\alpha(v) := -\alpha^{-1} \ln(-\alpha v)$ is strictly increasing on its domain $\{v \in \mathbb{R} : -\alpha v > 0\}$. Therefore, $\succsim$ is also represented by

$$h_\alpha(V(x \mid \mathcal{I})) = \sum_{i=1}^n \left(-\frac{1}{\alpha} \ln \mathbb{E}[e^{-\alpha u_{x,i}} \mid \mathcal{I}] \right) = \sum_{i=1}^n \text{CE}(u_{x,i} \mid \mathcal{I}) = \bar{U}(x \mid \mathcal{I}).$$

The normal moment-generating function immediately gives $\text{CE}(u_x \mid s) = \tilde{u}_x^s - \alpha\sigma_x^2/2$. For the ordinal formula, conditional on s let $X := u_x$ and $Y := u_x - u_r - v$. Then $\mathbb{E}[Y] = \tilde{u}_x^s - \tilde{u}_r$, $\mathbb{V}(Y) = \tau^2$, and $\text{Cov}(X, Y) = \sigma_x^2$. For jointly normal (X, Y) , completing the square gives

$$\mathbb{E}[e^{tX} \mathbf{1}\{Y \geq 0\}] = \mathbb{E}[e^{tX}] \Phi \left(\frac{\mathbb{E}[Y] + t\text{Cov}(X, Y)}{\sqrt{\mathbb{V}(Y)}} \right),$$

and the corresponding formula for $Y < 0$ replaces the cdf argument by its negative. Applying these identities with $t = -\alpha$ and dividing by $\Pr(Y \geq 0) = \Phi((\tilde{u}_x^s - \tilde{u}_r)/\tau)$ or $\Pr(Y < 0) = \Phi((\tilde{u}_r - \tilde{u}_x^s)/\tau)$, with $\tau := \sqrt{\sigma_x^2 + \sigma_r^2 + \sigma_o^2}$, yields the two cases defining $\tilde{\psi}_\alpha$. $\square$

E.18 Proof of Proposition 23

We first verify the object-side expressions. Let $X \sim \mathcal{N}(\mu_X, v_X)$ and $N \sim \mathcal{N}(\mu_N, v_N)$ be independent with $v_X, v_N \geq 0$ and $v_X + v_N > 0$, write $a := \mu_X - \mu_N$ and $\tau := \sqrt{v_X + v_N}$, and condition on

$X > N$. Proposition 22 implies

$$\begin{aligned} & \text{CE}(X \mid X > N) - \text{CE}(N \mid X > N) \\ &= a - \frac{\alpha}{2}(v_X - v_N) + \frac{1}{\alpha} \ln \frac{\Phi((a + \alpha v_N)/\tau)}{\Phi((a - \alpha v_X)/\tau)}. \end{aligned} \quad (44)$$

Indeed, the two certainty equivalents are

$$\text{CE}(X \mid X > N) = \mu_X - \frac{\alpha v_X}{2} + \frac{1}{\alpha} \ln \frac{\Phi(a/\tau)}{\Phi((a - \alpha v_X)/\tau)}$$

and

$$\text{CE}(N \mid X > N) = \mu_N - \frac{\alpha v_N}{2} + \frac{1}{\alpha} \ln \frac{\Phi(a/\tau)}{\Phi((a + \alpha v_N)/\tau)},$$

whose difference gives (44). Taking

$$(a, v_X, v_N) = (u_x, \sigma_x^2, \hat{\sigma}_x^2)$$

gives Γ_x , while taking

$$(a, v_X, v_N) = (u_x, \hat{\sigma}_x^2, \sigma_x^2)$$

gives Λ_x . We next verify the monetary expressions. Fix $p > 0$. When the DM receives p , the ordinal comparison $u_p > u_0$ is equivalent to $d_p > 0$. Conditional on this event,

$$d_p = pZ,$$

where Z has the distribution specified in the text. Because u_0 and d_p are independent and the conditioning event depends only on d_p , they remain independent after conditioning. Additivity of CARA certainty equivalents therefore gives

$$\text{CE}(u_p \mid u_p > u_0) = \text{CE}(u_0) + \text{CE}(pZ),$$

and hence

$$\text{CE}(u_p \mid u_p > u_0) - \text{CE}(u_0 \mid u_p > u_0) = -\frac{1}{\alpha} \ln \mathbb{E}[e^{-\alpha p Z}] = \mathcal{R}(p).$$

Similarly, when the DM pays p , the ordinal comparison $u_0 > u_{-p}$ is equivalent to $d_{-p} < 0$. Conditional on this event,

$$u_0 - u_{-p} = -d_{-p} = pZ.$$

The same independence argument yields

$$\text{CE}(u_{-p} \mid u_0 > u_{-p}) = \text{CE}(u_0) + \text{CE}(-pZ),$$

and therefore

$$\text{CE}(u_0 \mid u_0 > u_{-p}) - \text{CE}(u_{-p} \mid u_0 > u_{-p}) = \frac{1}{\alpha} \ln \mathbb{E}[e^{\alpha p Z}] = \mathcal{P}(p).$$

Thus, the common monetary component u_0 , its variance A_m , and any reduction in that variance due to an additional reference signal cancel from all three elicitation problems. For completeness, suppose $\vartheta > 0$, and let $Z_0 \sim \mathcal{N}(1, \vartheta)$. Completing the square gives, for every $t \in \mathbb{R}$,

$$\mathbb{E}[e^{tZ_0} \mid Z_0 > 0] = \exp\left(t + \frac{\vartheta t^2}{2}\right) \frac{\Phi((1 + \vartheta t)/\sqrt{\vartheta})}{\Phi(1/\sqrt{\vartheta})}.$$

Setting $t = -\alpha p$ and $t = \alpha p$ gives (25) and (26), respectively. When $\vartheta = 0$, $Z = 1$ almost surely, so $\mathcal{R}(p) = \mathcal{P}(p) = p$. Since the object-side and monetary variables, signals, and ordinal comparisons are independent, posterior utility components remain independent across dimensions. Proposition 22 therefore allows their certainty equivalents to be added dimension by dimension. This gives the three indifference conditions (27)–(29). We now establish existence and uniqueness. When $\vartheta > 0$, differentiation under the expectation gives

$$\mathcal{R}'(p) = \frac{\mathbb{E}[Ze^{-\alpha p Z}]}{\mathbb{E}[e^{-\alpha p Z}]} > 0, \quad \mathcal{P}'(p) = \frac{\mathbb{E}[Ze^{\alpha p Z}]}{\mathbb{E}[e^{\alpha p Z}]} > 0.$$

Both functions are continuous and satisfy $\mathcal{R}(0) = \mathcal{P}(0) = 0$. Since $Z > 0$ almost surely, dominated convergence implies

$$\mathbb{E}[e^{-\alpha p Z}] \longrightarrow 0 \quad \text{as } p \rightarrow \infty,$$

and hence $\mathcal{R}(p) \rightarrow \infty$. Likewise, monotone convergence implies $\mathbb{E}[e^{\alpha p Z}] \rightarrow \infty$, and hence $\mathcal{P}(p) \rightarrow \infty$. Thus, $\mathcal{R}$ and $\mathcal{P}$ are continuous strictly increasing bijections from $[0, \infty)$ onto $[0, \infty)$. The same conclusion is immediate when $\vartheta = 0$, since both functions then equal p . Under the posterior induced by the noiseless object comparison, $X > N$ almost surely. Hence,

$$e^{-\alpha X} < e^{-\alpha N} \quad \text{almost surely,}$$

so that

$$\text{CE}(X \mid X > N) > \text{CE}(N \mid X > N).$$

This applies under either reference configuration and therefore implies

$$\Gamma_x > 0, \quad \Lambda_x > 0.$$

Since $\mathcal{R}$ and $\mathcal{P}$ are bijections from $[0, \infty)$ onto itself, Equations (27)–(29) admit unique strictly positive solutions. It remains to rank them. Define

$$G(a) := a - \frac{\alpha}{2}\Delta_x + \frac{1}{\alpha} \ln \frac{\Phi((a + \alpha\hat{\sigma}_x^2)/\tau_x)}{\Phi((a - \alpha\sigma_x^2)/\tau_x)}.$$

Then

$$\Gamma_x = G(u_x), \quad \Lambda_x = G(u_x + \alpha\Delta_x).$$

Using the signed inverse Mills ratio $\psi(t, +) = \phi(t)/\Phi(t)$ defined above,

$$G'(a) = 1 + \frac{1}{\alpha\tau_x} \left[\psi\left(\frac{a + \alpha\hat{\sigma}_x^2}{\tau_x}, +\right) - \psi\left(\frac{a - \alpha\sigma_x^2}{\tau_x}, +\right) \right].$$

The two arguments of ψ differ by $\alpha\tau_x$. The mean value theorem therefore gives

$$G'(a) = 1 + \partial_1\psi(\xi_a, +)$$

for some intermediate ξ_a . For a standard normal random variable W ,

$$1 + \partial_1\psi(t, +) = \mathbb{V}(W \mid W > -t) > 0.$$

Thus, G is strictly increasing. Moreover,

$$\Delta_x = \sigma_x^2 - \left(\frac{1}{\sigma_x^2} + \frac{1}{\sigma_e^2} \right)^{-1} > 0,$$

so

$$\Lambda_x > \Gamma_x.$$

Equations (28) and (29), together with the strict increase of $\mathcal{R}$, therefore imply

$$p_{\text{WTA}} > p_{\text{Ch}}.$$

Finally, for every $p > 0$, Cauchy–Schwarz gives

$$\mathcal{P}(p) - \mathcal{R}(p) = \frac{1}{\alpha} \ln \left(\mathbb{E}[e^{\alpha p Z}] \mathbb{E}[e^{-\alpha p Z}] \right) \geq 0.$$

Equality holds if and only if Z is almost surely constant, which is equivalent to $\vartheta = 0$. Since $p_{\text{WTP}} > 0$,

$$\mathcal{R}(p_{\text{WTP}}) \leq \mathcal{P}(p_{\text{WTP}}) = \Gamma_x = \mathcal{R}(p_{\text{Ch}}).$$

The strict increase of $\mathcal{R}$ therefore gives

$$p_{\text{Ch}} \geq p_{\text{WTP}},$$

with strict inequality if and only if $\vartheta > 0$. Combining the two comparisons proves the result. $\square$